

When A Nation Moves to Opportunity:

Two Essays on Rural Population, Rural-Urban Migration and Income
Inequality in Post-Reform China

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Abstract

This thesis examines how the rural population and rural-urban migration have shaped income inequality in post-reform China. Existing research has focused predominantly on the urban labour market, leaving the rural population and the processes associated with its transformation largely unexamined. Through two empirical studies, I present population- and individual-level evidence on their role in shaping income inequality and mobility.

In the first study, using nationally representative survey data spanning 1996 to 2023 and a counterfactual decomposition framework, I show that estimates of the determinants and trends of income inequality differ substantially depending on whether the rural agricultural population is included. When the full population is examined, the rural-urban and agricultural-industrial divides emerge as among the most important structural axes of income inequality, and compositional change in employment sectors — driven by migration and industrialisation — exerted an equalising effect on income inequality between 1996 and 2015.

In the second study, using longitudinal data from the Chinese Family Panel Studies and the fixed effects counterfactual estimator, I estimate a substantial positive income return to rural-urban migration of approximately 40%, persisting for six years following migration. The effect is similar across gender, with suggestive evidence of stronger effects among high-school-educated migrants. I further find suggestive evidence that the transition from agricultural to non-agricultural employment accounts for one-fifth of this effect.

Together, the findings underscore the importance of incorporating the rural agricultural population in analyses of social stratification in contemporary China and more broadly.

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Chapter 1

Introduction

During his train trip to Anyang, Henan, in 1981 – the ancient capital of the Shang dynasty, where the prototype of Chinese characters was discovered on oracle bones – the American historian David A. Keightley wrote

The land is generally flat, monotonous, one village much like another ... Where are the gentry estates, the mansions, the great houses of England and France? What was it about the society that failed to produce such monuments to civilized aristocratic living? (cited in Hessler, 2007, p. 3)

The story began in these monotonous villages in Henan, China's most populated inland province, where underground blood sale – driven by entrenched poverty – triggered an AIDS epidemic in rural areas in the 1990s. In 1990, a young couple boarded the train from Zhengzhou, the capital city of Henan, to Qingdao, the former German-Japanese colony and an industrial hub in northern China. Their newborn daughter was still in the village with her grandmother, while the 12-hour train was taking them to the city 1000 kilometres away. Amid noises from the train engine and the overloaded crowd, they would steal some hours of sleep underneath the seats, perhaps dreaming of the coastal city, about which they had no idea that they and their children would call home for the rest of their lives. Nor did they

expect that, over thirty years later, their child would write about the monotonous village and the noisy train in the great houses of an English university.

Despite being exceptionally lucky with the outcomes, the experience of the young couple was not atypical of the Chinese population in the past decades. Since China's economic reform in the 1980s, the structural and institutional changes and the rapid economic developments have transformed the lives of millions of people, including hundreds of millions of farmers in the rural heartland. With the collective rural production system and extractive policies abolished, farmers gained more autonomy in agricultural production (Whyte, 2010b; H. Lu, 2010). Meanwhile, with regional movement restrictions lifted and rapid industrialisation and development in urban areas, hundreds of millions of people like our young couple left the villages seeking opportunities in urban factories and street-side shops, leading to unprecedented waves of internal migration (Z. Liang, 2016). This thesis is about how the changing lives of these people have shaped economic inequality in contemporary China.

Despite rapid development, a substantial proportion of the Chinese population remains rural. Before 2010, the majority of the Chinese population lived in rural areas, and by 2020, the rural population still accounted for 36% of the Chinese population. Despite accounting for a large share of the population and experiencing significant disadvantages and socioeconomic changes, the rural population has largely been overlooked in existing research on social change and inequality in contemporary China. Instead, it has focused almost exclusively on the urban industrial labour market through processes such as state-sector reforms and the emergence of the labour market (X. Wu, 2019; Walder, 2023). Similarly, most empirical studies have examined the trends and causes of inequality using only data from the urban population (X. Zhou, 2014; J. Lu, 2018; J. Lu and Feng, 2025). As a result, the rural population — and the consequences of its transformation for national income distribution — remain poorly understood (Bian, 2002).

Through two studies, the thesis examines the role of the rural population in shaping income inequality in contemporary China from complementary levels of analysis. The first

study, in Chapter 2, takes a population-level perspective. Drawing on nationally representative surveys spanning 1996 to 2023, it examines how the importance of income predictors and the contributions of socioeconomic changes to income inequality differ when the rural agricultural population is included alongside the urban industrial population. The results show that the rural-urban and agricultural-industrial divides are among the most important structural axes of income inequality in contemporary China, and that processes related to the rural sector — particularly changes in between-sector income gaps and sectoral compositional shifts driven by migration and industrialisation — reduced income inequality in ways that were previously overlooked in analysis of the urban population.

Having established the importance of rural-urban migration and the industrialisation of the rural labour force in shaping inequality, the second study in Chapter 3 further examines the economic consequences of rural-urban migration for rural workers from a micro-level life course perspective. Using individual-level longitudinal data from the Chinese Family Panel Studies, it estimates the average income effect of rural-urban migration on rural-origin workers using a fixed effects counterfactual estimator. Across specifications, the results consistently show a substantial positive income return to migration of approximately 40%, persisting for over six years. A further analysis of the mechanism finds suggestive evidence that the transition from agricultural to non-agricultural employment accounts for around one-fifth of the total income effect, and a larger share among workers initially employed in agriculture.

Taken together, the two studies form a multi-level account of how the movement of rural workers out of agriculture and into the urban industrial economy – as a structural and demographic process – shapes economic inequality in contemporary China. At the population level, this process reshaped the labour force structure and income distribution in post-reform China in ways that existing urban-focused research has overlooked. At the individual level, the movement generated substantial economic gains for the migrant workers, functioning as an important channel of upward mobility despite significant institutional obstacles. Overall,

the thesis underscores the importance of structural and demographic change, in addition to institutional or market factors, in shaping socioeconomic inequality, and demonstrates the necessity of incorporating the rural agricultural population in any account of social stratification in contemporary China.

More broadly, this study highlights the importance of the rural population in the analysis of social change and stratification in general. Social stratification research has predominantly focused on modern, industrial societies. Yet, as comparative and historical evidence has consistently shown, developments in the rural sector and the process of rural-urban, agricultural-industrial transition are among the most important drivers of development, inequality and mobility — from nineteenth-century Britain and the United States (Long, 2005; MacDonald, 2021; X. Song et al., 2020) to twentieth-century Europe (Breen et al., 2009; Maas and van, 2016; Ager, Goñi, and Salvanes, 2026). China’s experience, with its exceptional scale and institutionalised rural-urban segmentation, offers a powerful contemporary case in this body of literature, and one in which the rural-to-urban transition remains unfinished. As social scientists increasingly seek to understand stratification from longer-term and more globally comparative perspectives, the rural agricultural population and the processes of its transformation deserve more analytical attention, not merely as a historical precursor to industrial modernity, but as a shaping force in its own right.

Chapter 2

Income Inequality in Post-Reform China Revisited (1996-2023): Bringing in the Rural Population

Since the economic reform in the 1980s, China has undergone transformational development and socioeconomic changes that have attracted worldwide attention. However, alongside its widely admired, unprecedentedly rapid development, China has also witnessed a rapid increase in economic inequality. Xie and X. Zhou (2014) estimate that the household income Gini coefficient in China increased from 0.48 in 2005 to 0.53 in 2012, while the US figure was relatively stable at about 0.41 in the period. Corrected for top-earners' income, Piketty, Yang, and Zucman (2019) estimate that the top 1% income share in China increased from 6% at the start of the reform in 1978, to about 10% at the turn of the century, and then increased rapidly to and peaked at about 15% in 2008. While evidence has shown that income inequality in China has stabilised and gradually decreased since 2010 (Kanbur, Y. Wang, and X. Zhang, 2021; Piketty, Yang, and Zucman, 2019), it remains at a high level, surpassing many Western economies and is close to that in the US.

A large body of literature in sociology and economics has examined the causes of such

rapid increases in inequality. On the one hand, sociological research, since the beginning of China's economic reform, has examined the changing or persistent role of state institutions on individuals' life chances in the process of market transition (see X. Wu, 2019; Walder, 2023, for recent reviews). On the other hand, recent studies in economics have examined the distributional consequences of structural economic changes such as state-owned enterprise (SOE) reforms, globalisation and technological change (J. Lu, 2017; J. Lu, 2018; J. Lu and Feng, 2025; Han, R. Liu, and J. Zhang, 2012).

What is missing from this literature, however, is the rural population. Almost all empirical studies on the patterns and causes of income inequality, from both sociology and economics, focus on the urban labour market, while the rural economy is left to the separate literature on rural development and rural-urban inequality. This omission is surprising as the vast rural-urban division constitutes one of the most important dimensions of structural inequality in contemporary China (Whyte, 2010b; Xie and X. Zhou, 2014) and rural areas account for a significant proportion – in fact, until 2010, the majority – of the Chinese population (National Bureau of Statistics of China, 2021b). This omission, therefore, leads to an incomplete account of changes in inequality in contemporary China.

Furthermore, substantively, excluding the rural population also hinders understanding the structural socioeconomic changes and their implications, as embedded in an integrated system. While many of the socioeconomic changes in China's economic reform, such as ownership reforms and industrialisation, primarily occurred in and affected the urban industrial sector, the rural sector has played a critical enabling role, thus being equally affected in these processes. For example, the direct agricultural tax, which was abolished as late as 2006, effectively subsidised the urban industrial sector with primary resources (X. Wang and Shen, 2014). The massive waves of rural-urban migration also provided labour supply for the urban labour-intensive industries (Imbert et al., 2022; Z. Liang, 2016). Excluding the rural population in the analysis, therefore, limits our understanding of the inequality consequences of these socioeconomic changes.

This study, therefore, aims to provide a more complete and up-to-date account of the relationship between socioeconomic changes and income inequality in post-reform China by bringing the rural population into the analysis. Drawing on nationally representative surveys from 1996 to 2023, I examine the changing predictors of income and quantify the contributions of different socioeconomic changes to income inequality in 1996-2010, 2010-2015 and 2015-2023 using variance decomposition and counterfactual analysis. I conduct the analysis separately using the complete national sample and the sub-sample of urban industrial workers to examine the effect of including the rural population on the results.

First, R^2 -based measures show that rural-urban status and employment sectors are consistently two of the most important predictors of income in post-reform China, but only in the complete sample. This suggests that the rural-urban divide and the agricultural-industrial divide are two important dimensions of income inequality in contemporary China that are overlooked if focusing on the urban labour market. I also find that the importance of factors such as education and province varies when using the complete and urban samples.

Second, counterfactual analysis shows that changes in between-group income gaps and within-group income variation among education and employment sector groups are the most important contributors to income inequality in China between 1996 and 2010. While the change in returns to education increased inequality by similar magnitudes in the complete and urban samples, the changes in between-sector income gaps reduced inequality, with the effect being much stronger in the complete sample. Furthermore, counterfactual composition analysis shows that while the inequality-increasing effect of education expansion between 1996 and 2010 is similar in both samples, the effect of sector compositional change shows opposite signs in the complete and urban samples. While changes in the employment sector composition increased inequality in the urban sample, it actually reduced inequality in the complete sample by breaking the agricultural-industrial income segmentation. The inequality-reducing effect continued in 2010-2015.

Overall, the analysis shows that, both in terms of the predictors of income variation

and the contributors to changes in income inequality, the results differ significantly when examining the national population versus the urban-industrial sub-population. Therefore, it demonstrates the importance of incorporating the rural agricultural population in the analysis of income inequality. In particular, it highlights the role of changes in the agricultural sector – its improvements in relative income level and decline in employment share – in shaping income inequality. The result implies that processes such as agricultural reforms, rural-urban migration, and industrialisation could have been equally important in explaining changes in income inequality in post-reform China as institutional and market reforms.

More broadly, by demonstrating the importance of the rural population in shaping and understanding income inequality in contemporary China, the study contributes to research on social change and stratification. Since the discipline’s foundation, sociologists have predominantly focused on the division and stratification of labour in “modern” industrial society. And when the rural population is concerned, the vocabularies and methods developed for industrialised labour appear to be analytically inadequate (Kaufman et al., 1953). What is less often recognised is that during much of the time sociologists have studied industrial societies, considerable proportions of the population in today’s developed societies were still rural and agrarian and many social changes were driven by developments in the rural sector. Recent research has increasingly demonstrated that socioeconomic changes in the rural population in the twentieth century – such as improved access to nutrition and education and migration and entry into industrial occupations – substantially affected stratification patterns such as intergenerational mobility and gender inequality in US and European societies (Breen et al., 2009; X. Song et al., 2020; Ager, Goñi, and Salvanes, 2026). The Chinese experience, therefore, provides a valuable example demonstrating the importance of the rural population for understanding social change and stratification.

The rest of the study proceeds as follows. Section 2.1 provides a brief review of socioeconomic changes that affect income inequality in post-reform China and existing evidence. Section 2.2 discusses the rural population as an important yet often overlooked aspect of

Chinese development and income inequality. Section 2.3 explains the study’s empirical aims and plans for the analysis. Section 2.4 describes the survey data and Section 2.5 shows descriptive trends from the data. Section 2.6 evaluates the importance of predictors of income using R^2 -based measures. Section 2.7 describes the decomposition framework for counterfactual analysis in non-parametric terms, and Section 2.8 explains the empirical methods to conduct the counterfactual analysis. Section 2.9 presents the counterfactual results, with Section 2.10 discussing the limitations. Section 2.11 concludes with implications for future research.

2.1 Socioeconomic Changes and Inequality in Contemporary China

The institutional and structural changes in contemporary post-reform China have shaped income inequality through various mechanisms. In general, they can be grouped into three categories: 1. changing returns to individual characteristics, 2. structural changes in industry and the labour force, and 3. direct institutional and policy changes. This section reviews existing explanations and evidence on these mechanisms.

2.1.1 Returns to Individual Characteristics

The sociological market transition theory argues that, as China and other former socialist countries transition from a centrally planned, socialist society to a more market-oriented society, the fundamental principles of resource rewarding and distribution change. While in the socialist system, resources were institutionally managed and allocated, thus suppressing individual differences and favouring powerful groups such as party cadres and senior managers, the market economy allows more rewards to individual skills and entrepreneurship. Therefore, with market transition, the returns to education and skills would increase and thus increase inequality by amplifying individual differences in economic attainments (Nee,

1989; Nee, 1991; Nee, 1996).

Consistent with this prediction, empirical studies have found increases in the return to education in China over the transition period (Xie and Hannum, 1996; Hauser and Xie, 2005; X.-G. Zhou, 2000; X. Wu and Xie, 2003). More recently, labour economics studies have found that globalisation and technological changes have also increased the return to education (Han, R. Liu, and J. Zhang, 2012; J. Lu, 2017). Overall, decomposition analyses have identified the increase in the return to education as the most important driver of income inequality in post-reform China. For example, X. Zhou (2014) estimates that it can explain 45.2% of the total change in income variance between 1996 and 2010; while J. Lu (2018) estimates that it can explain 31% of the increase in the 90th/10th income quantile ratio between 1995 and 2002, and 40% between 2002 and 2013.

On the flip side, the market transition theory also predicts that, as the state's role in resource allocation diminishes, the power and economic advantages of party cadres and managers would also decrease. However, existing studies have found persistent economic advantages enjoyed by party cadres (X.-G. Zhou, 2000; Bian and Logan, 1996). The long-term trend and the overall contribution of the changes in positional power, however, currently lack systematic examinations.

Finally, the decomposition analysis by J. Lu (2018) highlights the role of changing gender wage gap in affecting income inequality. The study estimates that between 1995 and 2002, the narrowing of the gender wage gap decreased the 90th/10th ratio by 51%. However, between 2002 and 2013, the gender wage gap widened again and contributed 40% to the increase in the 90th/10th ratio. Though previously less theorised, the widening gender wage gap could be due to the removal of the protective effect of the socialist institution, which, at least in rhetoric, promoted gender equality and provided public childcare for female workers.

2.1.2 Structural and Compositional Changes

The structural reforms and socioeconomic development in China have also led to changes in the labour force and industry composition that affect overall income inequality. Three changes have been particularly noticeable: 1. the contraction of the state sector and the growth of the private sector, 2. education expansion and 3. massive waves of internal migration.

Rebalance of the State and Private Sector

State-owned enterprise (SOE) reforms and the opening-up of the private sector were two of the most important features of the Chinese economic reform. In the pre-reform socialist system, most non-agricultural workers were employed in SOE work units (*danwei*), which were characterised by rigid hierarchies but also good welfare – in many cases, including free childcare and healthcare – and job security (Walder, 1986). Under this system, while workers’ income and welfare heavily depended on the firms’ administrative level and profitability, there was relatively little job insecurity, uncertainty and income variation within workplaces (Xie and X. Wu, 2008).

However, due to their inefficiency and increasing indebtedness, from the mid-1990s, a series of reforms were implemented to reorganise SOE ownership and governance and shut down inefficient plants. Many workers were laid off, and the employment share of the state sector rapidly decreased. Meanwhile, the private sector was rapidly growing and absorbed many of the laid-off workers. Based on official statistics, J. Lu and Feng (2025) documents that the employment contribution of the non-financial SOE in urban China decreases rapidly from 40.5% in 1995 to 18.5% in 2002, and further to 9.8% in 2013. Meanwhile, the contribution of private firms increased from 19.6% to 56.3% between 2002 and 2013.

This compositional change in the employment structure could impact income inequality through two mechanisms. First, similar to the market transition process, as more workers enter the private sector, which allows higher and more varied returns to individual skills

and traits, between-group income variation would increase, especially between workers with different levels of skills. Second, because the market sector does not provide the same level of security and welfare as state-owned workplaces, workers now face greater insecurity and uncertainty, leading to increasing within-group inequality (X. Zhou, 2014).

Decomposition evidence generally suggests that sectoral compositional change mattered for inequality, although the direction and magnitude vary by period and inequality measure. For example, X. Zhou (2014) estimates that the increase in within-group income variation due to the compositional change between the state and market sector can account for 20.8% of the total increase in the variance of logged income between 1996 and 2010 in urban China. Using a different inequality measure, J. Lu (2018), however, paints a more nuanced picture. He estimates that the sectoral compositional change led to a 47% decrease in the 90th/10th ratio of logged income between 1995 and 2013, but a large 85% increase between 2002 and 2013.

Education Expansion

While the demand side of the Chinese labour market underwent significant transformation through market reforms and developments, the supply side was also changed by the rapid education expansion, in particular with the higher education expansion since 1999. Between 1995 and 1998, the average growth rate of college admissions was 4.7%; then, the figure jumped to over 40% in 1999 and 2000, and remained at about 20% until 2005. As a result, the total number of college graduates has increased from just about 1 million in 2000 to over 6 million in 2011, and further to over 8 million in 2018 (Che and L. Zhang, 2018; Hanushek, Y. Wang, and L. Zhang, 2025, based on official statistics). It should be noted, however, that the expansion has not equally benefited everyone. Research has found that intergenerational education mobility in China has decreased over time since the market reform and education expansion (Xie, H. Dong, et al., 2022; Hannum, Ishida, et al., 2019) and the rural population is particularly disadvantaged (Rozelle and Hell, 2022; Hannum, M. Wang, and Adams, 2010).

The rapid expansion of education could affect income inequality through two mechanisms: by changing the labour force composition and by changing the returns to education. First, the rapid yet unequal expansion of education has created a more dispersed distribution of education. Before the expansion, most people had relatively similar, lower levels of education; however, after the expansion, as more people gained higher education, the distribution became more dispersed. This compositional change, therefore, would mechanically increase the income dispersion. Empirically, X. Zhou (2014) estimates that the increase in between-group inequality due to compositional changes in workers' education could explain 21.5% of the increase in income variance between 1996 and 2010. However, J. Lu (2018)'s estimates based on the 90th/10th ratio are smaller: the compositional change in education contributed -2% to the increase between 1995 and 2002 and 5% between 2002 and 2013.

Second, the education expansion may decrease the returns to education by increasing the relative supply of skilled workers. While economic development, globalisation and technological changes in China have generated vast demands for skilled labour, thus leading to increasing returns to education, the expansion of education acts as a countervailing force that increases relative supply. For example, the recent study by Hanushek, Y. Wang, and L. Zhang (2025) has shown evidence for the reversal of the race between the supply and demand of education in China. It estimates that, between 2007 and 2018, while the return to high school has been relatively stable, the return to college degrees has decreased by over 20%. Other things equal, this would lead to a decrease in income inequality.

Internal Migration

The structural reform and development in China have been characterised by its vast spatial disparity. While the coastal provinces have benefited from favoured geographic and policy conditions and enjoyed rapid transformation into industrial powerhouses, inland provinces lagged considerably in their pace of institutional reform and economic development. Furthermore, development has mostly concentrated in urban areas while the rural areas were left

behind (Xie and X. Zhou, 2014; Kanbur and X. Zhang, 2005). Such unequal development, combined with the loosening of internal migration restrictions, has led to massive waves of internal migration, from inland to coastal provinces and from rural to urban areas. Based on the official census, Z. Liang, Z. Li, and Ma (2014) documents that the total size of the between-county migrant population almost doubled from 43.1 million in 1995 to 78.8 million in 2000, and further almost tripled to 221 million in 2010 (see also Z. Liang, 2016).

How do such dramatic demographic changes affect income inequality? On the one hand, internal migration could reduce inequality by providing workers from less developed areas with mobility channels to more opportunities and economic rewards. It therefore counterbalances the regional inequality in development. However, on the other hand, migrants, particularly those from rural areas, have been subject to persistent institutional and informal discrimination. For example, the *hukou* household registration system has institutionally excluded rural migrants from many public services. Empirical studies have found rural-urban migrants to be significantly disadvantaged compared to the urban residents, with lower income and higher downward mobility (X. Wu and Treiman, 2007; X. Wu and Zheng, 2018). The massive waves of migration could therefore lead to more people being subject to such disadvantages and thus more inequality.

Empirical studies have generally found positive contributions to income inequality by rural-urban migration. X. Zhou (2014) estimates that the increase in within-group income variance due to increases in the proportion of rural-urban migrants in urban labour markets could explain 5.6% of the increase in income inequality in urban China between 1996 and 2010. The estimate, though small, supports the latter inequality-enhancing mechanism. Focusing on regional composition, J. Lu (2018) estimates that the change in regional composition of the labour force contributed 8% to the increase in the 90th/10th income ratio between 1995 and 2002 and 6% between 2002 and 2013.

It should be noted, however, that because the empirical studies focused only on the urban population, they cannot capture the income-improving effect of migration on rural-

origin workers but only the difference between migrant and urban workers. Therefore, the results and conclusions are likely incomplete. I will return to this problem later.

2.1.3 Institution and Policy

A final, though less systematically studied, factor on inequality in contemporary China is market institutions and policies. During the early stages of economic reform and development in China, economic growth was by far the top policy priority, and there were limited policy efforts on market regulation and labour protection. However, with rising inequality and social tensions, market regulation and workers' rights have started to gain more policy attention since the 2000s and new legislation has been passed. For example, nationwide minimum wage standards were established in 2004, and several new laws on labour relationships were passed in 2008 (J. Lu and Feng, 2025).

More recently, the Xi Jinping administration has taken bolder actions to tackle economic inequality, such as anti-corruption investigations, crackdowns on lucrative industries such as technology and finance, poverty alleviation and elimination programmes, and expansions of the welfare system (Whyte, 2024). Since 2021, "common prosperity (*gong tong fu yu*)" has become a central policy agenda. Although the nature of the slogan is debatable, it gestures commitments to more inclusive growth and less inequality (G. Wu, 2022). However, the effects of these policy changes and actions on inequality have not been systematically examined and quantified (J. Lu and Feng, 2025).

The aforementioned *hukou* system is another policy area that has undergone substantial reforms with potentially significant impacts on inequality. A legacy from the Mao-era collective economy, the system has been a central tool for the state to control population movements and allocate resources, primarily between the rural and urban populations. The system effectively created a two-tier citizenship system in which people with rural *hukou* were excluded from many public services or even banned from living in cities without permits (Chan and L. Zhang, 1999; J.-M. Wu, 2010).

Since the early 2000s, with increasing internal migration, the social problems and injustice under the *hukou* system have drawn more public concerns. One pivotal incident was the death of a young college graduate and migrant worker, Sun, in 2003 due to physical abuse while being detained because of lacking a resident permit. Policy changes had since been pushed to reduce direct restrictions on internal migration and prevent direct discrimination and abuse against migrant workers (F.-L. Wang, 2004; Chan and Buckingham, 2008). Empirically, J. Sun (2017) finds that the returns to education for migrant workers substantially increased and the determinants of wages became more similar between 2001 and 2010, suggesting reduced discrimination and improved integration.

Since 2014, there have been further policy pushes to lower settlement barriers and expand the coverage of local public services. In 2014, the legal distinction between rural and urban *hukou* was abolished, and access to public services began to be based on individuals' place of settlement. While many cities opened up point-based settlement – with points awarded based on criteria such as education, age, and tax contributions – the barriers remain high for many migrants, who often have lower levels of education and less stable tax contributions, especially in first-tier megacities (T. Hu and Leahy, 2024). In the latest push to relax public service restrictions for migrant workers, on the 22nd of May 2026, the government announced new measures to ensure access to public services at people's residence rather than settlement locations (M. Chen, 2026). Overall, while institutional barriers to migrants' integration persist, recent policy changes should reduce migrant workers' disadvantages and help reduce inequality, though systematic evaluations are still lacking.

2.1.4 Summary

To summarise, income inequality in contemporary China has been shaped by multiple socio-economic changes. Table 2.1 summarises the relevant changes and related empirical evidence.

Table 2.1: Socioeconomic Changes Affecting Income Inequality in Post-Reform China

Change	Reasons	Mechanism	Evidence
<i>Returns to Individual Characteristics</i>			
Changing returns to education	Market transition; globalisation; skill-biased technological change; education expansion	Wider between-education income gaps; gaps reduced in more recent years due to the increase in supply	Explains 45% increase in income variance between 1996 and 2010 (X. Zhou, 2014); 31% and 40% increase in the 90th/10th ratio in 1995-2002 and 2002-13 (J. Lu, 2018)
Changing returns to political positions	Market transition; persistent state power	Reduced/persistent income premium to political power	Evidence of persistent party wage premium (Bian and Logan, 1996; X.-G. Zhou, 2000); no estimates on the systematic contribution
Changes in the gender wage gap	Removal of socialist protections/restrictions	Bigger/smaller gender wage gap	First reduced the 90th/10th ratio by 51% (1995-2002), but then widened it by 40% (2002-2013). (J. Lu, 2018)
<i>Structural and Compositional Changes</i>			
Movement from the state sector to the private sector	SOE reforms; market liberalisation	Exits from the state sector with job security and welfare into the private sector create more income uncertainty.	Explains 20.8% of the increase in within-group income variance between 1996-2010 (X. Zhou, 2014); explain 85% of the 2002-13 increase in the 90th/10th ratio (J. Lu, 2018).

continued on next page

(Table 2.1 continued)

Change	Reasons	Mechanism	Evidence
Education expansion	Rapid, unequal education expansion since 1999	More dispersed education distribution mechanically raises between-group dispersion	Explains 21.5% of the increase in between-group variance in 1996-2010 (X. Zhou, 2014); small effects on 90th/10th (J. Lu, 2018).
Internal migration	Regional/rural-urban development gaps; relaxed mobility	More migrant workers are subject to economic disadvantages and uncertainty.	Migration explains 5.6% of the increase in residual income variance in 1996-2010 (X. Zhou, 2014); regional composition accounts for 6 and 8% of the increase in the 90th/10th ratio in 1995-2002 and 2002-2013 (J. Lu, 2018)
<i>Institutions and Policy</i>			
Labour market regulation	-	Minimum wage and improved labour market protection and regulation reduce inequality	No systematic evidence.
Anti-inequality agenda (“common prosperity”)	-	Redistribution and sector regulation may compress top incomes; poverty alleviation	No clear empirical evidence.
Hukou reform	-	Reduced institutional discrimination reduces migrant disadvantages	Evidence of rising migrants’ returns to education and wage convergence with urban workers (J. Sun, 2017); lacks systematic evaluation.

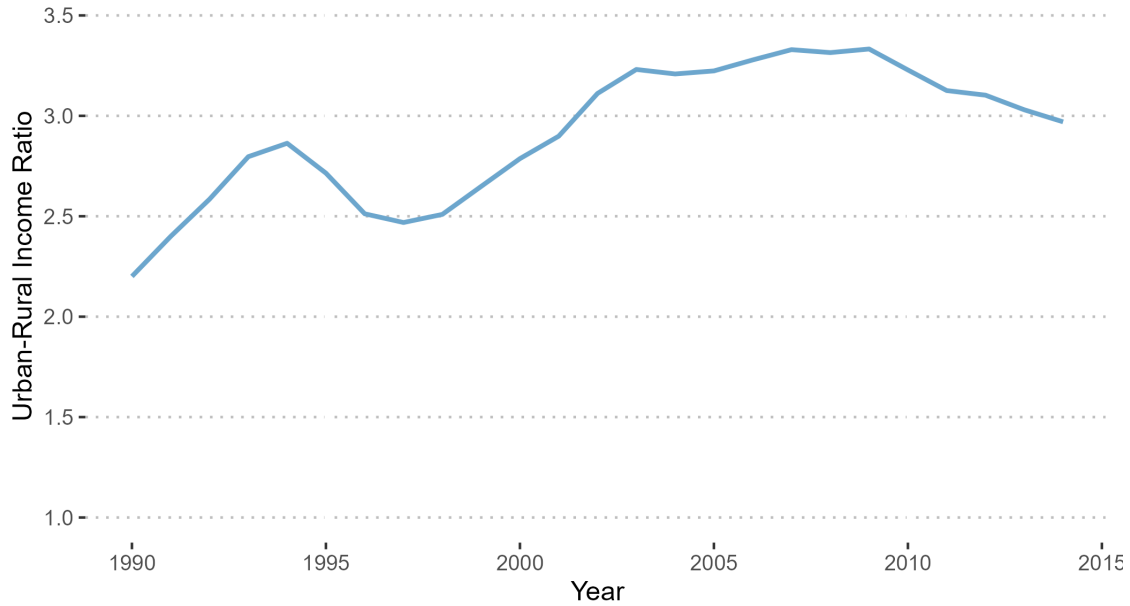
What is missing from these accounts and related empirical analyses, however, is the rural population. Accounting for a substantial proportion of the Chinese population, such an omission poses a serious limitation on our understanding of income inequality in contemporary China. The next section turns to this issue.

2.2 The Rural Population and Inequality

Despite rapid industrialisation and urbanisation, the rural areas account for a considerable proportion of the Chinese population. In fact, according to the official census, until 2010, the majority of the Chinese population still lived in rural areas. In 1990, only 26.4% of the Chinese population lived in urban areas. The proportion increased to 36.2% in 2000 and 50% in 2010. By 2020, 63.9% of the Chinese population lived in urban areas (National Bureau of Statistics of China, 2021b). Meanwhile, the urbanisation rate in 2020 was 73% in Europe and 82% in North and South America (World Bank, 2025).

The large rural population in China has also been characterised by substantial socio-economic disadvantages relative to the urban population due to unequal development and historical institutional discrimination. Figure 2.1 shows that, according to official statistics, the annual disposable income of urban residents has been consistently twice as high as that of rural residents since 1990. The gap expanded rapidly during the early reform years, between the mid-1990s and early 2000s, from 2.5 to over 3, and has remained so in the following decade. Estimates from survey data, adjusted for price differences and subsidies, show similar results (S. Li and C. Luo, 2010; Sicular et al., 2010). Compared to figures from other developing countries, which are usually around or below 2, the Chinese rural-urban gap is exceptionally large (Sicular et al., 2010).

In addition to the income gap, the Chinese rural population has been subject to institutional disadvantages with unequal access to public services and extractive policies. First, during the reform, the state reduced the provision of and expenditure on public services,



Notes: Data based on the 2015 China Statistical Yearbook (National Bureau of Statistics of China, 2015)

Figure 2.1: Urban-Rural Average Annual Disposable Income Ratio in China, 1990-2014

such as education and healthcare. The rural population was disproportionately affected because they were less able to access and afford alternative options. Even with the new welfare system being gradually established in the early 2000s, the rural areas still lagged considerably in both the pace and scale. Taken together, these disadvantages have led to worse education and health outcomes in the rural population, which further reinforce their economic disadvantages (Hannum, M. Wang, and Adams, 2010; Yip, 2010; Rozelle and Hell, 2022).

Second, under the socialist planned economy, the rural agricultural sector was heavily intervened in by the state through, for example, price suppression and direct agricultural tax. Under this system, the rural economy was effectively extracted to subsidise the urban, industrial sector. It was not until 2006 that these policies were formally abolished, and the policy objectives shifted to supporting and subsidising the rural sector. While the policy changes have significantly improved rural families' economic conditions, the extractive policies have resulted in cumulative disadvantages for the rural population (X. Wang and Shen,

2014).

Overall, the vast rural population and the large economic gap make the rural-urban divide one of the more prominent dimensions of inequality in contemporary China. For example, Xie and X. Zhou (2014) estimates that, in 2010, the rural-urban gap alone can explain over 10% of the total household income variance in China, while the corresponding figure for the US is negligible.

However, despite its significance, the rural population is often excluded in empirical analysis of social stratification in China. For example, almost all the empirical studies reviewed in the previous section are based only on the urban sample. As Bian (2002) summarises in a review of the literature on inequality in contemporary China at the *Annual Review of Sociology*, "there is relatively richer research output on urban than on rural China, despite the greater and more profound transformations that occurred in rural China." (p. 91). This omission substantially limits our knowledge about social stratification and inequality in contemporary China.

Excluding the rural population not only omits a large and disproportionately lower-income part of the income distribution but also limits our understanding of how socio-economic changes take place and shape inequality in an integrated rural-urban system. To see this, consider the effect of rural-urban migration and sectoral changes on income inequality. Decomposition studies with the urban industrial population, as reviewed above, have found that rural-urban migration has had a moderate positive effect on income inequality, while the sectoral change from the state to the private sector has been one of the biggest drivers of income inequality (X. Zhou, 2014; J. Lu, 2018). Substantively, these results reflect that in the urban labour market, 1. the direct income disadvantages of migrant workers are relatively minor, and 2. the transition to the private sector creates more heterogeneity and uncertainty.

However, the processes and their consequences may change when considering both the rural and urban populations. Considering rural-urban migration, while migrant workers are

disadvantaged when compared to local urban workers, they are advantaged compared to the rural stayers, with access to more opportunities. By breaking the rural-urban segmentation, rural-urban migration in this case could reduce income inequality. As for sectoral changes, while the transition between the state and the private sector was the biggest structural change in the urban industrial population, a much larger transformation was the transition from the agricultural to the industrial sector. As workers move from agriculture into industrial or self-employed work, these flows may reduce inequality by weakening the income segmentation between agricultural and non-agricultural sectors.

In fact, the increase in intergenerational upward mobility from the agricultural to the industrial sector due to rural-urban migration and industrialisation, amid a general decrease in social mobility, has been consistently identified as a unique feature of the mobility regime in post-reform China (X. Zhou and Xie, 2019; Xie, H. Dong, et al., 2022). Omitting the rural agricultural population in the analysis of income inequality, however, effectively prevents us from understanding the inequality consequences of such important changes in the stratification system.

Similar cases could be made for other socioeconomic changes. For example, with education expansion, given the large rural-urban gaps in resources and students' educational attainment, the education expansion may disproportionately benefits the urban population and thus exacerbate educational inequality. Excluding the rural population, therefore, may lead to underestimating the inequality-increasing effect of education expansion.

To summarise, the large rural-urban gap and significant rural-urban changes make the rural population a critical dimension for understanding social stratification and inequality in post-reform China. However, by omitting the rural population, existing studies have provided incomplete, if not misleading, accounts of the relationships between socioeconomic changes and income inequality in contemporary China, particularly on processes such as rural-urban migration and sectoral changes. The aim of this study, therefore, is to reassess the relationships by bringing in the rural population, and in doing so, provides a more

complete account of the relationship between socioeconomic changes and income inequality in contemporary post-reform China.

2.3 Research Design

To assess the relationship between socioeconomic changes and income inequality in post-reform China, I draw on two comparable nationally representative surveys: the 1996 Life History and Social Changes in Contemporary China (LHSC) survey and the Chinese General Social Survey (CGSS) from 2010, 2015 and 2023.

The three waves of CGSS are chosen for substantive reasons. While most studies have focused on the socioeconomic changes in early post-reform years in the 1990s and early 2000s, the Chinese economy since 2010 has undergone a new wave of transformations as the workforce becomes more skilled and urbanised, and the economy becomes more capital-intensive and service-oriented with higher value added (J. Lu and Feng, 2025; Hanushek, Y. Wang, and L. Zhang, 2025). As discussed above, existing research has found that income inequality has stabilised and declined since 2010.

Since 2015, with the Made in China 2025 initiative, the country has entered a new wave of technological development, as the government seeks to transform the economy from being driven by “low-end” labour-intensive manufacturing to automated production and digital economy; meanwhile, it also faces new challenges, such as slower growth, high debts and structural disruptions from technological development (Lei, 2023). These new changes could have affected income inequality in ways that have not been systematically examined. The LHSC and three waves of CGSS data, therefore, capture the state of the Chinese economy and society in the three distinct stages of development.

Using the four waves of surveys, the empirical analysis proceeds in two parts. In the first part, I provide descriptive evidence on trends in income level and inequality, and changes in the labour force composition. Then, I examine the changing importance of predictors of

income in post-reform China using R^2 -based measures (Xie and X. Zhou, 2014).

In the second part of the study, I decompose the change in income inequality and quantify the contributions of socioeconomic changes in different periods of China’s post-reform development. Following X. Zhou (2014)’s framework, I decompose the change in income inequality into four components: 1. changes in between-group gaps, 2. changes in within-group variation, 3. changes in between-group gaps due to compositional changes and 4. changes in within-group variation due to compositional changes. Then, I quantify the counterfactual contributions of changes in individual attributes and compositional changes to changes in income inequality in 1996-2010, 2010-2015 and 2015-2023 through the four channels.

To assess how the results vary when incorporating the rural agricultural population, I conduct the R^2 and decomposition analysis separately using the complete sample and the urban industrial sub-sample and compare the results. The next section describes the analytical sample and measurements.

2.4 Data

Both LHSC and CGSS are nationally representative cross-sectional surveys of adults aged 18 and above. The two surveys share similar sampling designs. Households were sampled using multi-staged stratified sampling, and one adult from each household was randomly selected to answer the survey. Both surveys include separate samples for rural and urban households based on their registration at rural or urban local administrative units, i.e., village committee (*cun wei hui*) or neighbourhood office (*jie dao*). Furthermore, the questionnaire structures of LHSC and the post-2010 waves of CGSS are similar. Finally, both surveys were conducted by the Department of Sociology at Renmin University of China. Overall, these similarities ensure that the two surveys are comparable in coverage and measurement.

For each wave of LHSC and CGSS, I include individuals aged between 20 and 69, employed in non-military sectors, with real annual income (adjusted to the 2010 base) no less

than 50 yuan and valid information on *hukou*, education and employment type. The province coverage varies somewhat across years, especially for more remote provinces. I drop observations from Tibet and Hainan because they were included only in the 2010 CGSS survey. Because the two provinces account for only a small proportion of the Chinese population, the omission should not significantly affect the data’s representativeness.¹ For each year, I create an urban sub-sample that includes only urban residents and migrants working in non-agricultural sectors.²

CGSS collects respondents’ total annual income, including wages, bonuses, and private business and/or agricultural profits. For the LHSC, because annual business and agricultural profits are reported at the household level, I follow X. Zhou (2014) and calculate individual total annual income by adding individual labour income to the sum of all household profits divided by the household size. For each year, the income is inflation-adjusted to the 2010 base using the official Consumer Price Index.

LHSC and CGSS include a consistent series of individual demographic and socioeconomic characteristics, including employment status, type and firm ownership. Based on the employment information, individuals are classified as self-employed or working in the agricultural, private, or state sectors. The agricultural sector includes mostly farmers, and the state sector includes SOE employees and civil servants. I also classify individuals as rural residents, urban residents or rural-urban migrants based on their residence location and *hukou* status: an individual is classified as a rural/urban resident if they live in the rural/urban area and have a rural/urban *hukou* and as a rural-urban migrant if they live in the urban area with a rural *hukou*. There is a small proportion of individuals with urban *hukou* living in the rural areas. For simplicity, I classify these individuals as rural residents. Because the effect of urban *hukou* in rural areas is limited and the size of this group is relatively small, such

¹According to the 2020 Census (National Bureau of Statistics of China, 2021a), Tibet and Hainan account for 0.26% and 0.72% of the total population, respectively.

²The sample selection procedure is adapted from X. Zhou (2014)’s original procedure. The sample sizes of the urban samples also overall match the original analytical samples, which have $N_{1996} = 2,019$ and $N_{2010} = 3,050$. Section A.1 describes the difference from X. Zhou (2014)’s original procedure in more detail.

classifications should not substantially distort subsequent results.

Table 2.2 reports the final sample sizes for each year. The sample size for 2023 is particularly small because the 2023 wave of the CGSS uses a mix of remote telephone and in-person interviews, and only the in-person interviewed individuals are included in the analytical sample because questions on rural-urban residence and *hukou* were otherwise not asked. A separate set of sampling weights is applied specifically for the in-person-interviewed sub-sample to maintain population representativeness. All subsequent analyses using the LHSC and CGSS data are weighted by the provided sample weight to achieve population representativeness.

Year	N_{total}	N_{urban}
1996	4,777	2,021
2010	6,198	3,106
2015	5,534	2,801
2023	2,378	1,292

Table 2.2: Analytical sample size by year and sub-samples.

2.5 Descriptive Trends

2.5.1 Income Level and Inequality

How have the average income level and inequality changed in post-reform China? Figure 2.2 shows the trends of mean annual income and the variance of logged income as a measure of inequality (Allison, 1978), separately for the complete sample and the urban industrial sub-sample, estimated from the LHSC and CGSS data. Additional data from the 2005 wave of CGSS is added using the same sample selection conditions described above.

Panel 2.2a shows the rapid growth in average annual income in post-reform China. The national average real annual income has increased sevenfold from 3,394 yuan in 1996 to 23,837 yuan in 2023, measured in the 2010 price level. The income level in the urban industrial sector is consistently higher than the national average, increasing from 6,424 in

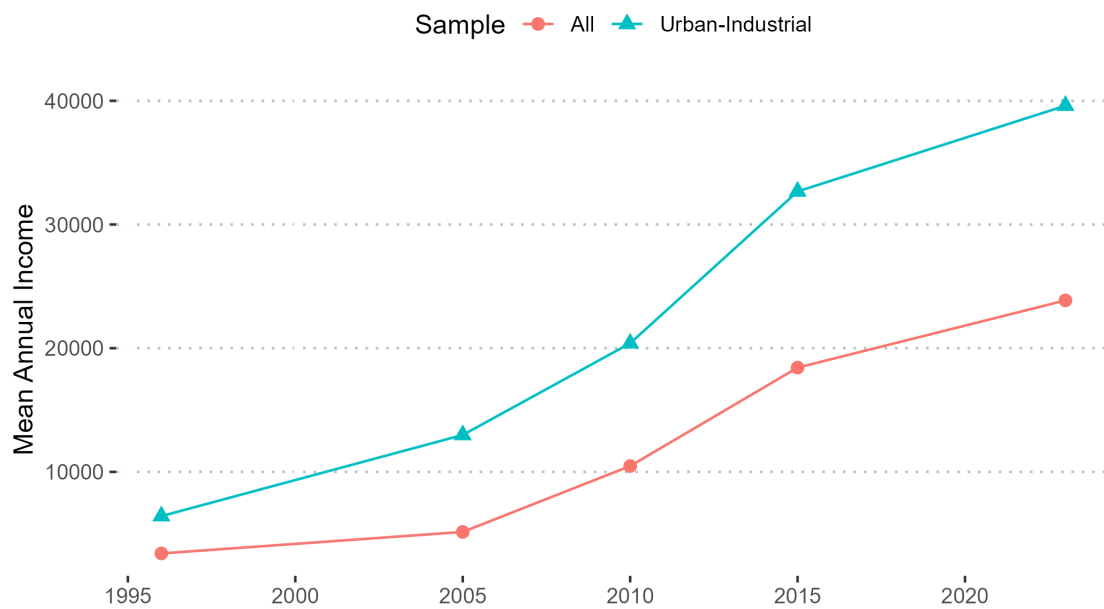
1996 to 39,604 in 2023. In terms of income inequality, Panel 2.2b shows that the variance of logged income in the entire national population has been much higher and experienced a larger increase than that of the urban industrial population. The trends of income inequality also differ in the two populations. While income inequality in the entire population increased steadily between 1995 and 2015 and declined in 2023, the income inequality increased most drastically in the urban industrial population between 2005 and 2010 and gradually decreased thereafter.

Overall, Figure 2.2 shows that the patterns and trends of income level and inequality in the urban industrial sub-population and the complete population in contemporary China are substantially different. Existing studies that have predominantly focused on the urban population, therefore, have painted an incomplete picture of economic inequality in contemporary China.

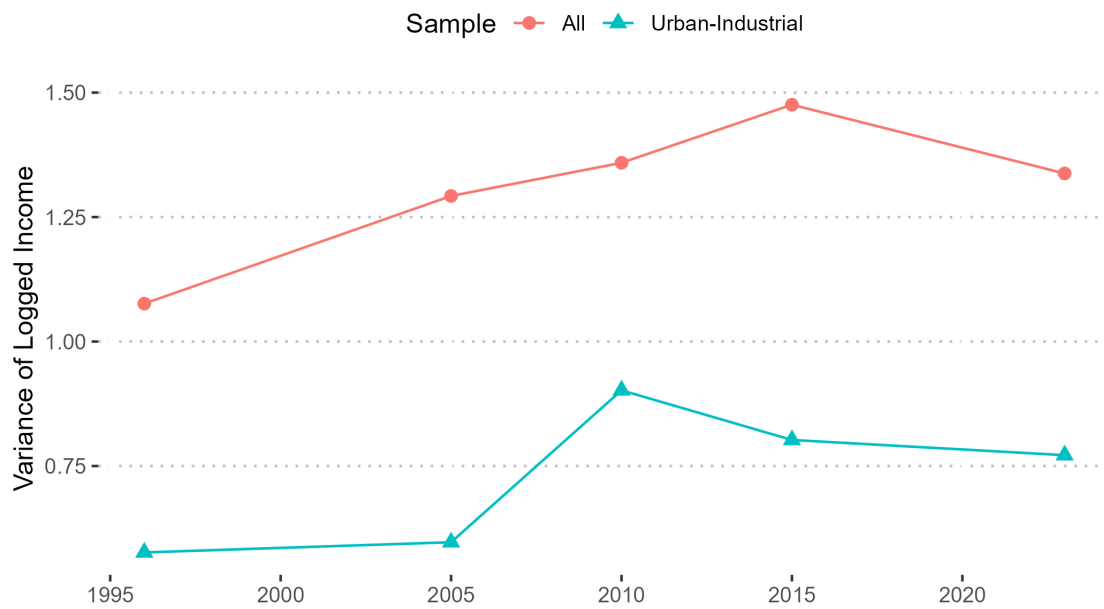
2.5.2 Labour Force Composition

Section 2.1.2 discusses the compositional changes in the labour force and their consequences for income inequality. The descriptive statistics, weighted by survey weights, in Table 2.3 reflect these changes.

For example, through rural-urban migration and urbanisation, the proportion of rural residents has substantially decreased from 57.3% in 1996 to 38.1% in 2023, while the proportion of rural-urban migrants has increased from only 5.9% to 32%. In terms of education distribution, the proportion of college graduates increased from only 2.2% in 1996 to 14.5% in 2023, while the proportion with primary education or lower decreased from about 44% in 1996 to less than 30% in 2023. Finally, due to marketisation and industrialisation, the sectoral employment structure has changed. The share of the state sector shrank from 26.9% in 1996 to 11.5% in 2023, while the private sector expanded from 14.5% to 38%. The farmer population also decreased from accounting for 46.7% of the working population in 1996 to less than 30% in 2023.



(a) Mean annual income



(b) Variance of logged annual income

Figure 2.2: Mean and Variance of Annual Income in China, 1996-2023

	1996 (N=4777)	2010 (N=6198)	2015 (N=5334)	2023 (N=2378)
Sex				
Female	43.5%	41.0%	40.3%	47.0%
Male	56.5%	59.0%	59.7%	53.0%
Age				
Mean	39.257	43.095	44.933	44.599
SD	11.778	11.483	12.630	11.829
Range	20.0–69.0	20.0–69.0	20.0–69.0	20.0–69.0
Rural–Urban Status				
Rural	57.3%	50.9%	42.2%	38.1%
Migrant	5.9%	15.3%	19.6%	32.0%
Urban	36.9%	33.8%	38.1%	29.9%
Ethnic Minority				
No	92.8%	88.6%	91.9%	90.2%
Yes	7.2%	11.4%	8.1%	9.8%
Education				
College Degree	2.2%	6.9%	11.7%	14.5%
Vocational College	3.6%	8.3%	10.3%	11.8%
Senior/Vocational High	16.7%	16.9%	18.4%	17.4%
Middle School	33.5%	27.2%	24.6%	26.6%
Primary	29.2%	30.5%	26.1%	24.1%
No Schooling	14.9%	10.2%	8.9%	5.6%
Marital Status				
Married	84.2%	89.1%	74.3%	74.1%
Never Married	11.7%	6.8%	15.5%	15.8%
Other	4.0%	4.1%	10.2%	10.1%
Employment Sector				
Farmers	46.7%	41.4%	32.2%	29.8%
Private	14.5%	31.4%	41.1%	38.0%
Self-employed	11.9%	14.3%	12.1%	20.7%
State	26.9%	13.0%	14.5%	11.5%
CCP Member				
No	89.1%	88.0%	89.3%	86.5%
Yes	10.9%	12.0%	10.7%	13.5%
Logged Annual Income				
Mean	8.134	9.256	9.822	10.080
SD	1.037	1.166	1.215	1.157
Range	4.315–13.237	4.094–15.607	4.243–15.978	4.731–14.124

Notes: Survey weights are applied to ensure population representativeness.

Table 2.3: Descriptive Statistics by Survey Year

As Table 2.3 shows, the rapid growth of the rural-urban migrant population has reshaped the population and workforce structure in contemporary China. How has this change related to the distribution of other characteristics? Figure 2.3 shows the changes in the education and sector composition in the rural, migrant and urban samples.

Panel 2.3a shows that rural-urban migrants have overall higher levels of education than rural workers, but substantially lower levels than urban workers. The patterns suggest that while there is positive education selection into migration among rural workers, their education and skills remain substantially lower than those of their urban counterparts. Meanwhile, Panel 2.3b shows that, compared to the rural population, of which the majority work in the agricultural sector, the majority of rural-urban migrants work in the private industrial sector or are self-employed. However, it should also be noted that compared to urban workers, they are much less likely to be employed in the state sector ³.

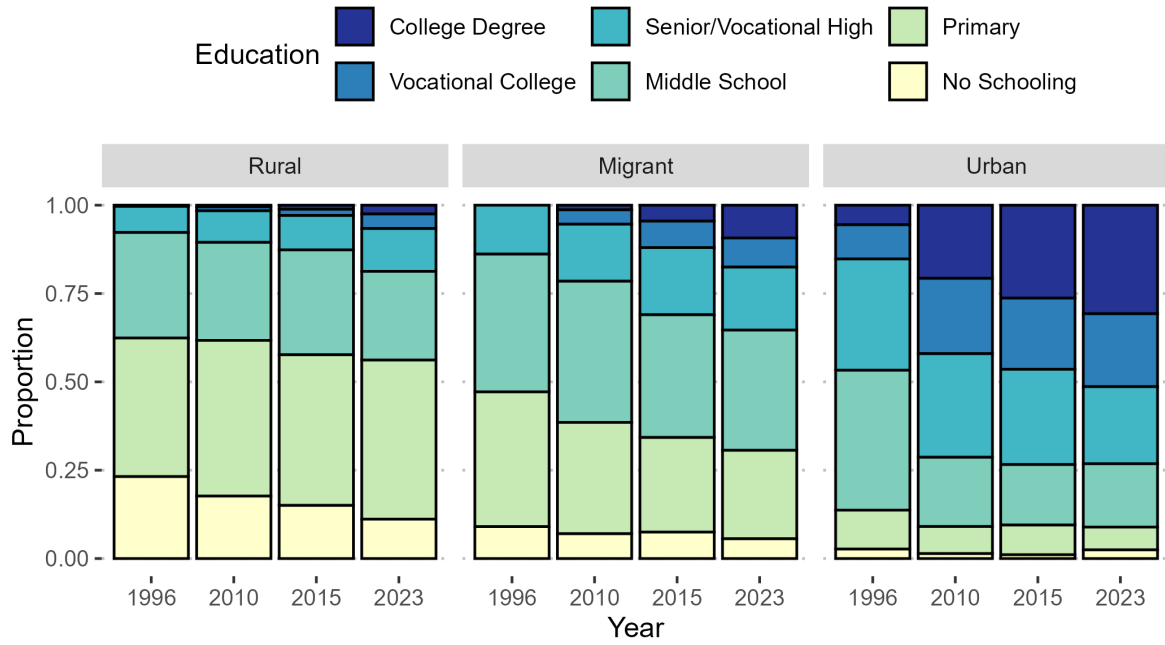
Taken together, Figure 2.3 suggests that, while the rural-urban migrants have similar, though on average higher, education levels to their rural counterparts, they are much more likely to be employed in the industrial sector or self-employed. Migration, therefore, could function as a channel for sectoral and occupational mobility (X. Zhou and Xie, 2019). This mobility effect is also reflected in the income distributions in 2.4, with migrants having significantly higher income than rural workers.

Figure 2.4 shows the year-specific income distribution in the three groups. The distribution for the migrants is consistently higher than that of the rural residents, suggesting that migration may act as a channel of upward income mobility for the rural population.

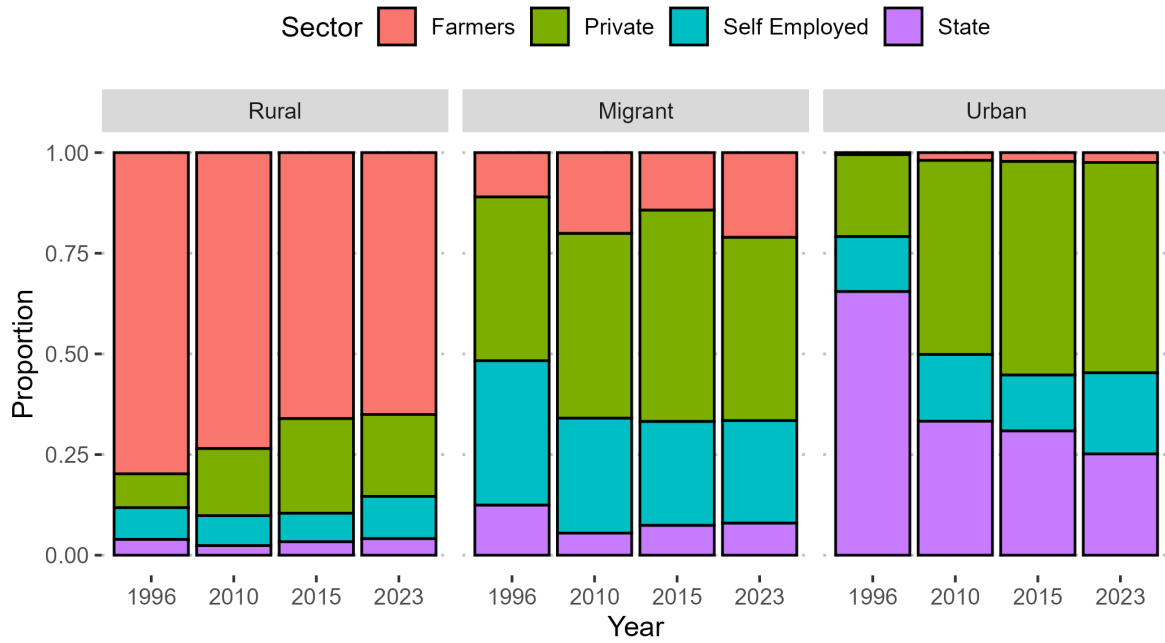
2.6 Predictors of Income

As discussed in Section 2.1.1, various theories of market transition and economic development in China have stipulated that the socioeconomic reforms would lead to changes in the

³While the difference may reflect barriers to entry, it could also reflect the fact that state sector employers provide better welfare, including urban *hukou* for migrants.



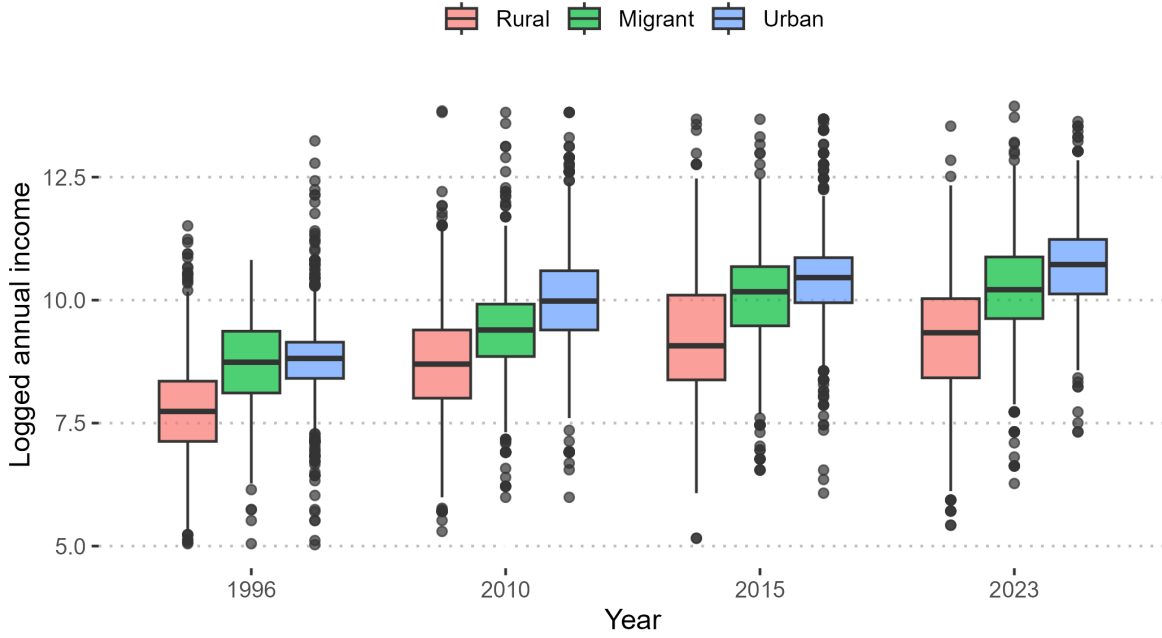
(a) Education



(b) Employment sector

Notes: Survey weights are applied to ensure population representativeness.

Figure 2.3: Education and Employment Sector Composition by Rural-Urban Status



Notes: Survey weights are applied to ensure population representativeness.

Figure 2.4: Income Distribution by Rural-Urban Status

determinants of life outcomes. So, how have the predictors of income changed in post-reform China? Following Xie and X. Zhou (2014), I use the bivariate and partial R^2 to quantify the explanatory importance of different factors.

For a predictor D of logged annual income Y , the bivariate R^2 when regressing Y on D , denoted as R_D^2 , captures the proportion of variation in Y that can be explained by the variation of D . The bivariate regression is equivalent to a one-way analysis of variance (ANOVA), and R_D^2 can also be read as the proportion of between-group (i.e., D) variance of Y .

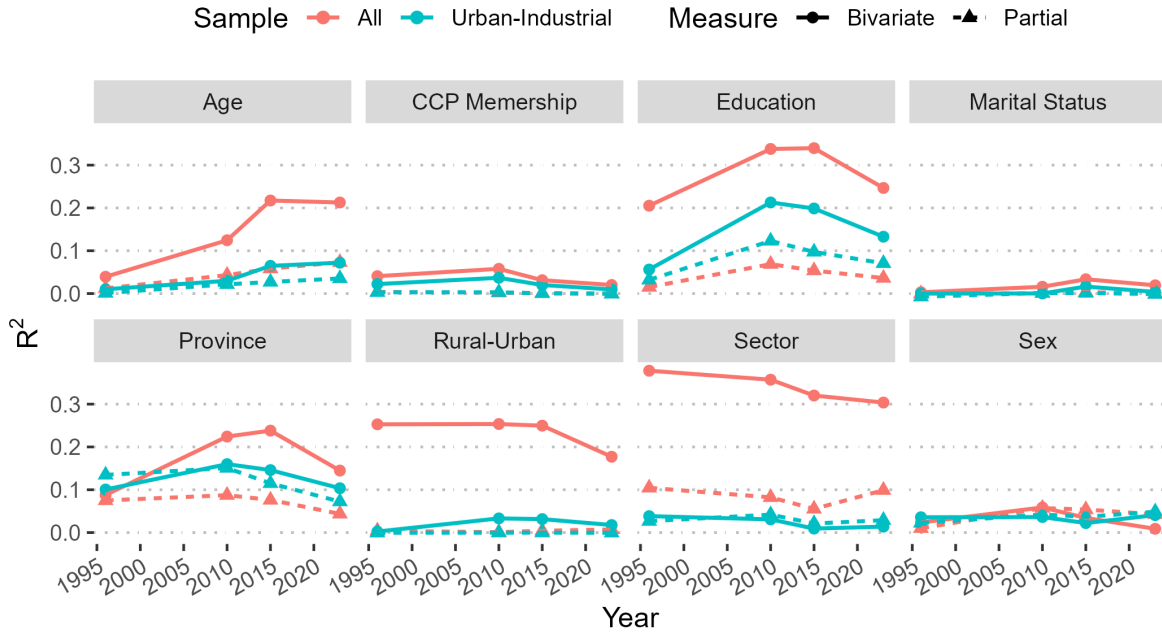
We can also define the partial R^2 of D on Y , denoted as R_{-D}^2 , given a set of additional predictors $\mathbf{X}$ as:

$$R_{-D}^2 = \frac{R_{D,\mathbf{X}}^2 - R_{\mathbf{X}}^2}{1 - R_{\mathbf{X}}^2}$$

where $R_{D,\mathbf{X}}^2$ is the R^2 value of a model that regresses Y on D and $\mathbf{X}$ and $R_{\mathbf{X}}^2$ is the R^2 value from a regression of Y on $\mathbf{X}$.

Therefore, R^2_D captures the proportion of the remaining variation of logged income that can be explained by D when all other factors $\mathbf{X}$ are taken into account.

Taken together, the bivariate and partial R^2 provide measures of the crude and conditional explanatory importance of a factor in explaining income variation. Empirically, I estimate the two measures for each of: age (natural cubic spline), CCP party membership, education, ethnicity, marital status, province, rural-urban status, employment sector and sex, using linear regression with the full model controls for all other variables, separately for data from each year and for the complete and urban industrial samples. All models are weighted by the survey weights.



Notes: Survey weights are applied in modelling to ensure population representativeness.

Figure 2.5: Bivariate and Partial R^2 of Predictors in Explaining Variation in Logged Annual Income

Figure 2.5 shows the bivariate and partial R^2 in each year, separately in the complete and urban industrial samples ⁴. Several patterns stand out. First, the predictors differ substantially in their crude explanatory power, measured by the bivariate R^2 . In the complete

⁴Ethnicity is excluded from presentation, but the values for all measures, samples and years are very close to zero.

sample, employment sector, rural-urban status, education and province explain the largest shares of income variation. By contrast, CCP membership, marital status and sex account for only small proportions of the total variation across all years. This does not, however, mean that these factors do not affect individuals' income. It means that, structurally, factors such as education, employment sector and rural-urban status constitute the most important axes of income variation in contemporary China.

Education is one of the most important predictors, consistently explaining over 20% of all income variation in the complete sample since 1996. Consistent with the predictions of the market transition theory and skill-biased technological change, the importance of education, measured by the bivariate R^2 , substantially increased between 1996 and 2010 and remained high in 2015 (Xie and Hannum, 1996; X.-G. Zhou, 2000; J. Lu, 2017). However, its importance decreased between 2015 and 2023, consistent with evidence on the decrease in the return to education due to increasing supply (Hanushek, Y. Wang, and L. Zhang, 2025). While the bivariate R^2 for education in the complete and urban industrial samples follows a similar trend, the magnitude in the complete sample is significantly larger, possibly due to the education difference between the rural and urban populations.

The partial R^2 of education exhibits a similar trend to the bivariate R^2 , with the increase between 1996 and 2010 and the slight decrease afterwards. Its magnitude remains substantial, explaining 5% to 10% of the income variation even after all other variables are accounted for. Unlike the bivariate measure, the partial R^2 is higher in the urban industrial sample, indicating greater returns to and income variation by education.

Employment sector and rural-urban status are the other two most important predictors of income, in terms of bivariate R^2 and the complete sample. Throughout the period, sectors alone can explain over 30% of the income variation, and rural-urban status can explain over 20%. However, the importance of both of them drastically decreases to almost negligible when focusing on the urban industrial sub-sample. This means that the explained variation is almost entirely driven by the difference between the rural agricultural and urban

industrial populations. Furthermore, in terms of partial R^2 , the employment sector is one of the most important predictors in the complete sample, but not in the urban industrial sample. This pattern suggests that the difference between the agricultural and industrial sectors constitutes an important dimension of income inequality, even after accounting for other variables. Meanwhile, however, the partial R^2 for rural-urban status is negligible in both samples, indicating that the difference between rural, migrant and urban workers is almost completely driven by the differences in other characteristics such as education and employment sector.

Province is another important predictor. Its bivariate R^2 increases from 1996 to 2015 before declining in 2023, reflecting the increasing importance of regional disparities in income inequality. The magnitude and the increase between 1996 and 2010 are particularly pronounced in the complete sample, reflecting the compounding effect of increasing regional and rural-urban disparity. Province is one of the most important predictors when considering partial R^2 , explaining about 10% of the remaining variation. Such high partial R^2 values confirm the persistent importance of structural regional disparity in shaping income inequality in contemporary China (Kanbur and X. Zhang, 2005; Xie and Hannum, 1996).

Finally, age has become increasingly important over time, especially in the complete sample. The bivariate R^2 for age rises markedly from about 0.5% in 1996 to 20% in 2015 and remains high in 2023. The partial R^2 also increases. This pattern suggests growing age or cohort differentiation in income, possibly reflecting changing life course dynamics such as cumulative (dis)advantages or cohort differences through market reforms, education expansion and structural economic change.

Taken together, Figure 2.5 shows that the determinants of income in post-reform China have changed in ways broadly consistent with existing theories and characterisations. On the one hand, structural factors such as employment sectors, rural-urban status, province and education constitute the most salient dimensions of income inequality. On the other hand, the importance of factors such as education has changed over time, in line with structural

socioeconomic reforms.

One particularly relevant pattern for the current study is that the factors' explanatory importance could substantially vary by the sample being used. For example, rural-urban status and employment sectors are two of the most important predictors of income in the complete sample, while being almost negligible in the urban industrial sub-sample. Such results highlight the importance of considering the rural agricultural population when analysing income inequality, as excluding it may substantially change the results and conclusions. In light of this finding, I now examine the drivers of income inequality in post-reform China using variance decomposition.

2.7 The Decomposition Framework

To quantify the effect of socioeconomic changes on income inequality, I follow the decomposition framework by X. Zhou (2014). Combining total variance decomposition and the Kitagawa-Oaxaca-Blinder (KOB) decomposition, X. Zhou (2014) decomposes the change in income inequality, measured by the variance of logged income, in urban China between 1996 and 2010 into four components: 1. changes in between-group gaps, 2. changes in within-group variation, 3. changes in between-group gaps due to compositional changes and 4. changes in within-group variation due to compositional changes. Then, through counterfactual analysis, X. Zhou (2014) quantifies the contributions of specific changes, such as changes in the return to education and education expansion, to the change in income inequality.

This section outlines and generalises X. Zhou (2014)'s regression-based framework in non-parametric terms and defines the target estimands for empirical estimation.

2.7.1 Variance Decomposition

Denote the logged income as Y . By the law of total variance, the variance of Y – as a measure of income inequality – can be decomposed into a between-group and a within-group

component, B and W :

$$V(Y) = \underbrace{V[E(Y|\mathbf{X})]}_B + \underbrace{E[V(Y|\mathbf{X})]}_W \quad (2.1)$$

where $\mathbf{X}$ is a set of predictors of logged income.

Assume that all predictors in $\mathbf{X}$ are categorical with $\mathcal{X} = \text{supp}(\mathbf{X})$. Let $\mu = E(Y)$, $\mu(\mathbf{x}) = E(Y|\mathbf{X} = \mathbf{x})$, $\sigma^2(\mathbf{x}) = V(Y|\mathbf{X} = \mathbf{x})$ and $\pi(\mathbf{x}) = P(\mathbf{X} = \mathbf{x})$, $\forall \mathbf{x} \in \mathcal{X}$.

B and W can be written as :

$$\begin{aligned} B &= V[E(Y|\mathbf{X})] \\ &= \sum_{\mathbf{x} \in \mathcal{X}} (\mu(\mathbf{x}) - E[\mu(\mathbf{x})])^2 \pi(\mathbf{x}) \\ &= \sum_{\mathbf{x} \in \mathcal{X}} \mu(\mathbf{x})^2 \pi(\mathbf{x}) - \mu^2 \end{aligned} \quad (2.2)$$

$$\begin{aligned} W &= E[V(Y|\mathbf{X})] \\ &= \sum_{\mathbf{x} \in \mathcal{X}} \sigma^2(\mathbf{x}) \pi(\mathbf{x}) \end{aligned} \quad (2.3)$$

Thus, the total variance can be expressed as a function of the conditional mean income $\mu(\mathbf{x})$, the conditional income variance $\sigma^2(\mathbf{x})$, and the marginal covariate distribution $\pi(\mathbf{x})$:

$$\begin{aligned} V(Y) &= \underbrace{\sum_{\mathbf{x} \in \mathcal{X}} (\mu(\mathbf{x}) - E[\mu(\mathbf{x})])^2 \pi(\mathbf{x})}_B + \\ &\quad \underbrace{\sum_{\mathbf{x} \in \mathcal{X}} \sigma^2(\mathbf{x}) \pi(\mathbf{x})}_W \end{aligned} \quad (2.4)$$

Decomposition of Change in Variance

Now consider two years, t and t' , assuming $\mathcal{X}_t = \mathcal{X}_{t'} = \mathcal{X}$. The difference in logged income variance, denoted as $\Delta_{t,t'}$, can be decomposed into changes in between- and within-group variances, $\Delta_{t,t'}^B$ and $\Delta_{t,t'}^W$:

$$\begin{aligned}\Delta_{t,t'} &= V_t(Y) - V_{t'}(Y) \\ &= \underbrace{B_t - B_{t'}}_{\Delta_{t,t'}^B} + \underbrace{W_t - W_{t'}}_{\Delta_{t,t'}^W}\end{aligned}\tag{2.5}$$

Substitute with Equations 2.2 and 2.3,

$$\begin{aligned}\Delta_{t,t'} &= \sum_{x \in \mathcal{X}} \mu_t(\mathbf{x})^2 \pi_t(\mathbf{x}) - \mu_t^2 - \\ &\quad \mu_{t'}(\mathbf{x})^2 \pi_{t'}(\mathbf{x}) + \mu_{t'}^2 + \\ &\quad \sigma_t^2(\mathbf{x}) \pi_t(\mathbf{x}) - \\ &\quad \sigma_{t'}^2(\mathbf{x}) \pi_{t'}(\mathbf{x})\end{aligned}$$

Apply the KOB decomposition

$$\begin{aligned}\Delta_{t,t'} &= \sum_{x \in \mathcal{X}} \mu_t(\mathbf{x})^2 \pi_t(\mathbf{x}) - \mu_t^2 - \mu_{t'}(\mathbf{x})^2 \pi_t(\mathbf{x}) + \mu_{t'}^2 + \\ &\quad \mu_{t'}(\mathbf{x})^2 \pi_t(\mathbf{x}) - \mu_{t'}^2 - \mu_{t'}(\mathbf{x})^2 \pi_{t'}(\mathbf{x}) + \mu_{t'}^2 - \\ &\quad \sigma_t^2(\mathbf{x}) \pi_t(\mathbf{x}) - \sigma_{t'}^2(\mathbf{x}) \pi_t(\mathbf{x}) + \\ &\quad \sigma_{t'}^2(\mathbf{x}) \pi_t(\mathbf{x}) - \sigma_{t'}^2(\mathbf{x}) \pi_{t'}(\mathbf{x})\end{aligned}\tag{2.6}$$

The terms in Equation 2.6 can be grouped and labelled as:

$$\delta_{t,t'}^B = \sum_{x \in \mathcal{X}} \mu_t(\mathbf{x})^2 \pi_t(\mathbf{x}) - \mu_t^2 - \mu_{t'}(\mathbf{x})^2 \pi_t(\mathbf{x}) + \mu_{t'}^2$$

$$\delta_{t,t'}^D = \sum_{x \in \mathcal{X}} \mu_{t'}(\mathbf{x})^2 \pi_t(\mathbf{x}) - \mu_{t'}(\mathbf{x})^2 \pi_{t'}(\mathbf{x})$$

$$\delta_{t,t'}^W = \sum_{x \in \mathcal{X}} \sigma_t^2(\mathbf{x}) \pi_t(\mathbf{x}) - \sigma_{t'}^2(\mathbf{x}) \pi_t(\mathbf{x})$$

$$\delta_{t,t'}^A = \sum_{x \in \mathcal{X}} \sigma_{t'}^2(\mathbf{x}) \pi_t(\mathbf{x}) - \sigma_{t'}^2(\mathbf{x}) \pi_{t'}(\mathbf{x})$$

$$\Delta_{t,t'} = \delta_{t,t'}^B + \delta_{t,t'}^W + \delta_{t,t'}^D + \delta_{t,t'}^A \quad (2.7)$$

Equation 2.7 suggests that the total change in logged income variance between t and t' can be decomposed into four components. δ^B captures changes due to the difference in between-group gaps; δ^D captures changes in between-group variance due to changes in the group marginal distribution, which X. Zhou (2014) named the distribution effect; δ^W captures changes due to the difference in within-group variation; and δ^A captures changes in within-group variance due to changes in the group marginal distribution, which X. Zhou (2014) named the allocation effect.

2.7.2 Detailed Decomposition and Counterfactual Variance

Equation 2.7 shows that the change in total income variance can be attributed to changes in three parameters: the conditional mean profile $\mu(\mathbf{x})$, the conditional variance profile $\sigma^2(\mathbf{x})$ and the covariate distribution $\pi(\mathbf{x})$, with the last affecting both between and within-group variance (i.e., the distribution and allocation effects).

However, while the decomposition identifies the channels through which inequality changes, it does not by itself show which specific socioeconomic changes – such as education expan-

sion, rural-urban migration, or employment structure changes – drive each channel. To obtain such detailed contributions, I construct counterfactual variances by replacing certain parameters in the variance-generating process in Equation 2.4 with counterfactual values – if certain changes did not happen – while holding the other parameters fixed at their observed values in the reference year.

For example, to quantify the effect of changes in the return to education on income inequality between 1996 and 2010, I estimate the variance in 2010 under the counterfactual conditional mean profile $\mu_{2010}^{\beta_{edu}=1996}(\mathbf{x})$, in which the income return to education is set at its 1996 level while all other components are kept at their 2010 levels:

$$V_{2010}^{\beta_{edu}=1996}(Y) = \sum_{\mathbf{x} \in \mathcal{X}} \mu_{2010}^{\beta_{edu}=1996}(\mathbf{x})^2 \pi_{2010}(\mathbf{x}) - E[\mu_{2010}^{\beta_{edu}=1996}(\mathbf{x})]^2 + \sigma_{2010}^2(\mathbf{x}) \pi_{2010}(\mathbf{x}) \quad (2.8)$$

Similarly, we could quantify the effect of changes in the education distribution by estimating the counterfactual variance in 2010 when the education distribution is fixed at that of 1996, by replacing $\pi_{2010}(\mathbf{x})$ with $\pi_{2010}^{edu=1996}(\mathbf{x})$:

$$V_{2010}^{\pi_{edu}=1996}(Y) = \sum_{\mathbf{x} \in \mathcal{X}} \mu_{2010}(\mathbf{x})^2 \pi_{2010}^{edu=1996}(\mathbf{x}) - \left[\sum_{\mathbf{x} \in \mathcal{X}} \mu_{2010}(\mathbf{x}) \pi_{2010}^{edu=1996}(\mathbf{x}) \right]^2 + \sum_{\mathbf{x} \in \mathcal{X}} \sigma_{2010}^2(\mathbf{x}) \pi_{2010}^{edu=1996}(\mathbf{x}).$$

We can further isolate the effect of the counterfactual distribution through between and within-group variance by applying it separately to each component:

$$\begin{aligned}
V_{2010}^{\pi_{edu}=1996,B}(Y) &= \sum_{\mathbf{x} \in \mathcal{X}} \mu_{2010}(\mathbf{x})^2 \pi_{2010}^{edu=1996}(\mathbf{x}) - \left[\sum_{\mathbf{x} \in \mathcal{X}} \mu_{2010}(\mathbf{x}) \pi_{2010}^{edu=1996}(\mathbf{x}) \right]^2 + \\
&\quad \sigma_{2010}^2(\mathbf{x}) \pi_{2010}(\mathbf{x}) \\
V_{2010}^{\pi_{edu}=1996,W}(Y) &= \sum_{\mathbf{x} \in \mathcal{X}} \mu_{2010}(\mathbf{x})^2 \pi_{2010}(\mathbf{x}) - \mu_{2010}^2 + \\
&\quad \sigma_{2010}^2(\mathbf{x}) \pi_{2010}^{edu=1996}(\mathbf{x})
\end{aligned}$$

More generally, for any counterfactual intervention $\mathcal{C}$ applied to the year- t income-generating process, the counterfactual variance can be written as

$$V_t^{\mathcal{C}}(Y) = V\left(\mu_t^{\mathcal{C}}(x), \sigma_t^{2\mathcal{C}}(x), \pi_t^{\mathcal{C}}(x)\right),$$

where the intervention may modify the conditional mean profile, the conditional variance profile, the marginal predictor distribution, or some combination of these components. The effect of the intervention is then measured by comparing the observed and counterfactual variances:

$$\Delta_t^{\mathcal{C}} = V_t(Y) - V_t^{\mathcal{C}}(Y). \quad (2.9)$$

A positive $\Delta_t^{\mathcal{C}}$ means that the total variance under the counterfactual condition $\mathcal{C}$ is lower than the observed values, therefore indicating that the corresponding observed change increases inequality. Vice versa, a negative $\Delta_t^{\mathcal{C}}$ indicates that the change reduced inequality.

2.7.3 Theoretical Estimands

The study aims to quantify the effect of socioeconomic changes on income inequality. Therefore, the theoretical estimands are $V_t^{\mathcal{C}}(Y)$ that quantify the level of income inequality under counterfactual conditions without the change, and $\Delta_t^{\mathcal{C}}$ that quantify the change's effect on

income inequality. For each of the periods 1996-2010, 2010-2015 and 2015-2023, I define the counterfactual condition $\mathcal{C}$ as the conditional mean $\mu(\mathbf{x})$, conditional variance $\sigma^2(\mathbf{x})$ or marginal distribution $\pi(\mathbf{x})$ of predictor X fixed at the earlier year, while other parameters remain at the later year's level. The next section explains the empirical procedure for estimating the quantities.

2.8 Models and Estimation

Equation 2.4 shows that the total variance can be calculated using 1. conditional means $\mu_t(\mathbf{x})$, 2. conditional variance $\sigma_t^2(\mathbf{x})$ and 3. marginal covariate distribution $\pi_t(\mathbf{x})$ and the counterfactual variance can be calculated by replacing these parameters with corresponding counterfactual values. The task, therefore, is empirically to estimate these quantities and construct counterfactual parameters.

2.8.1 Variance Function Regression

The conditional quantities $\mu_t(x)$ and $\sigma_t^2(x)$ can be estimated by various estimators (e.g., see Opacic, Wei, and X. Zhou, 2025, for a discussion of estimators for decomposition analysis). For interpretability and comparability with X. Zhou (2014), I use the variance function regression (Western and Bloome, 2009) and jointly estimate the following regression models for each year $t \in \{1996, 2010, 2015, 2023\}$:

$$\begin{aligned} \hat{\mu}_t(\mathbf{x}) = & \hat{\alpha}_t + \hat{\beta}_{1t}Age + \hat{\beta}_{2t}Age^2 + \hat{\beta}_{3t}Sex + \hat{\beta}_{4t}EthnicMinority + \\ & \hat{\beta}_{5t}RuralUrban + \hat{\beta}_{6t}Education + \hat{\beta}_{7t}Sector + \hat{\beta}_{8t}Party + \hat{\beta}_{9t}Province \end{aligned} \quad (2.10)$$

$$\begin{aligned} \hat{\sigma}_t^2(\mathbf{x}) = & \exp\left(\hat{\kappa}_t + \hat{\lambda}_{1t}Age + \hat{\lambda}_{2t}Age^2 + \hat{\lambda}_{3t}Sex + \hat{\lambda}_{4t}EthnicMinority + \right. \\ & \left. \hat{\lambda}_{5t}RuralUrban + \hat{\lambda}_{6t}Education + \hat{\lambda}_{7t}Sector + \hat{\lambda}_{8t}Party\right) \end{aligned} \quad (2.11)$$

EthnicMinority is a binary variable indicating whether the respondent is a non-Han ethnic minority; *RuralUrban* is a categorical variable indicating whether the individual is 1. a rural resident, 2. an urban resident or 3. a rural-urban migrant; *Education* is a categorical variable with 6 levels: No Schooling, Primary School, Middle School, High School/Vocational School, Vocational College and College Degree and Above; *Sector* is a categorical variable indicating individuals' employment sector; and *Party* is a binary variable indicating whether the individual is a Communist party member. Equation 2.11 excludes province fixed effects to ensure model convergence.

The mean regression in Equation 2.10 predicts the conditional mean of logged income, and the variance regression in Equation 2.11 takes the residuals from the mean regression and predicts the conditional mean of squared residuals (i.e., variance). Empirically, I fit these models using data from 1996, 2010, 2015 and 2023, all with sampling weights.

With $\hat{\pi}_t(x)$ estimated from the empirical sample-weighted covariate distribution from each year and $\hat{\mu}_t(x)$ and $\hat{\sigma}_t^2(x)$, I can apply the decomposition and estimate the components $\hat{\delta}_{t,t'}$ by plugging in corresponding values in Equations 2.7.

2.8.2 Counterfactual Conditional Mean and Variance

Next, to estimate counterfactual variance, I estimate the counterfactual conditional mean $\mu_t^C(x)$ and variance $\sigma_t^{2C}(x)$ by replacing coefficients in the mean and variance regressions. For example, based on Equations 2.10 and 2.11, the counterfactual conditional mean and variance in 2010, had the income returns to education and within-education income variation remaining at the 1996 level, can be estimated by:

$$\begin{aligned}
\widehat{\mu_{2010}^{\beta_{edu}=1996}}(\mathbf{x}) &= \hat{\alpha}_{2010} + \hat{\beta}_{1,2010}Age + \hat{\beta}_{2,2010}Age^2 + \hat{\beta}_{3,2010}Sex + \hat{\beta}_{4,2010}EthnicMinority + \\
&\quad \hat{\beta}_{5,2010}RuralUrban + \hat{\beta}_{6,1996}Education + \\
&\quad \hat{\beta}_{7,2010}Sector + \hat{\beta}_{8,2010}Party + \hat{\beta}_{9,2010}Province \\
\widehat{\sigma_{2010}^{2\beta_{edu}=1996}}(\mathbf{x}) &= \exp\left(\hat{\kappa}_{2010} + \hat{\lambda}_{1,2010}Age + \hat{\lambda}_{2,2010}Age^2 + \hat{\lambda}_{3,2010}Sex + \right. \\
&\quad \hat{\lambda}_{4,2010}EthnicMinority + \hat{\lambda}_{5,2010}RuralUrban + \\
&\quad \left. \hat{\lambda}_{6,1996}Education + \hat{\lambda}_{7,2010}Sector + \hat{\lambda}_{8,2010}Party\right)
\end{aligned}$$

Such coefficient replacement exercises suffer from the “omitted category” problem, as the results depend on the arbitrary choice of the reference category for categorical predictors (Oaxaca and Ransom, 1999). To resolve the issue, I use the normalisation method by Yun (2005) to calculate the average estimate across all choices of the reference category.

2.8.3 Reweighting for Counterfactual Distribution

Counterfactual distributions $\hat{\pi}_t^{\mathcal{C}}(x)$ are estimated by reweighting the data by a factor $\psi_{d|\mathbf{x}}$ so that the distribution of the variable of interest, D , matches that in the target year, while the distributions of other variables, $\mathbf{X}$, remain the same (see Lemieux, 2006; DiNardo, Fortin, and Lemieux, 1996):

$$\hat{\pi}_t^{\mathcal{C}}(x) = \hat{\pi}_t(x)\psi_{d|\mathbf{x}}$$

DiNardo, Fortin, and Lemieux (1996) show that, $\psi_{d|\mathbf{x}}$ can be calculated as:

$$\psi_{d|\mathbf{x}} = \frac{dF(d|\mathbf{x}, t')}{dF(d|\mathbf{x}, t)} = \sum_{d \in \mathcal{D}} \mathbb{1}(D = d) \frac{P(D = d|\mathbf{x}, t')}{P(D = d|\mathbf{x}, t)}$$

where $\mathcal{D} = \text{supp}(D)$ and t' denotes the target year that the data is reweighted to match

the distribution (i.e., the earlier year).

Estimating $\hat{\psi}_{d|\mathbf{x}}$ therefore requires predicting the propensity scores $P(\widehat{D=d}|\mathbf{x}, t)$ and $P(\widehat{D=d}|\mathbf{x}, t')$.⁵ For simplicity, I focus on the distribution of three variables: education, employment sector and rural-urban status, corresponding to the changes discussed in Section 2.1.2. For the predictors $\mathbf{X}$, I use the same set of variables in Model 2.10, except the target variable itself, in addition to the interaction between the other two target variables, and their interactions with sex. Then, to predict the propensity scores, I train an elastic net multinomial model with $\alpha = 0.5$ and 10-fold cross-validation and obtain the best predicted values. Post-weighting distributions show that the factor correctly adjusts the distribution of the target variable while maintaining the distribution of other variables, with the absolute difference in all variables below 2 percentage points (see Section A.2).⁶

With $\hat{\mu}_t^{\mathcal{C}}(x)$, $\hat{\sigma}_t^{2\mathcal{C}}(x)$ and $\hat{\pi}_t^{\mathcal{C}}(x)$, the counterfactual income variance $\hat{V}_t^{\mathcal{C}}(Y)$ can be estimated by plugging in corresponding estimated quantities. The standard errors of all final estimates are calculated from 250 bootstrap simulations.

2.8.4 Notes on Interpretations

Although the quantities above have been described as “counterfactual” variances and interpreted as indicating positive or negative *effects* on income inequality, it should be recognised that the estimation procedure and resulting estimates are model-dependent and cannot be directly interpreted as capturing causal effects.⁷ For example, to quantify the causal effect of changes in returns to education on income inequality, the regression coefficients being replaced need to capture the causal effect of education on income. This requires the unconfoundedness assumption, which is likely violated in our cross-sectional model with sparse

⁵The reweighting procedure described here is different from the original one used by X. Zhou (2014). He constructed the factor using the ratio of the unconditional distributions of the target variable, which does not maintain the distributions of other variables. He nonetheless also referred to Lemieux (2006) in outlining the method, which could be a methodological error.

⁶I also tested unregularised and LASSO multinomial models, but the post-weight distributions are poorly balanced.

⁷Some have instead described such results as synthesised outcomes to indicate their model-dependent nature (e.g., Kartsonaki, Jackson, and Cox, 2013)

adjustments.

Nonetheless, although the estimates cannot be interpreted in precise causal terms, they provide substantively meaningful descriptions of income inequality under counterfactual conditions and how corresponding changes account for changes in income inequality. Readers should read the results and interpretations in the rest of the study with this caveat in mind.

2.9 Results

2.9.1 Regression Results

Results from the mean and variance regressions reveal how the between-group income gaps and within-group income variation have changed in contemporary China. For brevity, I focus on three dimensions that have undergone particularly significant changes in post-reform China: education, employment sectors and rural-urban status. Full results from the regression models are reported in Section A.3.

Between-Group Income Gaps

Figure 2.6 shows the year-specific estimated annual income ratios between education, sector and rural-urban groups, separately using the complete sample and the urban industrial sub-sample. Panel 2.6a shows that the income return to education in the urban industrial population is slightly higher than in the national population. Estimates from both populations show that the return to education substantially increased between 1996 and 2010, confirming existing results on the increase in the return to education during the market reforms. However, consistent with Hanushek, Y. Wang, and L. Zhang (2025), the return to education gradually decreased after 2010. It is estimated that, relative to the completion of middle school, the average income premium of college education was about 38% in 1996 and it rapidly increased to 135% in 2010, then declined to 70% in 2023.

Panel 2.6b shows the between-sector income gaps. First, farmers persistently have sub-

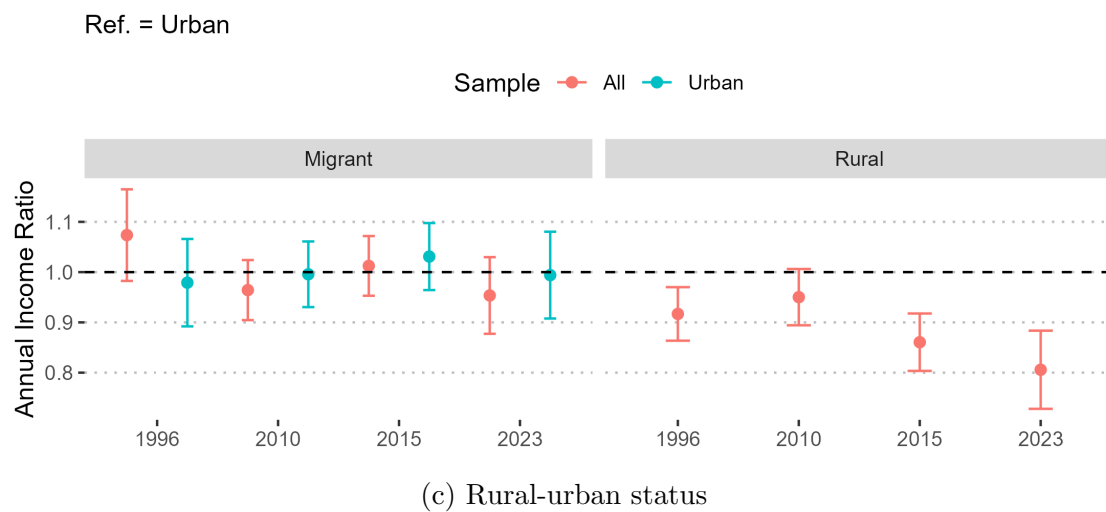
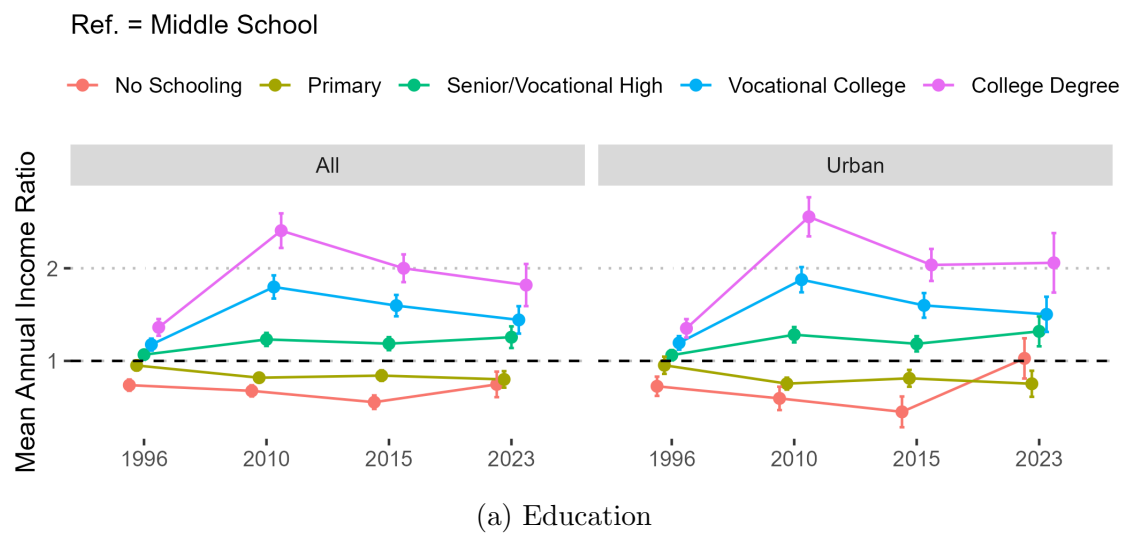


Figure 2.6: Estimated Between-Group Income Gaps

stantially lower income than state sector employees, controlling for other factors. Although their relative income disadvantage has improved between 1996 and 2015, it slightly rebounded in 2023. In 1996, adjusting for other characteristics, an average farmer earned only 32.4% as much as an average state sector employee. The figure increased to 58.6% in 2015. The post-reform improvement may reflect policy changes such as the abolition of agricultural taxes (X. Wang and Shen, 2014). The income gap between the state and the private sectors, on the other hand, has remained small and generally insignificant. What is more striking, however, is the change in the self-employed population's income positions. While in 1996, the self-employed on average experienced an over 20% income penalty relative to state and private sector employees, the gap changed sign in 2010, and the self-employed population earns on average about 30% more than otherwise similar employees. The gap remained stable between 2010 and 2023. This change may reflect increases in the space and rewards for entrepreneurship and private businesses in post-reform China.

Finally, Panel 2.6c shows the rural-urban income gaps. First, adjusting for other individual characteristics, the estimates suggest no significant income gaps between rural-urban migrants and urban workers throughout the studied period. The results thus suggest that, at least in terms of total income, there is no significant disadvantage associated with the rural *hukou* status. However, this could be because migrant workers tend to work for longer hours, and the disadvantages may be present in other outcomes, such as health. Second, the estimates suggest that the rural-urban income gap has expanded since 2010, from about 5% to 20%. This may reflect increasing resource and productivity concentration in urban areas.

Within-Group Income Variation

Figure 2.7 shows the year-specific estimated ratios between education, sector and rural-urban groups, separately using the complete sample and the urban industrial sub-sample. Panel 2.7a shows that in 1996, controlling for other characteristics, there was slightly less income variation within workers with beyond middle school education, suggesting greater

income security. However, such advantages were removed from 2010 onward. Instead, there is uncertain suggestive evidence that income variation among workers with no formal or only primary education increased between 1996 and 2015, especially in the urban industrial population. This trend may suggest increasing economic uncertainty for unskilled workers in post-reform China.

Panel 2.7b shows the income variation between different sectors. First, the income variation among the farmers relative to state sector employees has substantially decreased since 1996. It is estimated that, in 1996, adjusting for other characteristics, the income variation among farmers was over 3.5 times as high as that of state sector workers. The ratio decreased to 2.1 in 2010 and continued to gradually decrease, suggesting that economic uncertainty among farmers has improved since 1996. In the private sector, the estimates show that the income variation in 1996 was about twice as high as that in the state sector. The ratio decreased to about 1.3 in 2010, likely due to the SOE reforms, and has remained low and relatively stable thereafter. Again, the biggest disparity is between the self-employed and the employees. In 1996, the income variation among the self-employed was about 6 times as much as that among state sector workers. Then the ratio rapidly decreased to about 3 in 2010 and continued to gradually decrease to about 2 in 2023. The trend, therefore, suggests that while the self-employed population remained, by its nature, more prone to risk and uncertainty, its economic uncertainty has substantially improved in post-reform China.

Finally, Panel 2.7c shows that, adjusting for other characteristics, there is no significant difference in income variation and uncertainty between migrant and urban workers. Meanwhile, the rural areas show an increasing trend in income variation relative to the urban areas, though the estimates are small in magnitude and statistically insignificant.

Summary

Overall, the results from the mean and variance regressions indicate that, after the market reform, the income return to education in China initially increased rapidly between 1996 and

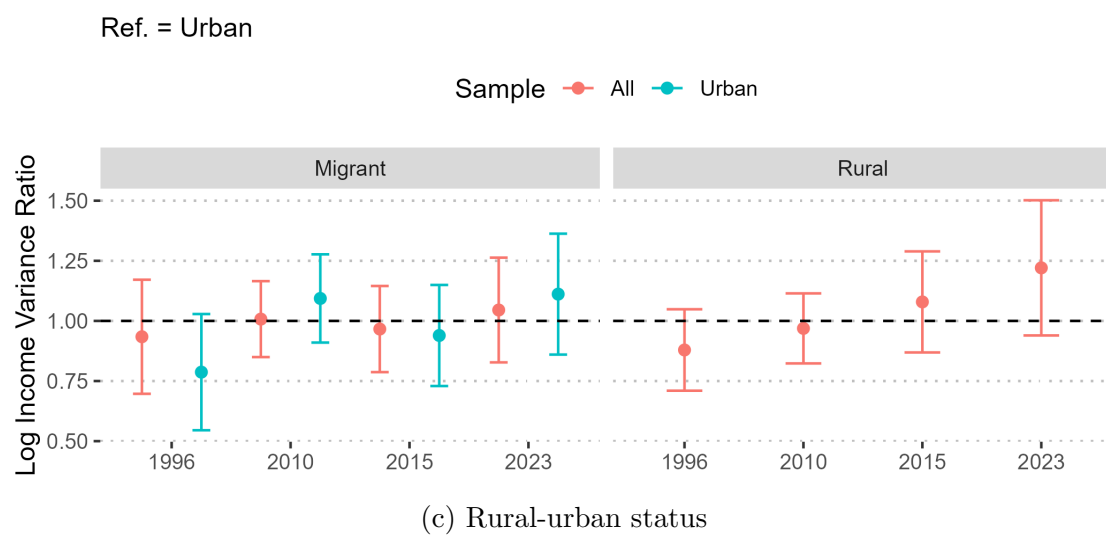
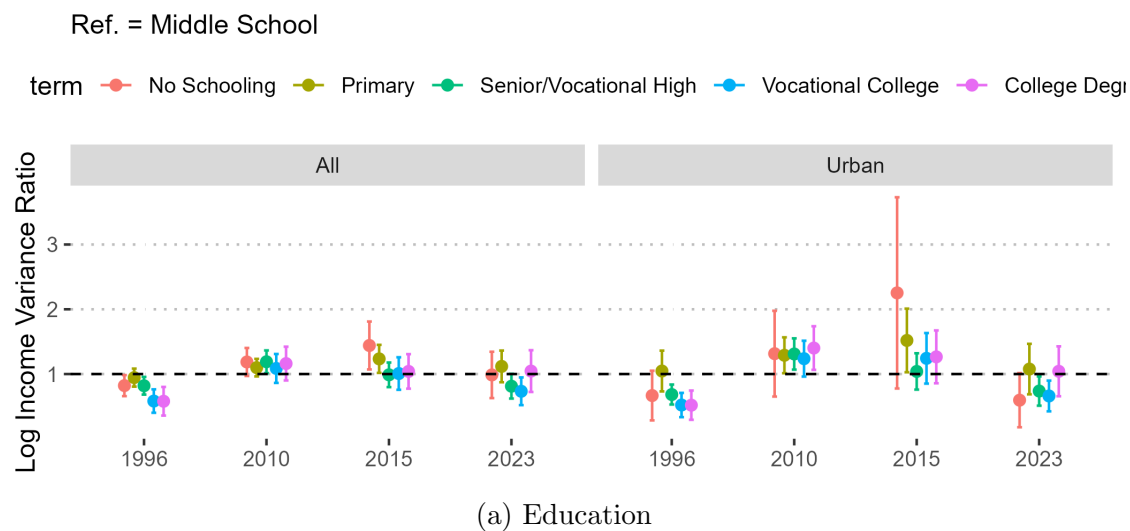


Figure 2.7: Estimated Differences in Within-Group Income Variation

2010 and has since then gradually decreased. In terms of income variation, there is, albeit weaker and more uncertain, evidence that the income security enjoyed by the more educated workers in 1996 was removed in the 21st century; meanwhile, workers with below secondary education experience increasing income variation and uncertainty.

Among different employment sectors, the differences between the state and the private industrial sectors have been relatively small and stable, except for the decrease in the state sector's income security advantage between 1996 and 2010. The more substantial changes occurred between the agricultural and industrial sectors, and between the employed workers and the self-employed. Both sectors have experienced significant improvements, both in their relative income positions and income variation and uncertainty. Finally, the direct effects of rural-urban status, net of other characteristics, on income level and variation have been relatively stable since 1996, except for the recent decrease in rural areas' relative income level.

Besides the general trends, the results also show that the agricultural sector in China is remarkably and persistently different from the industrial sector, with substantially lower income levels and higher income variation. Given that it accounts for 30% and even more of the Chinese working population, excluding it inevitably presents an incomplete picture of income distribution in post-reform China.

2.9.2 Counterfactual Analysis

Counterfactual Conditional Mean and Variance

How do the changes in between-group income gaps and within-group income variation described above contribute to the change in income inequality? Figure 2.8 shows the difference between the observed variance and the counterfactual variance in 2010, 2015 and 2023, normalised by the observed values, when between-group income gaps and within-group income

variations of different variables are fixed at the 1996, 2010 and 2023 levels.⁸

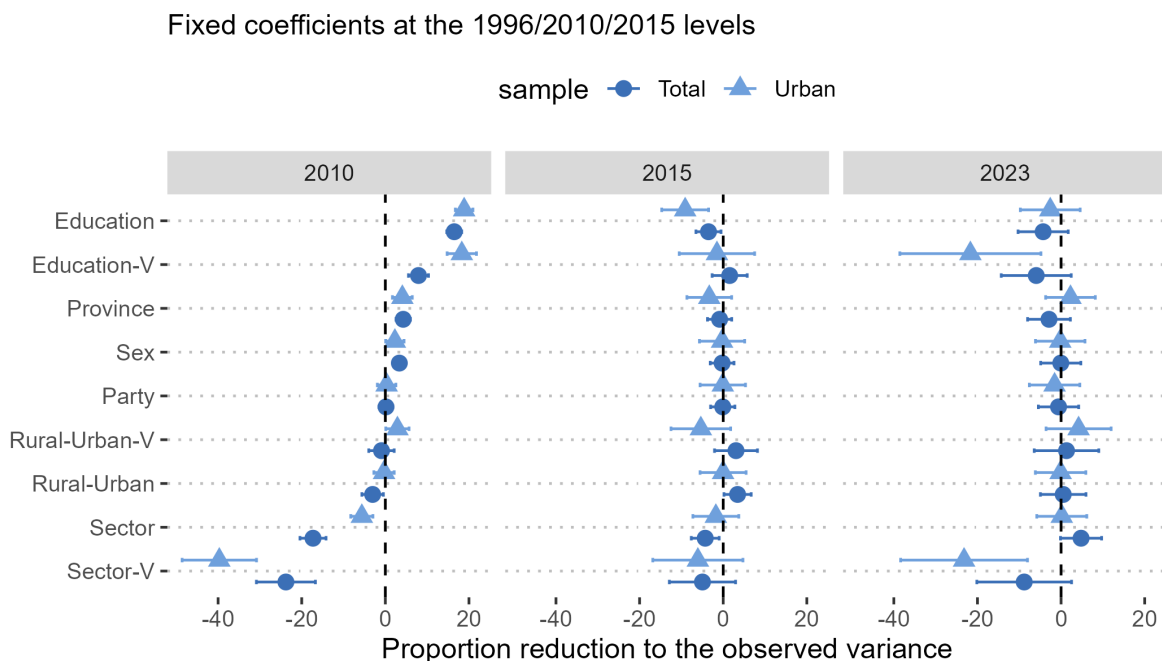
First, changes in the return to education were the biggest contributor to income inequality between 1996 and 2010. It is estimated that, other things being equal, if the returns to education were fixed at the 1996 level, income inequality in 2010 would have been about 17% lower, explaining about 70% of the total increase. The magnitudes in the two samples are similar, though slightly lower in the national sample. The results confirm existing findings that show that the increase in the return to education was one of the biggest drivers of income inequality in post-reform China (X. Zhou, 2014; J. Lu, 2018). What has previously not been documented, however, is the positive contribution of changes in within-education income variation to income inequality between 1996 and 2010. The contribution is particularly substantial in the urban industrial sub-sample, with the magnitude comparable to that of changes in between-education income gaps.

While the changes in income distribution between education groups contributed to the increase in income inequality, the changes between sectors act as a countervailing force. It is estimated that, other things being equal, if the sector income effects were fixed at the 1996 level, income inequality in the national population in 2010 would have been about 20% higher. The magnitude is significantly smaller when using the urban industrial sub-sample. In fact, when jointly fixing the income effects of sector and education, the inequality-increasing and inequality-decreasing effects almost perfectly offset each other in the national sample; whereas in the urban industrial sub-sample, the inequality-increasing effect of the increase in the return to education dominates. The strong inequality-decreasing effect, particularly in the national sample, could reflect the increase in the income of the self-employed and, more importantly, farmers, as shown in Figure 2.6b.

Another previously overlooked factor in the change in income inequality between 1996 and 2010 is changes in within-sector income variance. In the complete sample, fixing within-sector income variation at the 1996 level would lead to an over 20% increase in income

⁸Total decomposition results in Equation 2.7, where the values for all variables are fixed, are reported in Section A.4.

inequality in 2010. The effect is even stronger and almost doubles to about 40% in the urban industrial sample. The result could be driven by the decrease in income variation within the private sector and the self-employed. Other changes that have been argued to lead to changes in inequality, such as changes in the gender wage gap and the party cadre wage premium, are estimated to have relatively small counterfactual effects.



Notes: Proportion reductions are calculated as $\frac{\hat{\Delta}^c}{\hat{V}_t} \times 100 = \frac{V_t - V_t^c}{V_t} \times 100$. The error bars show the 95% confidence interval, where SEs are calculated based on 250 bootstrap simulations. Rows ending with "-V" represent counterfactual conditional variance.

Figure 2.8: Variance under Counterfactual Conditional Mean and Variance

While the results for the period of 1996 and 2010 clearly identify changes in between-education and between-sector income gaps, and changes in within-education and within-sector income variation as contributors to income inequality, the results for 2010 and 2015 are less clear. Between 2010 and 2015, the change in the return to education has moderately reduced income inequality. It is estimated that if the return to education were fixed at the 2010 level, income inequality in 2015 would have been about 8% higher in the urban industrial sample and about 3% higher in the complete sample. In the complete sample, changes in the rural-urban income gap and sector income effect contributed in opposite directions to

the change in income inequality, potentially reflecting the increase in the rural-urban income gap and the continuing improvement in farmers' relative income, as shown in Figures 2.6b and 2.6c.

Finally, the results for the period of 2015 and 2023 are mostly statistically insignificant, except for the changes in within-education and within-sector variance in the urban industrial sample.

Counterfactual Composition

How have the structural changes in labour force compositions documented in Section 2.5.2 contributed to the change in income inequality? Figure 2.9 reports the proportional difference between the observed variance and the counterfactual variance, with counterfactual between and within-group income variances under counterfactual variable distributions.

Panel 2.9a shows the difference when the between-group income variance is affected by compositional changes. Between 1996 and 2010, compositional changes in education significantly contributed to the increase in income inequality. Other things being equal, income inequality in China in 2010 would have been about 10% lower if between-education inequality had remained at the level under the 1996 education composition, explaining 42% of the increase in inequality. The magnitudes for the complete and urban industrial samples are similar, though slightly stronger in the complete sample. Overall, this result is consistent with X. Zhou (2014) and confirms that the rapid education expansion created a more dispersed distribution, which increased between-education inequality.

However, results from the two samples diverge on the effects of changes in sectoral composition. While in the urban industrial sample, the effect of the compositional change in employment sectors is statistically insignificant, the change significantly reduced inequality in the complete sample. Through between-group inequality, fixing the sectoral composition at the 1996 level would increase income inequality by about 3%. The null effect in the urban industrial sample is consistent with X. Zhou (2014), and suggests that the rebalance of the

state and private sector due to SOE reforms and marketisation did not directly affect income inequality through changes in between-group income gaps. However, the negative contribution of sectoral compositional changes in the complete sample is previously undocumented.

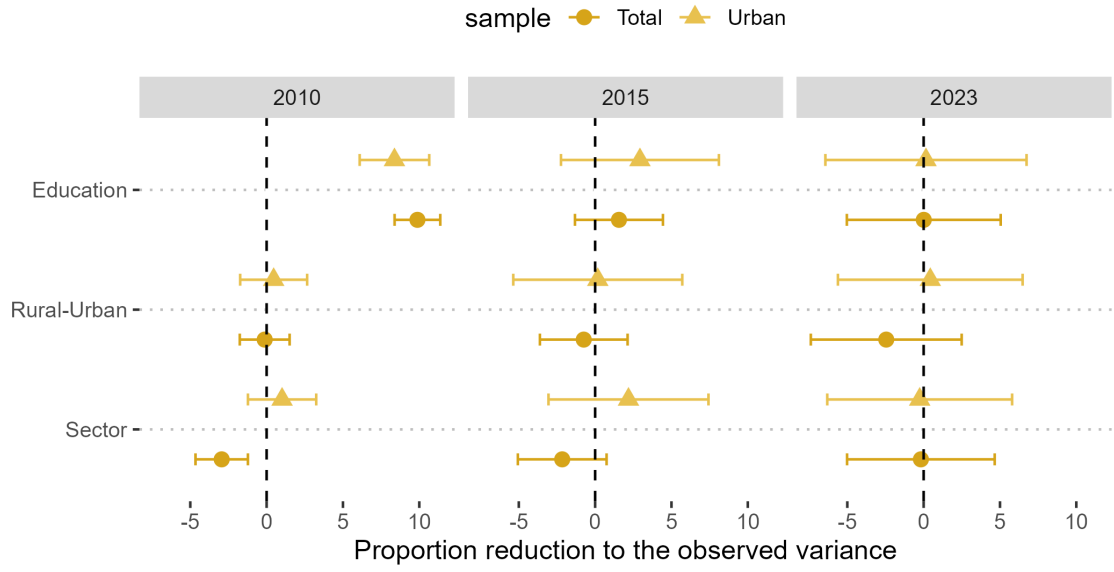
The contradicting results may be driven by the inclusion of flows of workers from the agricultural sector into the industrial sectors in the complete sample. As Table 2.3 shows, the employment contribution of the agricultural sector (i.e., farmers) decreased by 5 percentage points between 1996 and 2010. Given the substantial income disadvantage in the agricultural sector – over 50% less than the state and private sectors – the flows from the agricultural sector into the industrial sectors through migration and industrialisation would weaken the income hierarchy between the two sectors. The finding is consistent with sociological research that shows industrialisation acting as an important channel for social mobility (Maas and van, 2016; X. Song et al., 2020; X. Zhou and Xie, 2019).

Between 2010 and 2015, the estimates are more uncertain and mostly statistically insignificant. Nonetheless, the negative contribution of sectoral compositional change in the complete sample approaches significance, providing suggestive evidence for the continuation of the process described above, as the employment share of the agricultural sector continues to decrease by 9 percentage points in the period.

Finally, the estimates for 2023 are close to zero and insignificant, suggesting that compositional changes between 2015 and 2023 have no clear effects on the change in income inequality through between-group income gaps.

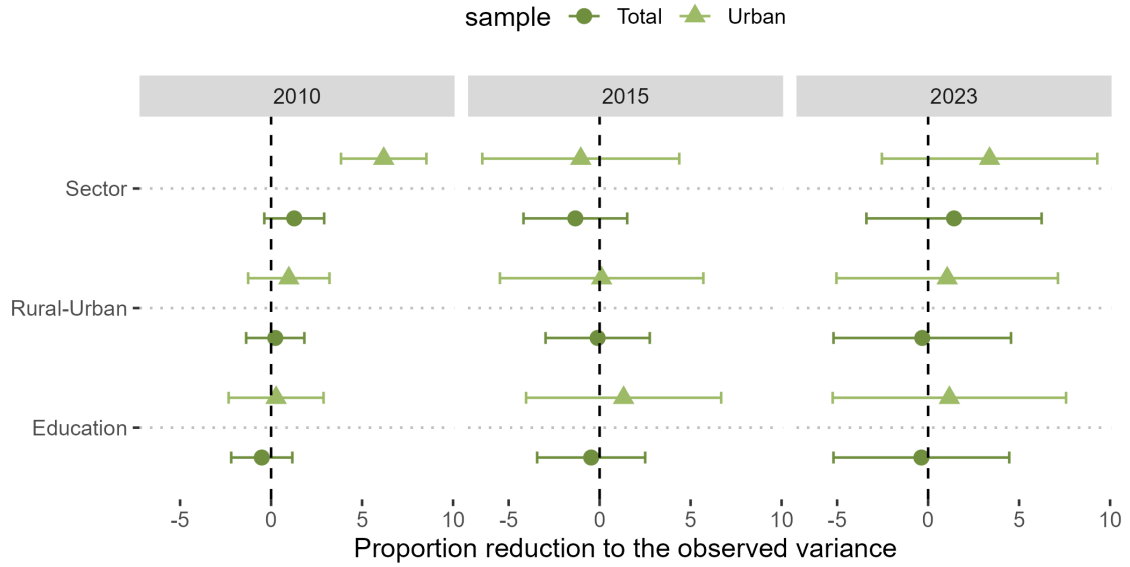
Figure 2.9b shows the counterfactual variance when the within-group income variance is changed by compositional changes. Between 1996 and 2010, the composition change in employment sectors positively contributed to income inequality in the urban industrial sample by increasing within-group income variation. The finding is consistent with X. Zhou (2014) and reflects the increase in income variation as former workers in the state sector entered the more insecure and unequal market sector. However, the effect is much smaller and statistically insignificant in the complete sample, as the rural agricultural population

Fixed compositions at the 1996/2010/2015 levels



(a) Between-Group Variance

Fixed compositions at the 1996/2010/2015 levels



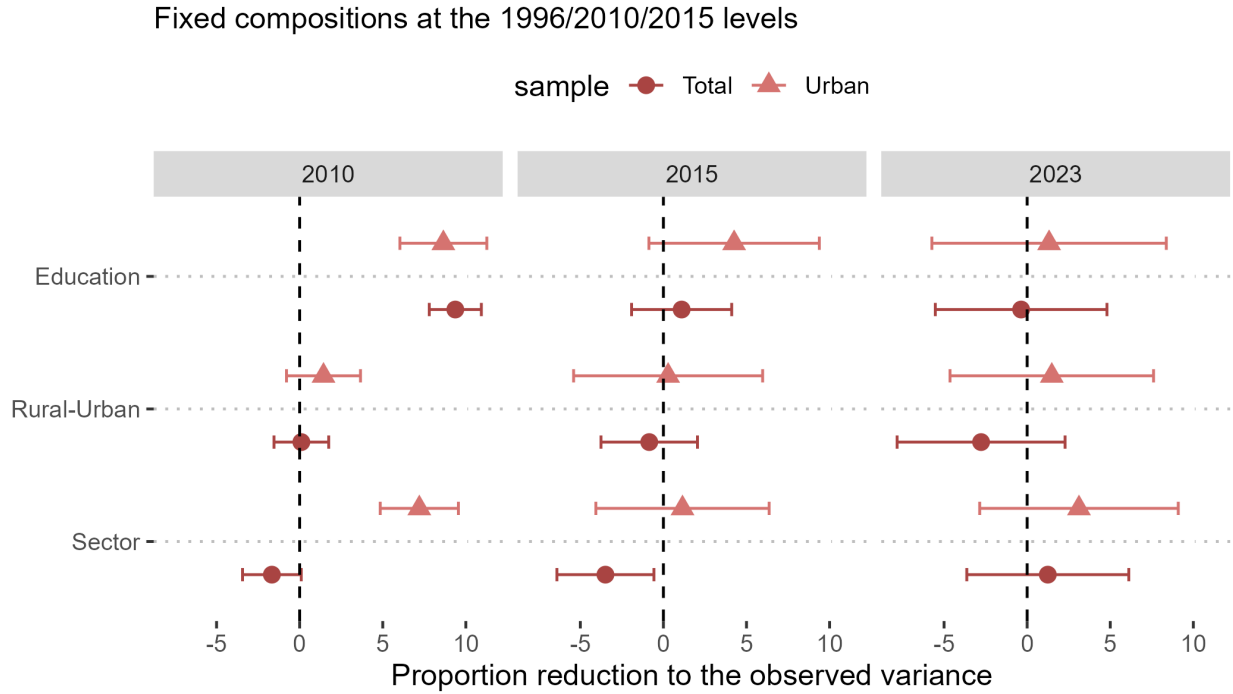
(b) Within-Group Variance

Notes: Proportion reductions are calculated as $\frac{\hat{V}_t^c}{\hat{V}_t} \times 100 = \frac{V_t - V_t^c}{V_t} \times 100$. The error bars show the 95% confidence interval, where SEs are calculated based on 250 bootstrap simulations.

Figure 2.9: Variance under Counterfactual Compositions

was largely unaffected by the SOE reforms. Therefore, while the compositional change in employment sectors induced by the SOE reforms significantly increased within-group income variation and inequality in the urban industrial labour market, its effect on national-level income inequality was much weaker. Focusing solely on the urban industrial sector may therefore lead us to overestimate the importance of the process on income inequality in post-reform China.

Except for the sectoral composition change between 1996 and 2010, the effects of other compositional changes on income inequality through within-group variation are close to zero and insignificant across the three periods. The results suggest that the post-reform rebalance of the state and private sector in urban China was a unique historical episode that affected income inequality in a special way.



Notes: Proportion reductions are calculated as $\frac{\hat{\Delta}^c}{\hat{V}_t} \times 100 = \frac{V_t - V_t^c}{V_t} \times 100$. The error bars show the 95% confidence interval, where SEs are calculated based on 250 bootstrap simulations.

Figure 2.10: Variance under Counterfactual Compositions: Total Effect

Taking the between-group and within-group channels together, Figure 2.10 shows the total contribution of compositional changes on income inequality. In 2010, if the education

distribution were fixed at the 1996 level, the income inequality would have been about 9% lower. The results are consistent in the complete and urban samples. However, the compositional change in employment sectors shows opposite effects in the complete and urban industrial sample. While in the complete sample, the inequality-reducing effect of the flow from the agricultural to the industrial sectors dominates, the increase in within-sector income variation from the rebalance of the state and private sectors dominates in the urban labour market.

Between 2010 and 2015, the inequality-reducing effect of sector compositional change in the complete sample persists and is even somewhat stronger. Except for the borderline significant positive contribution from the compositional change in education in the urban sample, the contributions of other changes are close to zero. There is also no significant effect from any changes between 2015 and 2023.

2.9.3 Summary

Taken together, two main patterns emerge from the counterfactual results. First, in terms of timing, the socioeconomic changes between 1996 and 2010 contributed most significantly to changes in income inequality, while the changes in 2010-2015 and 2015-2023 show much weaker effects. The pattern is consistent with the fact that most major structural changes, such as ownership reforms and education expansion, took place during the early stage of the reform in the early 2000s. It is consistent with the trend of income inequality in post-reform China, which stabilised and started to decline in 2010 (Kanbur, Y. Wang, and X. Zhang, 2021; Piketty, Yang, and Zucman, 2019).

Second, the effect of the same change on inequality may vary and even have opposite signs in the complete and urban samples. For example, the compositional change in employment sectors significantly increased income inequality in urban China, while reducing inequality at the national level. Such discrepancies highlight the importance of the rural agricultural population in constituting and shaping income inequality in contemporary China. Focusing

only on the urban industrial population, as most existing research on income inequality in China has done, could therefore lead to incomplete, if not inaccurate, accounts of the relationship between socioeconomic changes and income inequality.

Regarding the contributions of socioeconomic changes, between 1996 and 2010, changes in the between-group income gaps and within-group income variation by education levels and employment sectors were the most important drivers of income inequality. On the one hand, the increase in the return to education and within-education income variation positively contributed to inequality, particularly in the urban industrial sector. On the other hand, improvements in the relative income level and variation among farmers and the self-employed reduced income inequality. While the increase in the return to education and its inequality-increasing effect have been well theorised and documented, the income-reducing effect of changes associated with employment sectors has been much less studied (Nee, 1991; Xie and Hannum, 1996; X.-G. Zhou, 2000; X. Zhou, 2014; J. Lu, 2018). The result suggests that the economic reform entailed not only changes in the principle of reward allocation within the economic structure, but also structural changes that create more opportunities and certainty in previously disadvantaged sectors.

Compositional changes play a relatively less prominent, though still significant, role in shaping income inequality in China between 1996 and 2010. While the results for the urban sample are generally consistent with existing evidence by X. Zhou (2014), the results for sectoral compositional changes in the complete sample show different patterns. In the urban industrial sample, the sectoral compositional change between 1996 and 2010 increased within-group income variation, but had no significant effect on between-group income inequality; in contrast, in the complete sample, the compositional change reduced between-group inequality but did not affect within-group variation. The contrasting results offer new insights into the relationship between structural economic change, particularly industrialisation, and inequality in post-reform China. It also highlights the importance of incorporating the rural agricultural population in the analysis of social change and inequality.

2.10 Discussion

Because the counterfactual analysis is based on model-predicted group-specific means and variances, the results and interpretations are highly model-dependent. For simplicity, the discussion above uses terms such as “counterfactual” and “increase,” but these quantities should not be interpreted as causal effects on income levels, residual variation, or overall inequality, as they could be biased by unadjusted confounders or post-treatment adjustment (Opacic, Wei, and X. Zhou, 2025). For example, the total causal effect of migration on income could be mediated by changes in employment sectors (i.e, from the agricultural to the industrial sector or self-employed). However, because the model in Equation 2.10 includes the employment sector, the coefficient of rural-urban status is downward-biased relative to the true effect. Subsequently, this may lead to underestimations of the contribution of changes in rural-urban income gaps and compositional changes (i.e., migration).

Additional analysis in Section A.5 that excludes employment sectors from the variance function regressions and reweighting estimation confirms that, without controlling for employment sector, the rural-urban income gaps are larger and their changes more strongly reduce income inequality between 1996 and 2010. However, it does not affect the estimated contribution of rural-urban compositional change, possibly due to the under-representation of the rural population in the LHSC sample, which also leads to the underestimation of the compositional effect (see below).

Furthermore, for the counterfactual composition analysis, the results’ credibility relies on the sample’s representativeness. Although both LHSC and CGSS are designed to be nationally representative, their realised samples may still deviate from the target population because of access constraints and sampling variability. For example, 42.8% of the 1996 LHSC sample live and work in the urban area (including migrant and urban workers), while the official urban population share from the 1990 and 2000 Censuses is 21% and 26% (National Bureau of Statistics of China, 2021b). The LHSC data, therefore, over-represent the urban population, possibly due to limited access to the rural area. Practically, this means that the

change in rural-urban population composition – and possibly by extension, agricultural and industrial sector composition – and thus its contribution to income inequality, between 1996 and 2010, is underestimated. Future research should improve the analysis by augmenting the data using alternative sources, such as the 1% Population Sampling Survey (also known as the mini-Census) and the official Censuses.

Finally, even if the model estimates are credible, they only capture the partial equilibrium effect and do not consider the interdependent feedback effect among variables. For example, the estimated effect of counterfactual education composition does not consider the downstream effect on returns to education due to changes in labour supply. Future research could model such general equilibrium effects by specifying structural models linking different variables.

2.11 Conclusion

In China’s transformation from a planned, agricultural economy into a global industrial superpower, the lives of millions of people have been transformed, as they leave the fields for factories, construction sites, and service work. Such changes in their lives have had profound, yet scarcely examined, effects on income distribution in contemporary China.

This study provides an up-to-date account of the relationship between socioeconomic changes and income inequality in post-reform China that incorporates the rural agricultural population. Drawing on survey data spanning almost three decades from 1996 to 2023, I show that the importance of income predictors and the contribution of socioeconomic changes to income inequality substantially vary when using the complete and urban industrial sample. In particular, through counterfactual analysis, I show that changes in between-sector income gaps and employment sector compositions reduced income inequality between 1996 and 2010, with the effect of the latter persisting between 2010 and 2015. These new findings contrast with existing evidence based on the urban industrial sample (X. Zhou, 2014).

The results raise new questions on social change and inequality in China. Consistent with previous research, I find that income inequality in China stabilised and started to decline after 2010, and the effects of socioeconomic changes that shaped inequality in the early reform period mostly dissipated after 2010 (Kanbur, Y. Wang, and X. Zhang, 2021; Piketty, Yang, and Zucman, 2019). This means that, first, the prevalent impression in academic and public discourse about the rapid increase in income inequality in post-reform China is partly outdated. In a new stage of stable, high inequality, with signs of moderate decreases, future research should, therefore, look more closely at potential changes in different sections of the income distribution and drivers of inequality decline (see e.g., Aeppli and Wilmers, 2022).

Second, the stabilised and even decreasing trend in inequality decouples from the increasingly pessimistic public perception of inequality and mobility in China (Alisky, Rozelle, and Whyte, 2025). Future research, therefore, could analyse the new socioeconomic developments that give rise to the contradictory trends. For example, the recent decrease in returns to education may, at the same time, reduce income inequality and diminish people's confidence in social mobility (Hanushek, Y. Wang, and L. Zhang, 2025). In that case, the force that drives income inequality may paradoxically act as a source of confidence in meritocracy and equality in a period with rapidly increasing inequality (Whyte, 2010a; Waight and Goldstein, 2025).

More importantly, the study demonstrates the importance of the rural agricultural population for understanding socioeconomic changes and inequality in post-reform China. In doing so, it highlights mechanisms, such as the compositional changes in employment sectors, that shaped income inequality in contemporary China but have been previously understudied. At the conceptual level, the findings underscore the fact that all socioeconomic changes in China – as an economy with a substantial rural population and agricultural sector that are closely connected to the urban industrial sector through population and material flows – affect both the rural and urban populations. For example, changes in trade policy that affect the export-oriented industrial sector would also affect the rural agricultural sector by

drawing labour and capital into the industrial sector (Facchini et al., 2019; Erten and Leight, 2021). Any analysis concerning population outcomes such as income inequality, therefore, is incomplete if the rural population is not included. Similarly, changes in the rural population and agricultural sector should also be examined as potentially equally consequential for income inequality (e.g., rural pension change in Gai et al., 2025). Currently, with developments such as robotisation, artificial intelligence and agricultural contracting, China is undergoing a new wave of socioeconomic transformations that affect both the rural and urban populations. Future research should examine how the historical and ongoing changes affect the rural and urban populations through different mechanisms and their implications for income inequality.

Finally, beyond the Chinese case study, the study highlights the important relationship between the rural agricultural population, socioeconomic changes and inequality and mobility, especially in historical and developmental contexts. With most developed countries and many developing countries in the post-industrial economy and society, and with novel technological developments, it is easy for social scientists to overlook the pre-industrial world as categorically irrelevant. However, as classic theories and recent evidence from economics and sociology have shown, socioeconomic changes in the agricultural sector – such as education expansion, technological changes and industrialisation – have been among the most important factors shaping inequality and mobility (Ager, Goñi, and Salvanes, 2026; Breen et al., 2009; X. Song et al., 2020; Gollin, 2014; Hällsten, 2025). As social scientists try to understand inequality and mobility from more long-term and global perspectives, the rural agricultural population deserves more attention. After all, for the better part of human history, rural society and agricultural production were – and in many parts of the world, remain — the most important form of social and economic organisation and understanding them is key to understanding the trends and causes of inequality and mobility.

Chapter 3

The Labour Market Returns to Rural-Urban Migration in China

Since the beginning of its economic reform in the 1980s, China has witnessed one of the biggest waves of internal migration in human history. The size of the so-called “floating population” – individuals whose residence location for no less than six months is different from their household registration (*hukou*) location – has increased by over ten times from 21 million in 1990 to 221 million in 2010 (Z. Liang, Z. Li, and Ma, 2014; Z. Liang, 2016). The latest 2020 Chinese Census documents that the size of the floating population is 492 million, equal to an 88.5% increase from the 2010 Census (National Bureau of Statistics of China, 2021b). That is a population of over 7 times the UK population on the move.

The unprecedented scale of internal migration – mostly between rural and urban areas – constitutes one of the most unique dimensions of China’s economic reforms and developments. The economic reforms that began in the 1980s relaxed the previous rigid division of rural and urban sectors and strict controls over migration under the socialist planned economy, which have been described as a form of serfdom (Whyte, 2010b; H. Lu, 2010). The newly opened mobility opportunities and development in urban areas in coastal provinces have prompted millions of people from rural areas to move to cities seeking economic oppor-

tunities. The process accelerated after China joined the World Trade Organisation in 2001, as the rapid development of the export-oriented labour-intensive manufacturing industry demanded and attracted large numbers of migrant workers (Facchini et al., 2019; Erten and Leight, 2021).

While rural-urban migration has fuelled China’s rapid industrialisation and development, and provided channels of upward mobility for many people, it has also created significant social problems – some due to the institutional and everyday discrimination against migrant workers and their children and some due to the demographic decline in origin villages. A large body of literature in sociology and demography has documented migrants’ systematic disadvantages and the negative consequences on left-behind elderly and children (Friedman, 2022; Z. Liang, 2016; Zhuo and Z. Liang, 2015; Yue et al., 2020; X. Wu and Zheng, 2018; J.-M. Wu, 2010; He and Ye, 2014; Q. Song, 2017).

What is surprisingly less studied, however, is the labour market consequences of migration on workers’ life chances. On the one hand, the movement from the rural agricultural sector to the more productive industrial sector could increase workers’ income (Gollin, 2014). Through exposure to the more skilled and productive urban economy, migrants could also acquire skills that lead to long-term income growth (Glaeser and Mar, 2001; Andersson and Molinder, 2025). However, on the other hand, the institutionalised rural-urban segmentation creates high barriers to migrants’ integration. Due to the rigid rural-urban division institutionalised by the *hukou* household registration system, the rural population and migrants have been systematically excluded from the urban public services system. Until 2003, migrants were not even allowed to live in the urban areas without a permission certificate; they would otherwise be arrested and deported (J.-M. Wu, 2010). Furthermore, due to their relatively low skills and education and unfitting manners, migrant workers are often perceived as low-quality people by urban residents (Friedman, 2022; Z. Liang, 2016). Due to these institutional and everyday discriminations, migrant workers may face significant economic disadvantages, such as income penalty and higher risks of downward mobility (X. Wu and Treiman, 2007; X. Wu

and Zheng, 2018).

Furthermore, in recent years, as China’s economic growth slows down and its economy shifts away from labour-intensive manufacturing toward automation, opportunities for migrant workers have narrowed (Yuan, 2023; Lei, 2023). A recent statement from the Ministry of Agriculture and Rural Affairs in December 2025 even states that the government must “prevent a large-scale return and idling of migrant workers in their hometowns” (D. Zhou, 2025; Langley, 2026).

So does rural-urban migration lead to genuine upward mobility? Or does it simply lead rural-origin workers to another dead end? This study, to the best of my knowledge, provides the first analysis of the effect of rural-urban migration on rural-origin workers’ economic outcomes. Using individual-level longitudinal data from the Chinese Family Panel Studies (CFPS), I estimate the average treatment effect on the treated (ATT) of rural-urban migration on migrants’ annual income. Across specifications, estimates from the new two-way fixed effect counterfactual estimators, proposed by L. Liu, Y. Wang, and Xu (2024), consistently suggest a substantial positive income return to rural-urban migration of about 40%, although baseline estimates may be upward-biased by pre-migration income selection. Heterogeneity analysis reveals that the effect magnitudes are similar for male and female migrants, though there is suggestive evidence that the effect is stronger for migrants with at least high school education. I also find suggestive evidence that about one-fifth of the average income effect is explained by the workers moving from the agricultural to the non-agricultural sector. The mediating effect is more salient among migrants initially working in the agricultural sector, accounting for one-third of the total effect.

Overall, by demonstrating the importance of rural-urban migration as an upward mobility channel, the study contributes to the literature on social change and inequality in contemporary China and the wider literature on the geographic dimension of inequality. The rest of the study proceeds as follows. Section 3.1 draws on research from sociology, demography and economics and outlines a model of the income return to rural-urban mi-

gration. Then, Sections 3.2 and 3.3 explain the construction of the analytical panel sample from the CFPS data and the empirical design and methods to estimate the income effect of migration. Section 3.4 presents the results with the limitations discussed in Section 3.5. Finally Section 3.6 concludes with implications for future research.

3.1 Literature Review

Rural-urban migration affects individuals' life chances as both a life course event and an economic action. As a life course event, individuals' migration decisions are conditioned by and affect their own and family members' life course trajectories and conditions, as they balance different life course constraints, expectations and needs. Meanwhile, as an economic action, migration shapes individuals' economic outcomes by changing their positions and experiences in the labour market. In this section, I review existing theories and evidence on the relationships between migration, life course and economic outcomes.

3.1.1 Migration as a Life Course Event

Health

In terms of health, existing research has consistently documented significant positive health selectivity among migrants and negative selectivity among return migrants in China (Tong and Piotrowski, 2012; X. Hu, Cook, and Salazar, 2008; Y. Lu and Qin, 2014; S. Wang and Y. Hu, 2019). The finding is consistent with the “healthy migrant effect” and the “salmon hypothesis” observed in other international and internal migration contexts. In particular, Tong and Piotrowski (2012) shows that health selectivity is particularly salient in early reform years, suggesting interactions between individual life-course decisions and macro-social contexts.

However, despite the positive health selectivity, research has also examined the health disadvantages faced by rural-urban migrants due to factors such as hazardous working con-

ditions, unhealthy lifestyle and limited access to healthcare under the *hukou* system (X. Hu, Cook, and Salazar, 2008; Bi et al., 2016; Shi, 2022). For example, using panel data between 2011 and 2014, Shi (2022) estimates that interprovincial migration leads to a 3% decrease in migrants' self-reported health score. Such negative health effects, further, could translate into negative income effects as they limit migrants' work capacity.

Marriage and Fertility

The relationship between marriage and migration is more complex. On the one hand, for married spouses, research has found that the spatial separation due to migration is detrimental to marriage quality and individuals' health (F. Chen et al., 2015; Tong, F. Chen, and Shu, 2019). However, on the other hand, for the young unmarried migrants, migration brings new norms and prospects for intimate relationships and marriage. For example, Xiong (2023) shows that, given the marriage market composition and mating norms, while women face better marriage prospects after migration, migrant men face much worse prospects, except for those with high education. Furthermore, Choi and Peng (2016) show that the living and working arrangements in cities create new gender norms that allow greater autonomy and control for women and force men to negotiate the traditional presentation of masculinity and domination. Such cultural changes could also affect migrants' marriage intentions and prospects.

A similar shift can be observed in fertility intention and decision. Based on life-story interviews with migrants, You (2026) finds that the spatial separation from the origin communities and exposure to new uncertainties and conditions through migration disrupt people's fertility plans and previously socialised fertility norms. The disruptions of fertility plans and changes in norms are consistent with existing research that has found negative effects of migration on women's and households' fertility intentions (Y. Liang, Y. Yi, and Q. Sun, 2014; M. Zhou and Guo, 2021; Y. Yi and Shangguan, 2024; W. Luo et al., 2026).

Overall, the evidence suggests that rural-urban migration reshapes migrants', especially

female migrants’, attitudes and behaviours on marriage and fertility. Although the directions of the effects are less predictable, these changes would alter migrants’ marriage and fertility decisions, and thus affect their economic prospects.

Linked Lives

Migration as a life course event affects not only the migrants themselves but also the linked lives of intergenerational family members dependent on them (Carr, 2018). The massive waves of rural-urban migration in China, combined with lagging social services, have created a care crisis in rural areas where a large population of so-called “empty-nested” elderly and “left-behind” children lack family support and care.

Existing research has shown that, with lagging and limited pension and social services in rural areas, the migration of children has significantly undermined the family support for the elderly, subjecting them to greater material, physical and mental stress and leading to worse health (He and Ye, 2014; Q. Song, 2017). Meanwhile, for the left-behind children, the lack of parenting and parental care has led to worse developmental outcomes in nutrition and cognitive development and deteriorating intergenerational relationships (L. Wang and Mesman, 2015; Murphy, 2020; Yue et al., 2020; Rozelle and Hell, 2022).

These negative effects could trigger different responses from migrants. On the one hand, they may quit certain economic opportunities or even return to the villages to balance their care responsibilities. Such responses have been particularly common among female migrants (see e.g., Y. Dong, 2023). On the other hand, more prevalent among male workers, migrants may seek to compensate for the absence of care by sending more remittances, which motivates them to earn more (Q. Cai, 2003; Pan and W. Dong, 2020; Z. Wang, 2026). The two types of response, therefore, lead to contrasting and gendered effects on migrants’ economic prospects.

Finally, with the special importance of kinship and family in Chinese society, individuals’ migration decisions are more fundamentally linked with the life course outcomes of their household members beyond direct impacts. Based on longitudinal interviews, Fan (2024)

show that individuals' migration decisions and timings are often driven by household members' – especially children's – life course events, such as education and marriage, as they seek to provide for the family. For example, as marriage in China typically requires a large amount of gifts and a house, parents may migrate to earn and save money for their children's marriage. In this way, workers' migration decisions are deeply embedded in the linked life courses of family members.

3.1.2 Migration as an Economic Action

For rural-urban migrants seeking job opportunities and higher income, migration is fundamentally an economic action that leads them from the rural agricultural sector to the urban industrial and service sectors. In this process, migrants' economic conditions could be affected by the changes in employment sectors, locations and institutions under China's unique *hukou* system.

Employment Sector

As workers migrate from rural to urban areas, their economic conditions change as they leave the agricultural sector and enter the industrial manufacturing and service sectors. This process is described in the classic Lewis two-sector model in development economics (Lewis, 1954; Gollin, 2014). The model stipulates that, in developing countries, the economy is divided into a traditional agricultural sector – where productivity and the marginal return to labour are low, and the labour supply is excessive – and a modern sector where the marginal return to labour is higher. Economic development occurs when excess labour in the traditional sector enters the modern sector. In turn, workers earn higher incomes from higher marginal returns to labour and provide labour to the modern sector.

Despite limitations in Lewis (1954)'s original formulation, the distinction between the traditional agricultural and the modern industrial sector and the movement of labour from the former to the latter have been supported by evidence from developing countries and

historical cases such as the 19th-century US. (Gollin, 2014; MacDonald, 2021). The Chinese economy has, to a large extent, also been consistent with the two-sector characterisation, with a large agricultural sector with low marginal return to labour and a fast-developing industrial economy in urban areas, especially in coastal provinces (Huang, 1985; Kanbur and X. Zhang, 2005). Rural-urban migration, under such circumstances, functions as a mobility channel for rural workers to earn higher incomes in the more productive modern sector.

Location Effect

Besides changes in employment sectors, the change in location itself – from rural to urban areas – could have direct effects on workers’ income. Research in urban and labour economics has extensively documented and theorised the income difference between urban metropolitan and non-metropolitan areas.

There are two main explanations for the urban income premium. First, cities could foster workers’ skill development and long-term wage growth through intellectual interactions and spillover (Glaeser and Mar, 2001; Roca and Puga, 2017). For example, Andersson and Molinder (2025) examine the rural-urban migration in Sweden between the 1880s and 1990s and estimate that about two-thirds of the net urban wage premium (i.e., net of migrant selectivity) can be explained by the accumulation of skills and workers moving into higher-skilled occupations.

Second, firms in cities could have higher productivity due to better infrastructure and the agglomeration effect that directly increases workers’ income (Glaeser and Gottlieb, 2009). The recent study by Card, Rothstein, and M. Yi (2025) that corrects for movers’ sorting into firms shows that about half of the income variation between commuting zones in the US can be attributed to differences in direct place effects. Given the large regional and rural-urban disparity in development and economic compositions, and the larger sizes of Chinese cities, the direct effect of urban areas could be even larger.

The *Hukou* System

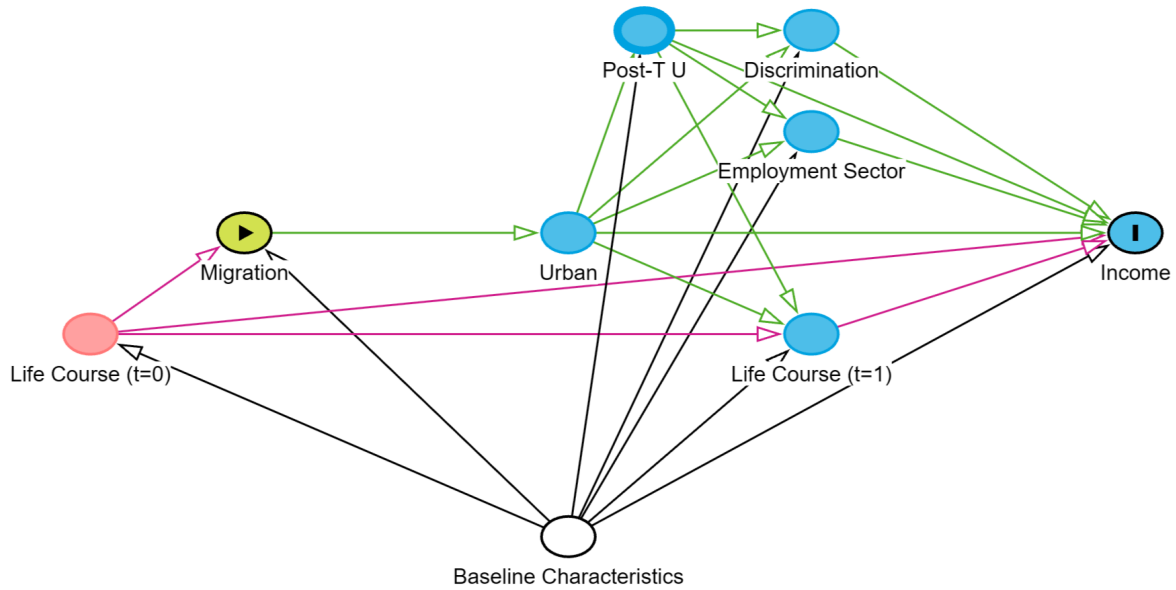
While cities could provide economic opportunities for migrant workers, the Chinese *hukou* system poses an institutional obstacle to migrant workers' life chances in the cities. A legacy from the socialist planned economy era, when population movements were strictly controlled, the system created a de facto two-tier citizenship system in which people with rural *hukou* were excluded from many public services in cities, and were not even allowed to live in cities without permission certificates (J.-M. Wu, 2010).

Despite multiple pushes to reform the system since the 2000s, many discriminatory policies and barriers remain. For example, in recent years, many local governments have adopted the point-based settlement system – in which non-local residents with enough points, awarded based on traits like education, age and tax contributions, can settle in the cities and access local public services – to lower the integration barriers for migrants. However, by institutionalising restrictions on settlement and access to public services, this seemingly meritocratic system also puts more implicit discrimination on rural-urban migrants, especially those who are older and have lower education and skills, as undeserving of public services (Friedman, 2022; Lei, 2023).

Existing research based on comparisons between migrants and urban workers has found mixed results on the disadvantage experienced by migrant workers with rural *hukou*. On the one hand, X. Wu and Treiman (2007) show that migrant workers without urban *hukou* are particularly prone to downward mobility back to the agricultural sector. Even if migrant workers could obtain urban *hukou*, its positive effect on earnings is limited to relatively advantaged workers with better education and a higher probability of obtaining the *hukou* (X. Wu and Zheng, 2018). On the other hand, research has found small and insignificant direct income differences between workers with and without urban *hukou*, adjusting for individual characteristics like education (e.g., X. Zhou, 2014). The returns to education and the determinants of income between migrant and urban workers also converge over time (J. Sun, 2017).

3.1.3 A Life Course Model of the Income Returns to Migration

To summarise, existing research in sociology, demography and economics has shown that rural-urban migration, as a life course and economic event, is embedded in the life course conditions of individuals and their family members and can shape individuals' life chances by affecting their life course outcomes and economic conditions. On the one hand, rural-urban migration could improve rural-origin workers' income prospects by entering more productive industries and areas. On the other hand, these positive effects could be discounted by institutional discrimination and negative effects on, for example, health.



Notes: **Post-T U** denotes post-treatment unobserved confounders. The red nodes and arrows represent confounders and confounding paths. The green arrows represent the causal paths.

Figure 3.1: Directed Acyclic Graph for the Effect of Migration on Income

The directed acyclic graph (DAG) in Figure 3.1 summarises the hypothesised causal relationship between rural-urban migration and income for rural-origin workers. The green causal paths show that rural-urban migration – by definition – changes workers' rural-urban condition, which then affects income both directly through location effects and indirectly through the changes in employment sectors, *hukou* discrimination and life course conditions

such as health, childbearing and care responsibilities. The **Post-T U** node captures unobserved post-treatment variables such as attitudes and skills that are affected by migration, which affect both income and other mediators.

On the left of the graph, the red node shows that people’s migration decisions are shaped by their existing life course conditions ($t=0$), including those of their own and their family members. These existing conditions affect subsequent conditions after the migration ($t=1$). They also directly affect individuals’ income, for example, by restricting their work capacity due to health issues or care responsibilities, or motivating them to earn more income to provide for family members’ life events. These life course conditions, therefore, confound the relationship between migration, subsequent life course conditions and income.

At the bottom of the graph, individuals’ baseline characteristics, such as sex, family background and education (assuming that the individuals have left school and education is constant), affect both their migration decision and income levels.

Given the hypothesised causal structure, identifying the total causal effect of rural-urban migration on income, therefore, requires adjusting for the time-constant baseline characteristics and the time-varying pre-migration life course conditions. The following sections explain the empirical data and methods used to estimate the causal effect.

3.2 Data

3.2.1 Analytical Sample

To estimate the effect of migration on individuals’ income, I use data from seven waves of the Chinese Family Panel Studies (CFPS) between 2010 and 2022. Conducted by the Institute of Social Science Survey at Peking University, CFPS is a biannual, nationally representative longitudinal survey that began in 2010. The 2010 baseline wave includes a nationally representative sample of households and household members, who are followed in subsequent surveys. Later waves include and follow new individuals with direct kinship relationships

with the baseline individuals through, for example, marriage and birth. The survey collects information at the community, household and individual levels, focusing primarily on economic well-being and family relationships. In each wave, individuals aged 16 and above answer the self-report adult questionnaire, which collects information on their demographic and socioeconomic conditions, including occupations and income.

For the current study, I use data from self-report questionnaire responses from individuals aged 16 to 55 in the 2010 baseline sample who were also interviewed in 2022. To focus on the working population with rural origins, I further restrict the sample to individuals who are in rural areas in the first observed wave, are not in school in any wave, and are not economically inactive for two consecutive observed waves. After dropping person-year observations with missing values in rural-urban status and annual income, and then dropping individuals with no repeated observations, the sample includes 37,049 person-year observations from 6,487 individuals. The 2022 wave of CFPS provides panel weights that adjust for survey sampling and attrition, relative to the 2010 baseline population. After dropping 1,403 individuals without valid weights, the sample reduces to 30,875 person-year observations from 5,084 individuals. The panel is unbalanced due to missing values.

3.2.2 Measurement

For the treatment variable, rural-urban status, I use the type of respondents' current residence location, classified based on the type of local administrative units, i.e., village committee (*cun wei hui*) v.s. neighbourhood office (*jie dao ban*). The information is directly recorded by the interviewers in each wave.

The measurement of income, however, is more complicated. In 2010 and 2012, the self-report adult questionnaire collected individuals' total annual income from wages, bonuses, benefits and transfers. From the 2014 wave onward, the questionnaire focuses only on annual post-tax wage income. The data in 2016, however, were missing for individuals whose main job did not change between 2014 and 2016 and were retrospectively imputed in the 2018

wave. Due to these complications, the personal income data are not strictly comparable across years. However, these between-year inconsistencies should be partly adjusted for by the year fixed effect in subsequent modelling, which could mitigate the bias to the estimates (see below).

For rural farmers and self-employed migrants, wage income does not adequately capture their economic conditions. Therefore, I also consider individuals' annual household income, collected in the household economy questionnaire. In each wave, CFPS separately collects four components of the household income: wages, agricultural and business profits, asset returns and transfers. There were also changes in component definitions between 2010 and 2012. In the analysis, I supplement the individual wage income with household per-capita non-wage income. The final individual annual income for person i in household j in year t , therefore, is calculated as:

$$Income_{ijt} = Wage_{ijt} + \frac{HouseholdTotal_{jt} - HouseholdWage_{jt}}{HouseholdSize_{jt}}$$

I adjust the income in each year to the 2010 price level using the official Consumer Price Index. For subsequent analysis, I take the natural logarithm of the adjusted real annual income plus one.

In addition to the treatment and outcome variables, I also construct a binary indicator of whether the respondent's main occupation is in the agricultural sector. The 2010 wave includes a designated item for this question. From the 2014 wave onward, the data include a variable classifying the type of workers' main occupation, and I classify observations as working in the agricultural sector if they work on household farms or are agricultural employees. In the 2012 wave, however, there is no directly related item, so I infer individuals' agricultural employment status from their occupation code. Because the measure was not standardised in 2010 and 2012, there were more missing values in the two waves (see Section B.1).

Finally, it should be noted that, although the sample specifically targets rural-origin

workers who are not in school throughout the period of study, the recorded education level of many individuals changes across waves. This could be due to measurement or recall error (see e.g., Obermaier and Hastings, 2025). In the main analysis, I code individuals' education as a time-constant variable using the first observed value. As a robustness check, I also create a sample excluding the 924 individuals with changes in their recorded education level.

3.2.3 Panel Data Structure

Figure 3.2 shows the treatment histories of the 5,084 individual observations in the analytical sample. The 2020 wave has noticeably more missing values because the data were collected during the COVID-19 pandemic, with many interviews conducted remotely through telephone. The response rate was therefore lower than in other years.

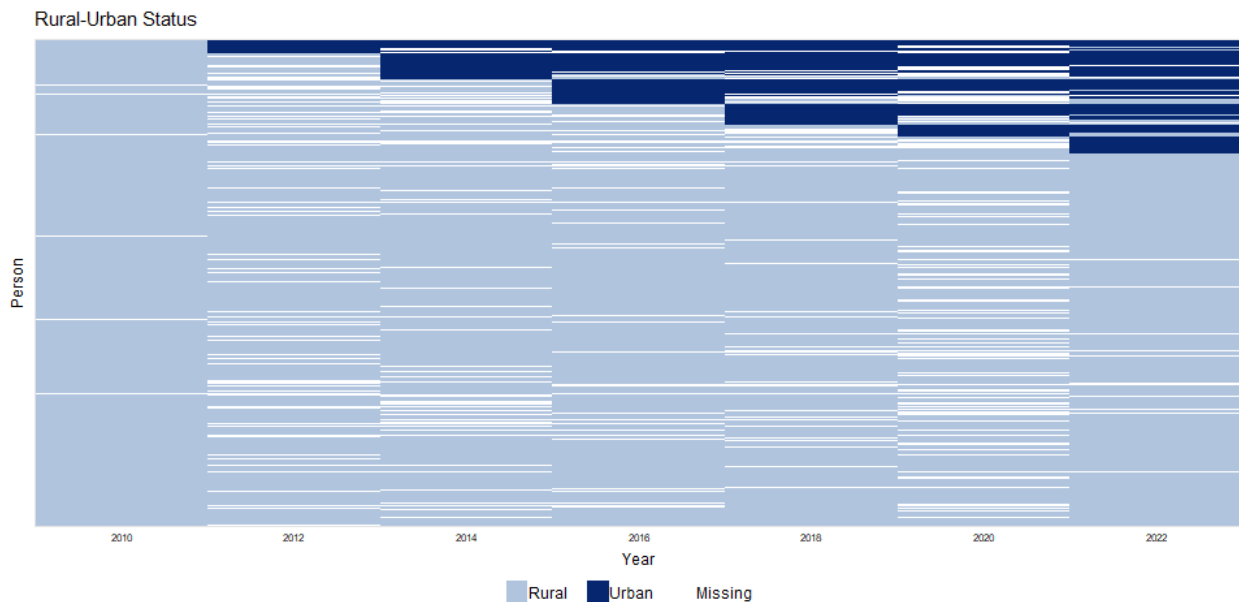


Figure 3.2: Treatment Histories of Rural-Urban Status

Three patterns can be observed from the treatment histories. First, perhaps unsurprisingly, most rural-origin people aged 16-55 in 2010 did not migrate between 2010 and 2022. 1,471 individuals, or about 29% in the sample, ever migrated. Second, the migration timings vary, indicating a staggered treatment setting. Finally, there are 487 individuals the sample

with reversals in their treatment histories through return migration. Overall, the data have sufficient variation to identify the effect of rural-urban migration; however, the staggered treatment histories and reversal in treatment need to be handled in modelling (see below). As a robustness check, I create a sample that excludes individuals with treatment reversal.

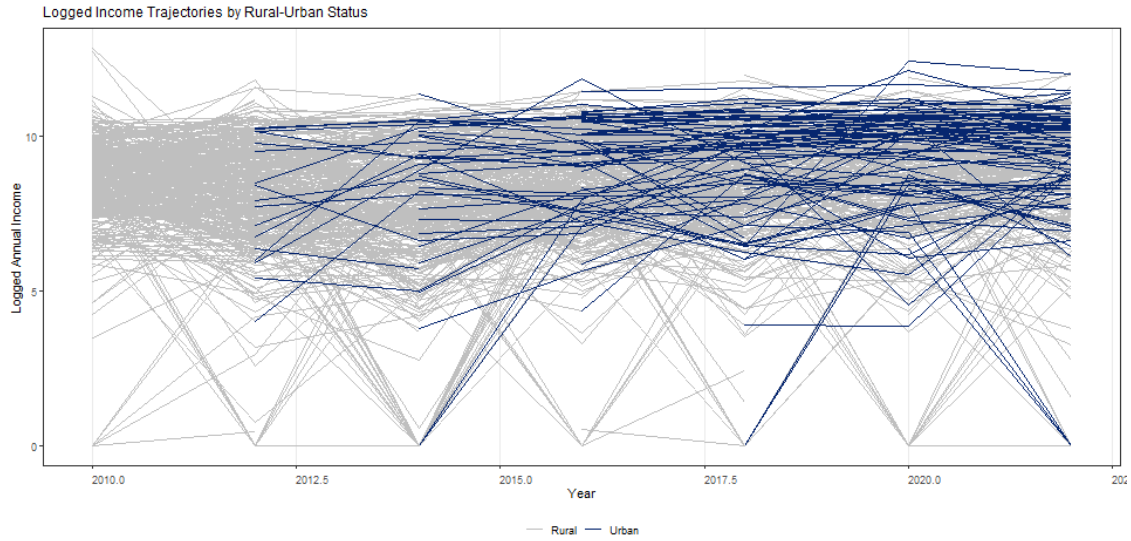
Figure 3.3 shows the descriptive trajectories of logged real annual income of 500 randomly sampled individuals, coloured by treatment status. The top panel 3.3a shows the trajectories from the main analytical sample. It shows that the migrated (i.e., treated) individuals tend to have higher income. They also have an overall somewhat positive trend. However, there are visible clusters of observations with very low income, including those with zero annual income. To mitigate their impacts as potential outliers, I create a separate sample that trims the logged annual income at the year and rural-urban-specific 1st and 99th percentiles. The bottom panel shows the trajectories of 500 randomly sampled individuals from the sample, with fewer observations with very low incomes.

3.3 Empirical Design and Methods

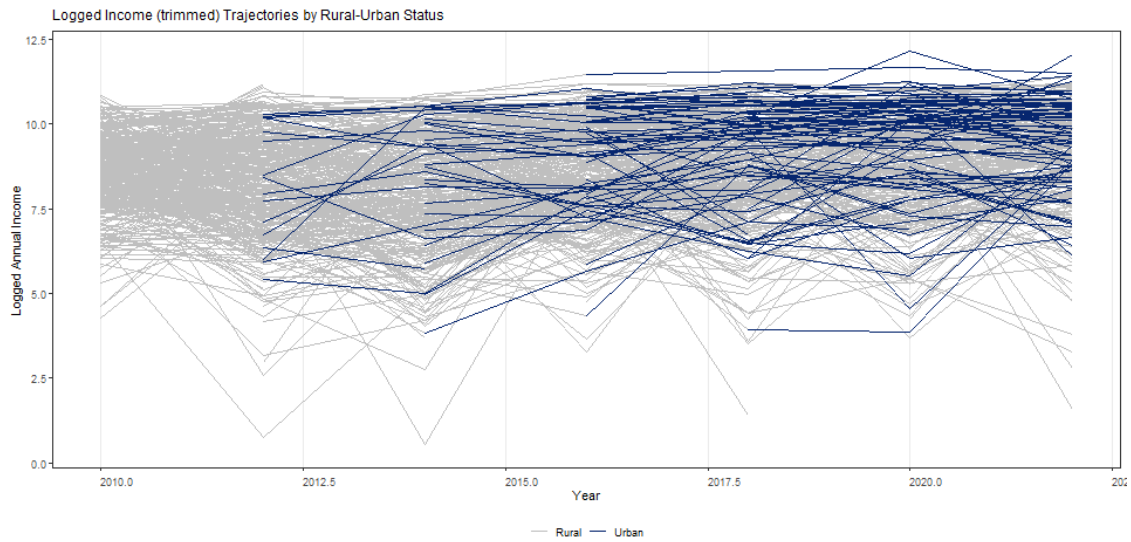
3.3.1 The Fixed Effects Counterfactual Estimator

To estimate the causal effect of rural-urban migration on income, I exploit within-person variation in the occurrence and timing of migration, which allows us to identify the effect of migration on the ever-migrated units. The most common approach for such analysis is the two-way fixed effects (TWFE) regression model, which regresses the outcome on the treatment indicator while adjusting for individual fixed effects, time fixed effects and optional time-varying variables. In the simple case of two groups and two periods, it reduces to the familiar difference-in-differences (DiD) estimator. Under the strict exogeneity assumption – also known as the parallel trend assumption – the coefficient for the treatment variable captures the average treatment effect on the treated units (ATT).

However, with multiple time periods and variation in treatment timing – as is the case



(a) Main Sample ($N=30,875$)



(b) Trimmed Sample ($N=29,915$)

Notes: 500 individuals were randomly sampled from each sample. Person-year observations are trimmed at the year and rural-urban-specific 1st and 99th percentiles of logged real income in the trimmed sample.

Figure 3.3: Income Trajectories by Treatment Status

in the panel data shown in Figure 3.2 – TWFE regression produces estimates that are difficult to interpret and could be biased. Goodman-Bacon (2021) and de Chaisemartin and D’Haultfoeulle (2020) show that in such staggered adoption settings, the TWFE estimate is a weighted average of all pairwise comparisons between groups adopting the treatment at different times, where the weights are determined implicitly by the estimation procedure and can be negative. Negative weights arise when early adopters – individuals who migrated earlier – are used as implicit controls for late adopters. Such comparisons are problematic if treatment effects are heterogeneous across individuals or change over time after migration, which is plausible and indeed suggested by the dynamic results below.

To address this problem, I use the fixed effects counterfactual (FEct) estimator proposed by L. Liu, Y. Wang, and Xu (2024). Instead of estimating the ATT from multiple contrasts, the FEct estimator directly imputes the missing counterfactual among treated units: what would the person’s income have been in this year had they not migrated? It then calculates the individual treatment effect and aggregates them to calculate the ATT using uniform or designated weights.

The estimator proceeds in three steps. First, using only the untreated observations, it fits the response surface of the potential outcome under control $Y_{it}(0)$ with a TWFE functional model:

$$Y_{it}(D_{it} = 0) = \alpha_i + \xi_t + \mathbf{X}'_{it}\boldsymbol{\beta} + \varepsilon_{it}, \quad \text{for all } (i, t) \text{ s.t. } D_{i,t} = 0 \quad (3.1)$$

where α_i is an individual fixed effect, ξ_t is a time fixed effect, $\mathbf{X}_{it}$ are time-varying control variables and ε_{it} is an idiosyncratic error term.

Second, the estimated model is used to predict the counterfactual outcome $\hat{Y}_{it}(0)$ for each treated observation — that is, what the individual’s income would have been in the post-migration period had they remained in the rural area. Third, the individual treatment effect is estimated as the difference between the observed outcome and the predicted counterfactual:

$$\hat{\delta}_{it} = Y_{it} - \hat{Y}_{it}(D_{it} = 0), \quad \text{for all } (i, t) \text{ s.t. } D_{i,t} = 1 \quad (3.2)$$

The ATT and the period-specific ATT at s periods since migration are then estimated by averaging the individual treatment effects:

$$\widehat{\text{ATT}} = \frac{1}{|\mathcal{M}|} \sum_{(i,t) \in \mathcal{M}} \hat{\delta}_{it}, \quad \widehat{\text{ATT}}_s = \frac{1}{|\mathcal{S}_s|} \sum_{(i,t) \in \mathcal{S}_s} \hat{\delta}_{it} \quad (3.3)$$

where $\mathcal{M}$ denotes the set of all treated observations; $\mathcal{S}_s$ denotes the set of treated observations at s periods since migration; and $|\mathcal{M}|$ and $|\mathcal{S}|$ denote the number of observations in the sets. L. Liu, Y. Wang, and Xu (2024) shows that, under the strict exogeneity assumption, $\widehat{\text{ATT}}$ and $\widehat{\text{ATT}}_s$ are unbiased and consistent to the true ATT.

The FEct estimator resolves the implicit weighting problem of TWFE in two ways. First, because treated observations are excluded from the model-fitting stage, early migrants never serve as implicit controls for late migrants, which is the fundamental cause of negative weights in the staggered adoption setting. Second, the estimator explicitly imposes uniform (or designated sampling) weights on all treated observations when computing the ATT. Furthermore, unlike TWFE regression estimators that estimate the ATT and period-specific ATTs based on contrasts with a base pre-treatment period, the FEct estimator estimates the ATT based on direct comparisons between the observed and counterfactual outcomes at each period. The estimates are, therefore, more directly connected to the theoretical estimand.

3.3.2 Main Specification

As Figure 3.2 shows, identifying the causal effect of migration on income requires adjusting for the time-constant baseline characteristics and the time-varying life course conditions. The former is adjusted by the individual fixed effects in the TWFEct estimator, while adjustment for the latter is more complicated. Controlling for the pre-treatment life course condition ($t = 0$) mechanically controls for the post-treatment conditions ($t = 1$), biasing the estimate

in two ways. First, because post-treatment life course conditions are mediators of migration’s effect on income, controlling for them leads to over-control bias, in which parts of the total effect are absorbed. Second, because post-treatment life course conditions are also affected by post-treatment unobserved confounders, controlling for them induces collider bias (Elwert and Winship, 2014). While we cannot quantify the combined magnitude of the over-control and collider bias from adjustment relative to the unadjusted confounding bias, it could substantially bias the estimate.

Given these considerations, I do not adjust for time-varying life course conditions in the main baseline estimate, which thus includes only the individual and year fixed effects. I estimate a separate model that controls for a set of life course variables: whether any parents are over age 70, whether the mother/father died, the number of children aged 0-20, and the number of unmarried children, as a robustness check. I evaluate the credibility of the strict exogenous assumption by inspecting the pre-trend of with dynamic treatment effects (i.e., estimated “effects” in pre-treatment periods, which should be around zero with no clear trend given parallel trends) and conducting formal placebo tests.

Panel weights are used in model fitting (Equation 3.1) and estimate aggregation (Equation 3.3) to ensure representativeness to the 2010 baseline population. The standard errors of all estimates are simulated using block bootstrap clustered at the individual level with 500 iterations.

3.3.3 Extensions

The baseline model estimates the ATT of rural-urban migration on income. I extend it in two ways to further examine the mechanism and heterogeneity of the effect.

Mediation through Employment Sector Change

As explained in Section 3.1.2, the two-sector model suggests that as workers enter the modern industrial sector from the traditional pre-industrial sector through rural-urban migration,

their income increases as their marginal output increases. This suggests that, as Figure 3.1 shows, parts of the income effect of migration are mediated by the change in employment sectors, from the agricultural to the non-agricultural. As it entails a structural change in the returns to workers' labour, it could even be seen as the most important mechanism for the income effect.

To quantify the importance of the sectoral change mechanism, we need to obtain the average causal mediation effect on the treated units (ACMET) of non-agricultural sector employment on income:

$$\text{ACMET} = \mathbb{E}[Y_{it}(D_{it}, M_{it} = 1) - Y_{it}(D_{it}, M_{it} = 0)], \quad \text{for all } (i, t) \text{ s.t. } D_{i,t} = 1 \quad (3.4)$$

where M_{it} indicates where the worker is employed in the agricultural sector. The quantity captures, among the migrated individuals, how much their income would change if, counterfactually, they did not change their employment sector.

Imai, Keele, and Tingley (2010) show that, under the sequential ignorability (i.e., no unadjusted post-treatment confounders between the mediator and the outcome) and no-interaction (i.e., no mediation effect heterogeneity by treatment status) assumptions, the mediation effect can be consistently identified by differencing the treatment effect estimates from models with and without adjusting for the mediator. The former estimates the total treatment effect, while the latter estimates the direct effect net of mediation. Therefore, I evaluate the mediation effect by comparing the estimated ATTs with and without adjusting for workers' agricultural employment status.

However, Figure 3.1 shows that there are potential unobserved post-treatment confounders – such as attitudes and skills – that are affected by rural-urban migration, and affect both workers' employment sector and income. The presence of these confounders violates the sequential ignorability assumption, and adjusting for the employment sector, in

this case, induced collider bias to the estimated direct effect. Although there exist methods to address the post-treatment confounding and evaluate the sensitivity of estimates to assumption violations, such methods are often not integrated with the FEct estimator (e.g., Imai, Keele, and Yamamoto, 2010; X. Zhou and Wodtke, 2019; Hsia et al., 2025). Relevant results should therefore be interpreted as suggestive evidence, rather than precise estimates, of the mediation effect.

Heterogeneity by Gender and Education

The causal mechanisms in Figure 3.1 may operate differently in different groups, thus leading to heterogeneous causal effects. Gender and education constitute two of the most important dimensions of such heterogeneity.

First, under the strong familial and patriarchal culture in China and the care crisis created by migration, female migrant workers are disproportionately burdened to balance their productive and reproductive roles. Existing research has shown that female migrant workers face more hegemonic industrial relationships, more precarious working conditions and lower wages due to these conflicting responsibilities (Lee, 1995; Y. Dong, 2023). Furthermore, the action of migration and working itself may be perceived as defying traditional gender expectations and draws pressure from older generations, which eventually pushes female migrants to return to villages (Chuang, 2016). Taken together, the gender norms and care responsibilities could cause the income prospects of female migrant workers to be significantly worse.

In terms of education, migrant workers with higher levels of education are better prepared with the skills and cultural capital to integrate into the urban society and economy, thus having better income prospects after migration. Such advantages are further consolidated by the urban *hukou* institution, which, as described in Section 3.1.2, systematically favours migrants with better skills and education (Friedman, 2022; Lei, 2023).

To examine the heterogeneous effects by gender and education, I estimate the conditional

ATTs using stratified samples by gender and whether the workers have at least high school education.

3.4 Results

3.4.1 Descriptive Results

Table 3.1 reports the descriptive statistics of the individuals' characteristics, separately for those who ever migrated (migrants) and never migrated (stayers) between 2010 and 2022. In terms of demographics, migrants are generally younger and have higher levels of education than stayers. The proportion of migrant workers with above-high-school education is almost twice as high as that of the stayers. The non-Han ethnic minority is also under-represented among the migrants, possibly reflecting greater migration barriers for them. Finally, migrants are also less likely to be married than stayers

In terms of economic conditions, migrants are significantly less likely than stayers to be initially employed in the agricultural sector. In fact, the majority of migrants initially work in non-agricultural sectors, indicating positive selection into migration by employment conditions. The gap widens in the last observations. While the proportion of agricultural occupations decreases by only about 8% among stayers compared to the initial observations, it decreases by 14% among migrants. In other words, more migrants shift from agricultural to non-agricultural occupations. The change is also reflected in the change in income levels. Between the first and last observations, on average, the logged real annual income among migrants increased by 0.58, corresponding to a 78% increase in the original scale; meanwhile, the average increase among stayers is only 0.21, corresponding to a 23% increase.

The descriptive statistics show that migrants experience more substantial improvements in their economic conditions over time. To what extent do such improvements reflect the causal effect of rural-urban migration? I now turn to the FEct estimates.

	Migrants (N = 1,471)	Stayers (N = 3,613)	Total (N = 5,084)
Sex			
Female	721 (49.1%)	1700 (47.1%)	2421 (47.7%)
Male	747 (50.9%)	1912 (52.9%)	2659 (52.3%)
N-Miss	3	1	4
Birth Cohort			
1950-1960	113 (7.7%)	445 (12.3%)	558 (11.0%)
1961-1970	445 (30.2%)	1349 (37.4%)	1794 (35.3%)
1971-1980	537 (36.5%)	1187 (32.8%)	1724 (33.9%)
1981-1990	342 (23.2%)	560 (15.5%)	902 (17.8%)
1991-2000	35 (2.4%)	71 (2.0%)	106 (2.1%)
Ethnicity			
Ethnic Minority	127 (8.6%)	531 (14.7%)	658 (12.9%)
Han	1345 (91.4%)	3082 (85.3%)	4426 (87.1%)
Education			
Above high school	230 (16.0%)	303 (8.5%)	533 (10.6%)
High school	530 (36.8%)	1176 (32.8%)	1705 (34.0%)
Middle school	444 (30.8%)	1139 (31.8%)	1583 (31.5%)
Primary or below	237 (16.4%)	964 (26.9%)	1201 (23.9%)
N-Miss	31	30	61
Initial Marital Status			
Divorced/Widowed	15 (1.0%)	68 (1.9%)	83 (1.6%)
Married	1281 (87.0%)	3309 (91.6%)	4590 (90.3%)
Single/Cohab	176 (11.9%)	236 (6.5%)	411 (8.1%)
Age at First Migration			
Mean (SD)	42.478 (9.536)	NA	42.478 (9.536)
Range	18.000–67.000	NA	18.000–67.000
N-Miss	0	3613	3613
Agricultural Occup.– Initial			
No	788 (53.8%)	1144 (31.7%)	1932 (38.1%)
Yes	676 (46.2%)	2467 (68.3%)	3143 (61.9%)
N-Miss	8	2	10
Agricultural Occup.– Last			
No	987 (67.5%)	1441 (39.9%)	2428 (47.9%)
Yes	476 (32.5%)	2170 (60.1%)	2646 (52.1%)
N-Miss	8	2	10
Δ Logged Real Annual Income¹			
Mean (SD)	0.579 (2.560)	0.211 (2.427)	0.317 (2.472)
Range	-10.116–11.917	-11.411–11.349	-11.411–11.917
No. of Observations			
Mean (SD)	6.089 (1.134)	6.301 (1.016)	6.240 (1.056)
Range	2.000–7.000	2.000–7.000	2.000–7.000

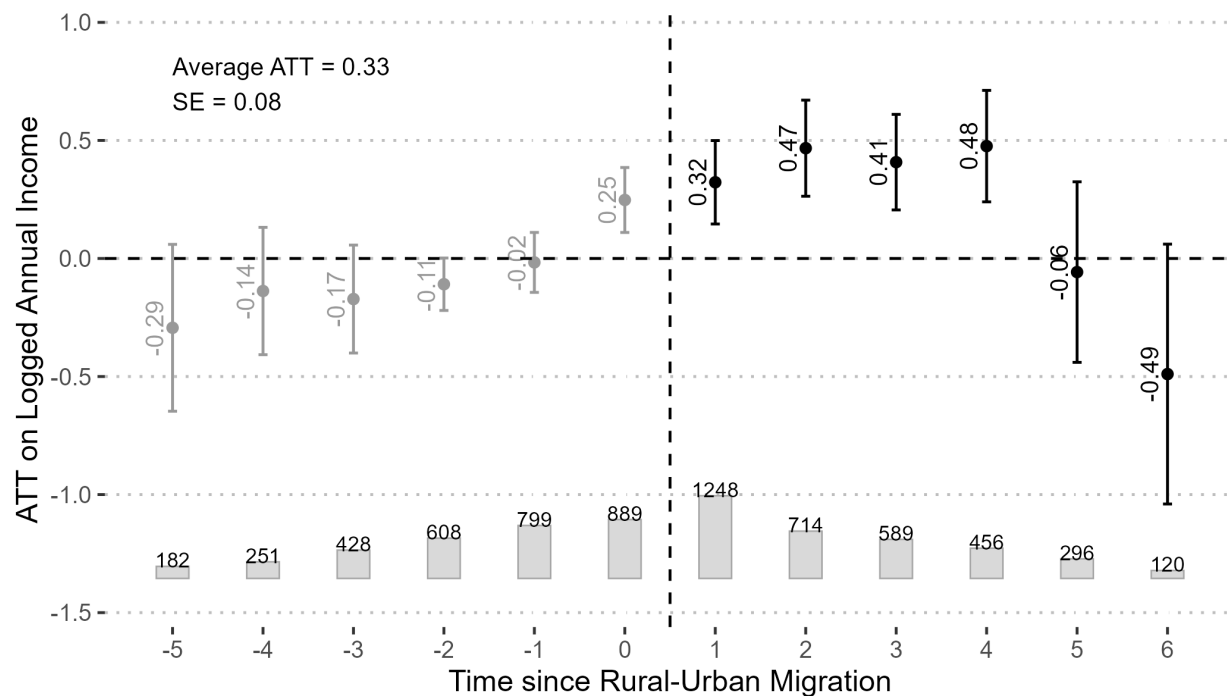
Notes: ¹The difference between the first and last observations.

Results are weighted by panel weights to adjust for sampling and attrition, relative to the 2010 population.

Table 3.1: Baseline Characteristics of Migrants and Stayers

3.4.2 Income Effect of Rural-Urban Migration

Figure 3.4 shows the dynamic treatment effect of rural-urban migration on logged annual income estimated from the baseline model that includes only individual and year fixed effects. First, in the pre-treatment periods, the estimates for periods -5 to -1 are small and statistically insignificant, suggesting that there is no systematic difference in the income level between future migrants and other workers. However, at one period before migration ($t = 0$), migrant workers have significantly higher income than the predicted counterfactual, indicating selection that may bias the causal effect estimates.



Notes: $t = 1$ indicates the first treated period. Error bars show the 95% confidence interval with S.E. simulated from bootstraps with 500 iterations. The bars and numbers at the bottom show the number of units used to estimate the period-specific effect. Panel weights are used in model estimation and result aggregation.

Figure 3.4: Dynamic ATT of Rural-Urban Migration on Logged Annual Income, Baseline Model

After migration, the estimated income effects are immediately positive and remain stable until the fourth period, or six years after migration, because the survey is biannual. In the first period after migration, the ATT on logged income is 0.32, corresponding to a substantial

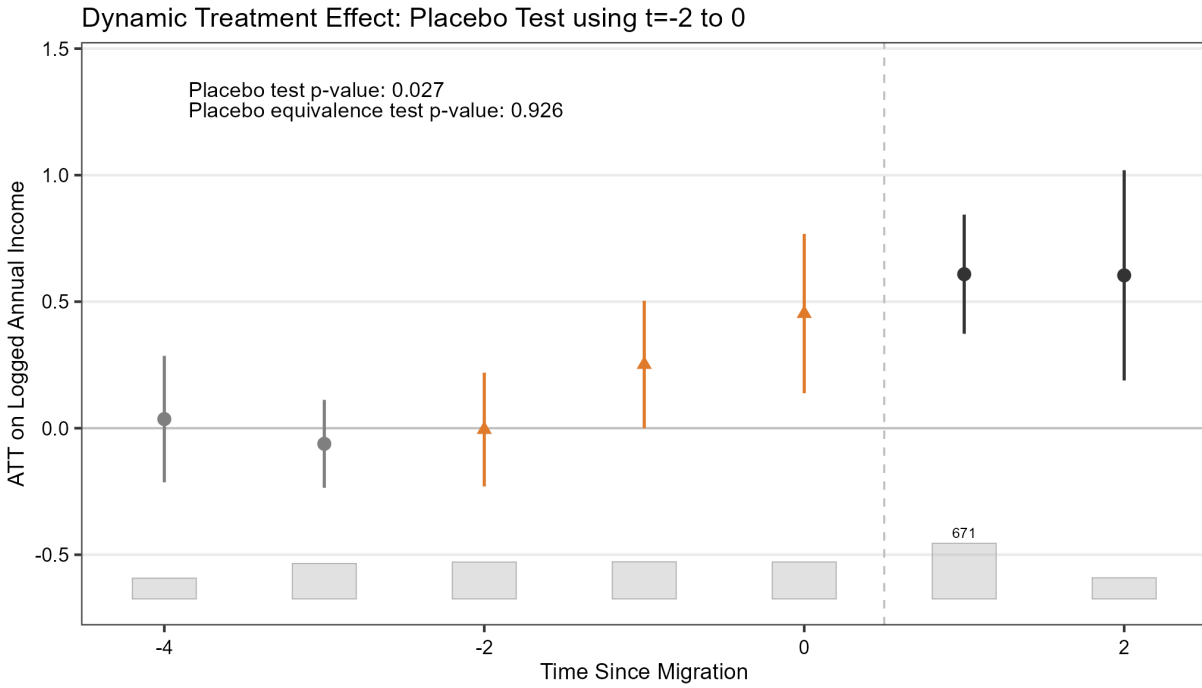
37% increase in income. The effect increases to between 0.41 and 0.48 in 2 to 6 years after migration, corresponding to about 50% to 60% increases in income. However, the effect abruptly collapses to near zero eight years after migration (5th period), and even approaches significantly negative after ten years. Taking all units in post-treatment periods together, the FEct estimates indicate that the ATT of rural-urban migration on logged income is statistically significant at 0.33, corresponding to a 39% increase.

Overall, results from the baseline FEct estimator show that the rural-urban migration has an immediate, substantial positive effect on workers' income, which persists for at least 6 years. Across ten years after migration, migration leads to, on average, a 39% increase in annual income. The credibility of the results, however, is threatened by the positive selection at one period before the treatment, which suggests that the post-treatment income difference could be driven by the positive selection of migrants and hence violates the strict exogeneity assumption for causal identification.

To further test the assumption violation, I conduct the placebo test suggested by L. Liu, Y. Wang, and Xu (2024) by excluding observations from periods -2 to 0 in the counterfactual model fitting stage and test whether their "effects" are significant. Figure 3.5 shows that the estimates at period -1 and 0 are significantly positive, and the placebo test p-value is below 0.05. The results confirm that the strict exogeneity assumption is likely violated and the ATT could be positively biased.

Alternative Specifications

The strict exogeneity violation in the baseline model could be due to unadjusted time-varying confounders. However, estimates from the model that adjusts for whether any parents are over age 70, whether the mother/father died, the number of children aged 0-20, and the number of unmarried children yield virtually the same estimates (see Section B.2). While the model could still omit time-varying confounders, it at least suggests that the baseline estimates are not sensitive to the inclusion of additional variables.



Notes: $t = 1$ indicates the first treated period. The placebo counterfactual model is fitted using untreated observations before $t - 2$. Error bars show the 95% confidence interval with S.E. simulated from bootstraps with 500 iterations. The bars and numbers at the bottom show the number of units used to estimate the period-specific effect. Panel weights are used in model estimation and result aggregation.

Figure 3.5: Placebo Effect Test for the Baseline Model, using $t = -2$ to 0

Another reason for the pre-trend violation, especially near the onset of the treatment, is that people may actively coordinate their migration and income conditions. For example, worker may plan their migration to times when their economic conditions improve, or deliberately draw more income in advance, to prepare for the up-front cost of migration (e.g., travelling and rent). In fact, research has shown that the liquidity constraint to the up-front cost of migration is a significant reason limiting rural workers from migrating, and improvements in banking credit access enable rural-urban migrations, especially for workers from poorer villages (S. Cai, 2020).

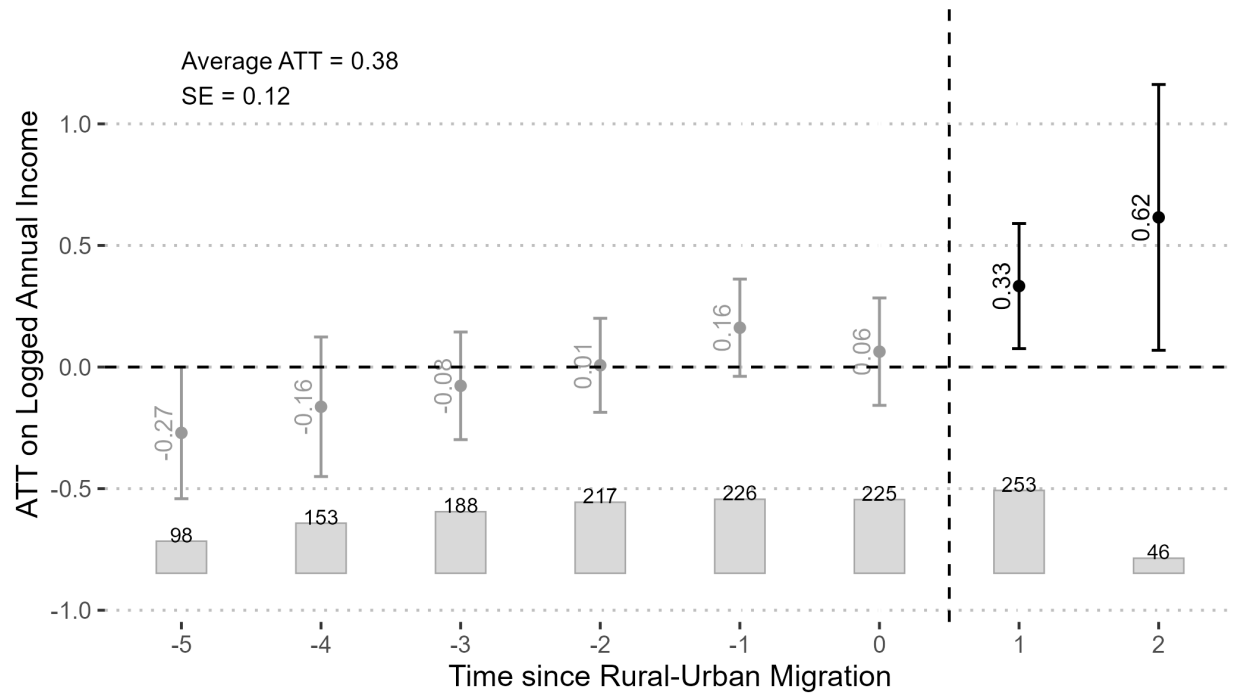
Given such planning behaviour, the pre-trend violation could reflect selection based on the existing income trend itself, instead of omitted variables. To account for this possibility, I estimate an additional model with individual and period fixed effects and unit-specific linear time trends:

$$Y_{it}(D_{it} = 0) = \alpha_i + \xi_t + \kappa'_i Q_t + \varepsilon_{it}, \quad \text{for all } (i, t) \text{ s.t. } D_{i,t} = 0$$

where $\kappa'_i Q_t$ captures unit-specific loadings on known time trends. For simplicity, I specify the time basis Q_t to be linear, and the individual-specific loadings κ_i are estimated using the expectation-maximisation (EM) algorithm. However, due to the increased computational complexity, the model only uses 3,801 individuals with at least 5 untreated observations.

Figure 3.6 shows that, after introducing the individual-specific time trend, the significant effects before the migration onset disappear. However, due to the limited number of pre-treatment periods, I cannot apply the placebo test to test the pre-trend violation. The ATT in the year of migration is 0.33, and it almost doubles to 0.62 two years later. The latter estimate, however, is driven by only 46 observations and should be interpreted cautiously. The overall ATT is 0.38, corresponding to a 46% increase in annual income.

Overall, by removing the significant pre-treatment effects, the individual-specific trend model provides more credible estimates of the ATT of rural-urban migration on annual income. The estimates are broadly consistent with the baseline estimates, confirming the



Notes: $t = 1$ indicates the first treated period. The model is estimated using 3,801 individuals with at least 5 untreated observations. Error bars show the 95% confidence interval with S.E. simulated from bootstraps with 500 iterations. The bars and numbers at the bottom show the number of units used to estimate the period-specific effect. Panel weights are used in model estimation and result aggregation.

Figure 3.6: Dynamic ATT from Model with Individual-Specific Time Trends

substantial positive income return to rural-urban migration. However, due to the restricted sample, the model cannot inform dynamic effects over longer time frames, and the estimates could be less representative of the target population.

Alternative Samples

Furthermore, I test the baseline model’s robustness using alternative samples. As described in Section 3.2, the panel data contain treatment reversals due to return migration. It also has clusters of observations with low to zero annual income. Finally, many individuals’ recorded education levels change across waves, even though the sample is restricted to individuals who have finished schooling.

To account for each of these issues, I estimate the baseline models using three alternative samples that 1. exclude individuals with changes in recorded education; 2. exclude individuals with reversals in rural-urban status; and 3. trim observations based on year and rural-urban-specific 1st and 99th annual income percentiles.

	$N_{individual}$	ATT	S.E.	t
Baseline TWFEct	5,084	0.33	0.08	4.1
<i>Alternative Specifications</i>				
Baseline – Unweighted	5,084	0.21	0.06	3.5
Lagged Treatment	5,036	0.19	0.09	2.1
+ Family Controls	4,615	0.33	0.09	3.7
+ Unit-specific Trends	3,801	0.38	0.12	3.2
<i>Alternative Samples</i>				
No Education Change	4,009	0.30	0.10	3.0
No Return Migration	4,808	0.34	0.09	3.8
Trimmed Income	5,073	0.29	0.05	5.8
Range	0.19–0.38			

Notes: ATT denotes the estimated average treatment effect on the treated. Standard errors are simulated from block bootstraps with 500 iterations.

Table 3.2: Estimated ATT of Rural-Urban Migration on Logged Annual Income using Different Model Specifications and Samples

Table 3.2 reports the estimated ATT using these alternative samples, alongside estimates from alternative model specifications using the main sample. Qualitatively, the estimates

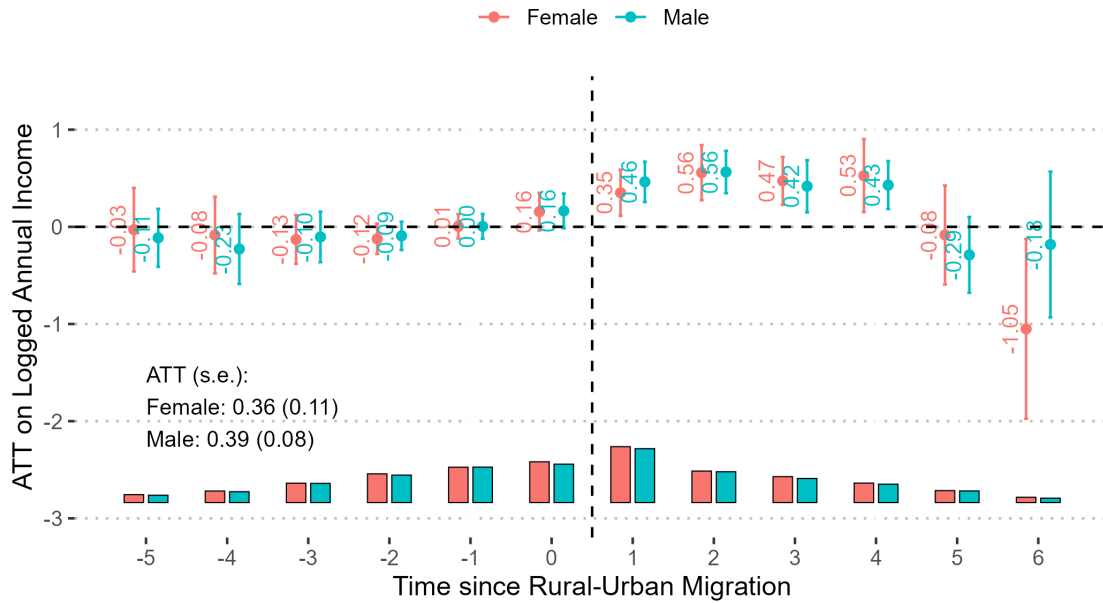
consistently suggest a significantly positive effect of rural-urban migration on annual income. Quantitatively, except for the unweighted estimate from the baseline model and the model with lagged treatment, which lead to different estimands, the ATT estimates are consistently around 0.33. The estimate from the individual-specific trend model is higher at 0.38, but it is driven by a small number of observations, as shown in Figure 3.6, while the ATT from $t = 1$ is consistent with 0.33.

Overall, the results suggest that, while we cannot rule out potential strict exogeneity violations, the ATT estimate from the baseline model is quantitatively and qualitatively robust to different model and sample specifications, thus lending confidence to the estimate. For rural-origin Chinese migrant workers in 2010, rural-urban migration, on average, leads to a 40% increase in their annual income.

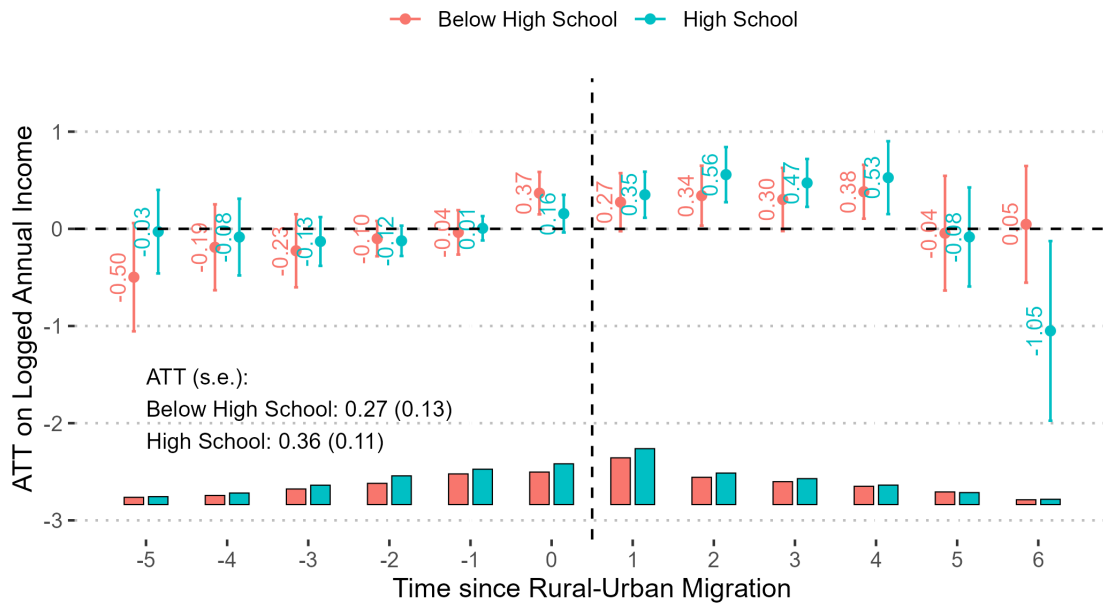
3.4.3 Heterogeneity Analysis

How does the income effect of rural-urban migration vary by gender and education? Figure 3.7 shows the baseline model estimates obtained using stratified samples by gender and high school education. The top panel shows no significant difference in the dynamic treatment effect trajectories or overall ATTs between male and female migrants. Although at $t = 6$ or 10 years after migration, the income effect is estimated to be significantly negative for female migrants, the effect is driven by only a few observations, thus may not be generalisable. Overall, despite various challenges faced by female workers, the result suggests no significant gender difference in the income returns to rural-urban migration.

The heterogeneity by education, shown in the bottom panel, shows somewhat more differentiated patterns. In each post-treatment period, the estimated ATT for workers with at least high school education is higher than that of those without, though the differences are not statistically significant. The point estimate of average ATT for high-school-educated workers is 0.09 higher than that of workers without high school education, corresponding to a 9-percentage-point higher proportional income effect. However, the difference is again



(a) By Gender



(b) By Education

Notes: $t = 1$ indicates the first treated period. Error bars show the 95% confidence interval with S.E. simulated from bootstraps with 500 iterations. The bars and numbers at the bottom show the number of units used to estimate the period-specific effect. Panel weights are used in model estimation and result aggregation.

Figure 3.7: Dynamic ATT of Rural-Urban Migration on Logged Annual Income by Gender and Education

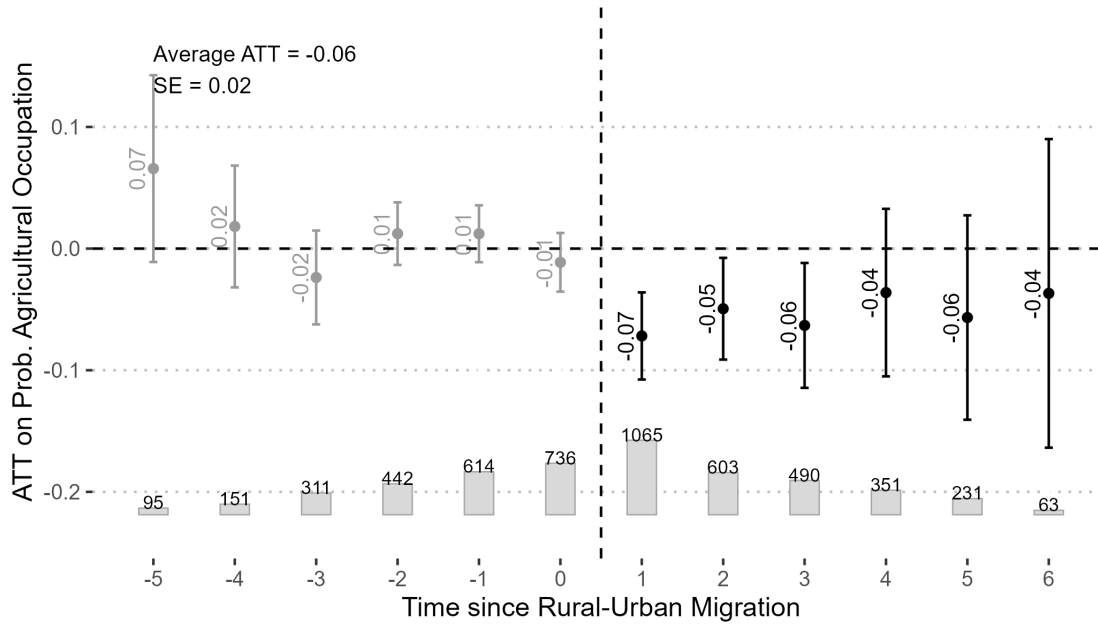
insignificant. Overall, the result suggests that, while there is some evidence showing that the income return to migration is higher for more educated workers, it remains too uncertain for definitive conclusions.

Two additional patterns in the bottom panel are worth noting. First, the positive effect at $t = 0$ is stronger and significant among workers without high school education, while the effect for high-school educated workers is statistically insignificant, suggesting that the planning behaviour is stronger among the former group. It could reflect the fact that workers with lower levels of education and skills are subject to greater uncertainty and thus higher up-front costs from migration, while workers with higher levels of education could more easily find jobs and integrate into the urban economy. Second, the effect at $t = 6$ is significantly negative for workers with high school education but not for those without. As with the negative effect for female workers in the top panel, the estimate is driven by a few observations with potential selection issues.

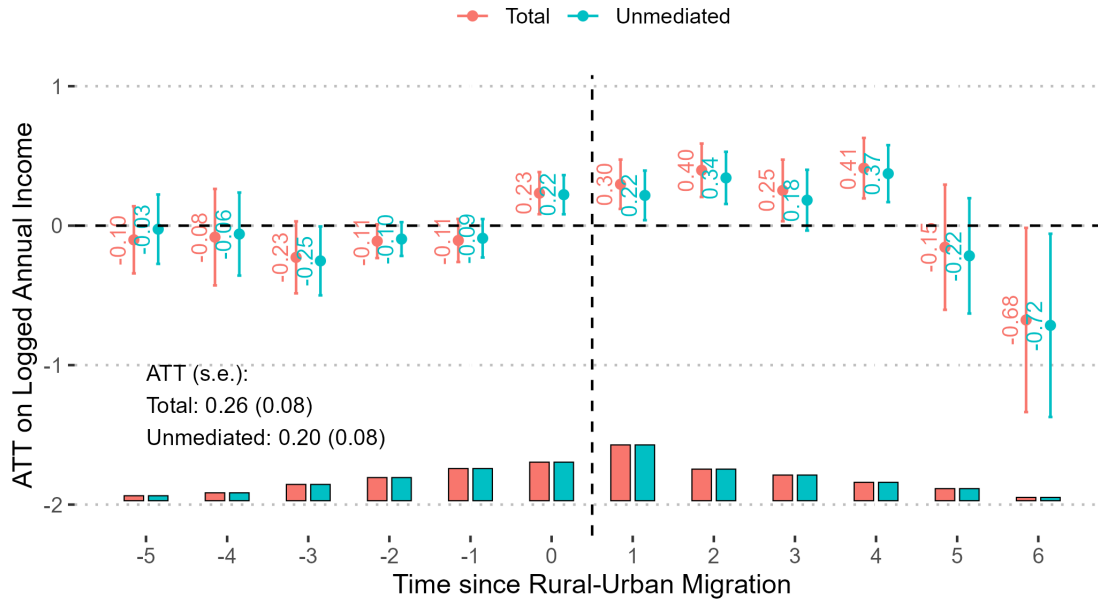
3.4.4 Mediation through Sectoral Change

The two-sector model stipulates that workers' income increases as they move from the labour-excessive traditional sector to the modern industrial sector. It therefore implies that the change from the agricultural to the non-agricultural sector is an important mechanism driving the income effect of rural-urban migration. To test the importance of this mechanism, I test the mediation effect by comparing the total ATT of rural-urban migration on workers' probability of working in the agricultural sector, and the unmediated ATT on annual income, adjusted for changes in employment sectors.

Figure 3.8 shows the ATT estimates using the baseline TWFE specification and a restricted sample excluding missing values in occupation. The top panel 3.8a shows an immediate 7 percentage-point decrease in the probability of working in the agricultural sector in the year of migration. The effect persists over time with stable magnitudes, though the estimates become more uncertain in later periods due to smaller sample sizes. The persis-



(a) ATT on Agricultural Occupation



(b) Total and Unmediated ATT

Notes: $t = 1$ indicates the first treated period. The model is estimated using 26,668 observations with occupation information from 4,962 individuals with at least one untreated observation. Error bars show the 95% confidence interval with S.E. simulated from bootstraps with 500 iterations. The bars and numbers at the bottom show the number of units used to estimate the period-specific effect. Panel weights are used in model estimation and result aggregation.

Figure 3.8: Dynamic ATT of Rural-Urban Migration on the Probability of Agricultural Main Occupation, and Total and Unmediated ATT on Logged Annual Income

tent negative effect is consistent with the expectation that workers enter the non-agricultural sector through rural-urban migration.

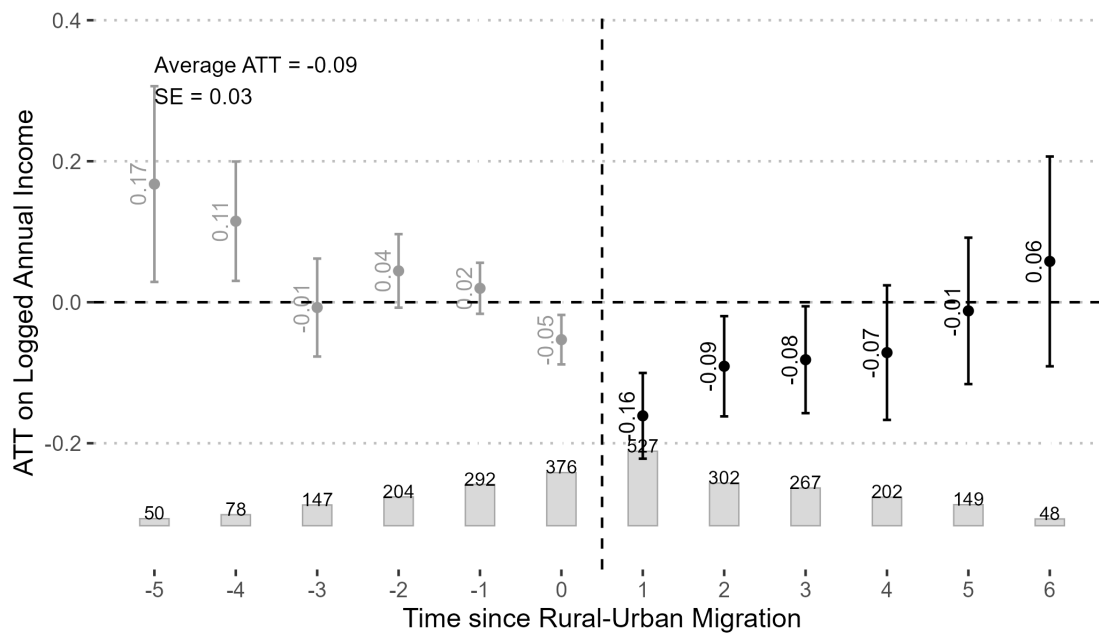
How does the shift away from the agricultural sector contribute to the income effect? The bottom panel 3.8b shows the ATT on income with and without controlling for agricultural occupation. Adjusting for mediation through changes in agricultural occupation status reduces the estimated ATTs in all periods consistently by about 0.06. In terms of the overall ATT, it corresponds to a 20% decrease in the total effect. Nonetheless, the difference between the total and unmediated estimates is not statistically significant. Although, as discussed in Section 3.3.3, the unmediated effect estimates cannot be interpreted as causal effects due to unobserved post-treatment confounders, the estimates provide suggestive evidence that the change from the agricultural to the non-agricultural sector is an important mediating mechanism driving the positive income effect of rural-urban migration, accounting for one-fifth of the total effect.

Effect on Agricultural Workers

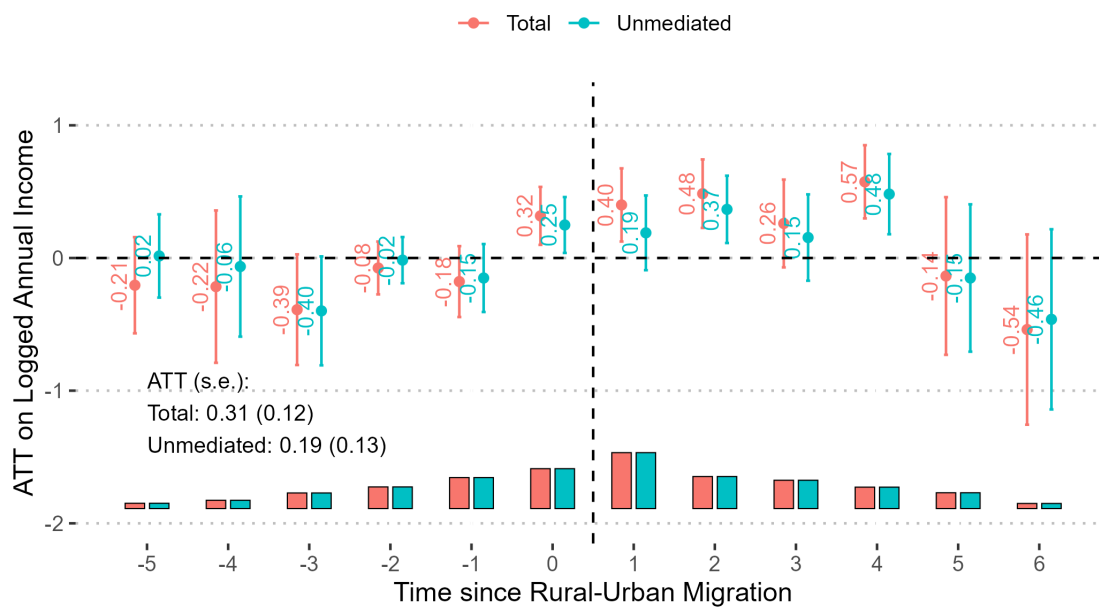
The result from Figure 3.8, however, could understate the importance of sectoral change as a mechanism of income mobility due to the occupational composition of migrants. As Table 3.1 shows, the majority of rural-urban migrants initially have non-agricultural occupations, and they may not be affected by sectoral change. Aggregating these workers with those who actually move from the agricultural to the non-agricultural sector, therefore, dilutes the true mediating effect of sectoral change. To test this possibility, I repeat the mediation analysis, this time using the subsample of workers initially working in the agricultural sector in their first observation.¹

The top panel in Figure 3.9 shows that the immediate negative effect on the probability of having agricultural occupations at the year of migration is substantially stronger than the previous estimate, decreasing the probability by 16 percentage points. However, the effect

¹Note that this sample is different from workers who change occupation at migration, as their occupations may change in subsequent pre-treatment periods.



(a) ATT on Agricultural Occupation



(b) Total and Unmediated ATT

Figure 3.9: Dynamic ATT of Rural-Urban Migration on the Probability of Agricultural Main Occupation, and Total and Unmediated ATT on Logged Annual Income, for workers whose initial occupations are agricultural

somewhat dissipates over time and approaches zero in periods 5 and 6, possibly reflecting downward mobility to the agricultural sector (X. Wu and Treiman, 2007).

The bottom panel shows that, among workers who initially worked in the agricultural sector, the difference between estimates with and without adjusting for mediation is larger than in Figure 3.8, being about 0.1 or one-third of the total effect. In fact, after adjusting for agricultural occupation, the average ATT becomes statistically insignificant, although it could reflect a type-II error due to reduced statistical power from a smaller sample. In terms of the point estimates, the total ATT estimate is 0.05 units larger, while the unmediated effect through other mechanisms is similar at 0.2, suggesting that the stronger mediation effect through sectoral change drives the stronger total effect compared to the complete sample. The result, therefore, provides stronger evidence for the mediating effect of sectoral change in the positive income effect of rural-urban migration, as predicted by the two-sector model.

3.5 Discussion

Overall, the results above have demonstrated a robust positive effect of rural-urban migration on annual income, and the importance of the transition from the agricultural to the non-agricultural sectors in mediating this income effect. Nonetheless, the results need to be interpreted in light of several methodological and substantive limitations.

Causal Identification

First, as explained in Section 3.3, identifying the causal income effect of migration using the TWFEct estimator requires the strict exogeneity (i.e., parallel trends) assumption. However, dynamic effect estimates using various model specifications and samples have persistently shown a significant effect at the period before migration ($t = 0$), which suggests positive planning and selection into migration, and violation of the parallel trend in the pre-treatment

periods. The estimates, therefore, could be biased and should not be directly interpreted as the precise causal effect. While advanced modelling methods, such as the unit-specific trend model, can offer statistical solutions that remove the positive effects, the estimate is based on a restricted sample and model-dependent. Therefore, it does not fundamentally resolve the endogenous selection problem at the substantive design level.

However, instead of treating the positive selection only as a nuisance to causal identification, we could interpret it as a substantially meaningful pattern that informs our understanding of the consequences of migration as a life course event. As life course research scholars have argued, life course trajectories and outcomes are shaped by individuals’ agency, including the deliberations and planning of life course events (see e.g., Elder, Johnson, and Crosnoe, 2003). This creates inherent endogeneity in the onset and timing of life course events, such as migration (Williams, 2025). In other words, there is no hypothetical “baseline randomisation” which we can leverage to identify the effect of migration, as individuals constantly plan and adjust their life course decisions. Therefore, a more productive perspective is to view the trend divergence not just as a failure of causal identification but as information about individuals’ income trajectories in the causal process, including the planning and consequences of migration. Future research should disentangle the inherent endogenous causal process using both more sophisticated statistical analysis and qualitative evidence that traces individuals’ decision-making processes and consequences (Williams, 2025).

Measurement Errors

Even if the causal assumptions are valid, the results could be subject to deeper sources of bias from measurement errors, both in the outcome and the treatment. As described in Section 3.2, the measurement of annual income in the CFPS survey has changed multiple times across waves. While responses from within each wave are subject to potential systematic measurement errors, the inconsistent measurements across waves add more uncertainty. Although the period fixed effect could account for period-specific common error,

individual-specific errors could still bias the results.

What is more problematic, however, is the measurement of rural-urban migration. As also described in Section 3.2, I infer individuals' migration status based on changes in the administrative type of their local community of residence. However, the change from rural to urban communities could occur through both rural-urban migration or urban expansion. With the rapid urban development in China, many rural communities in the urban fringes have been – sometimes coercively – transformed into urban areas (Hsing, 2010). The consequences of such negative urbanisation are different from those of migration. On the one hand, people may receive resettlement compensation if their properties and lands are expropriated, which, in the short run, increases their income; on the other hand, they do not experience changes in socioeconomic conditions through migration and may experience long-term disadvantages due to land loss. Confusing these two processes would, therefore, bias the estimated effect of rural-urban migration. Future research should construct more precise measures of rural-urban migration using more detailed data, such as community-level identifiers available in the restricted-access version of CFPS, to differentiate active migration and passive urbanisation.

Missing Data

Finally, because the analysis uses list-wise deletion to handle missing values in relevant variables, the results may be biased if the missing data is systematically related to the treatment and outcomes. For example, when using the restricted sample of observations with valid agricultural occupation information in the mediation analysis, the ATT estimated using the baseline model in Figure 3.8 is lower than the estimate using the main analytical sample in Figure 3.4 by 0.07. This discrepancy could be due to the non-random missing of occupation data, especially in the 2010 and 2012 waves, as discussed in Section 3.2. Further research should consider handling the missing values using multiple imputations.

Substantive Limitations

Besides methodological limitations, the result based on annual income may not fully reflect the economic effect of rural-urban migration. First, the annual income measure does not consider the number of working hours, and since migrant workers tend to work for longer hours, the results may overstate the income return to migration. Future research should supplement the analysis using measures of the number of working hours or hourly income. Second, the analysis does not account for the price difference between rural and urban areas. Since the urban areas have substantially higher price levels and costs of living, the increase in nominal income may be offset by the higher price levels, leading to overstatements of the real income return to migration (see e.g., Moretti, 2013). Future analysis should adjust for the regional differences in price levels.

3.6 Conclusion

Contemporary China has been a nation on the move. As the country moves on from its socialist past and becomes a state-capitalist industrial powerhouse, millions of people leave their rural hometowns in search of opportunities in the cities. This study provides the first analysis of the effect of rural-urban migration on migrants' economic outcomes.

Using longitudinal survey data from CFPS and two-way fixed effect counterfactual estimators, I estimate a robust positive effect of rural-urban migration on workers' annual income of about 40%, persisting for over 6 years. The effect magnitudes are similar for male and female workers, though there is some evidence showing that the effect is stronger for workers with at least high school education. Furthermore, I provide suggestive evidence that about one-fifth of the total income effect is mediated by the change from the agricultural to the non-agricultural sector. The mediation effect is strong when focusing on workers initially working in the agricultural sector, accounting for one-third of the total effect. Overall, the results suggest that, despite structural disadvantages and institutional exclusion against

migrant workers, rural-urban migration has brought substantial economic benefits to rural-origin workers and thus functioned as a channel of upward economic mobility.

The study contributes several novel insights to the study of socioeconomic change and inequality in contemporary China. First, previous research on Chinese migrant workers has predominantly focused on the structural disadvantages and exclusion that they face relative to urban residents, and the negative effects on their and their family members' life outcomes (X. Wu and Treiman, 2007; X. Wu and Zheng, 2018; X. Wu, 2019; Friedman, 2022). While such research has yielded valuable insights into an important institutional mechanism of inequality, it overlooks the other side of migration as workers' active decisions to pursue economic opportunities and upward mobility. By demonstrating the positive income effect of rural-urban migration as an economic mobility channel, the study invites future research to examine the empowering effect of migration on rural workers and the beliefs and strategies migrants develop in response to the structural opportunities and challenges it presents.

Second, existing research on social change and stratification in post-reform China has mainly focused on structural and institutional changes, especially in the urban industrial sector, that have changed the determinants of life chances (see Bian, 2002; X. Wu, 2019; Walder, 2023, for reviews). By demonstrating the substantial income return to migration for rural-origin workers, this study highlights the importance of rural-urban migration as a determinant of rural workers' life chances, and at the population level, as a demographic process that shapes economic inequality and mobility. Future research should investigate the effects of migration on other outcomes, such as occupational class and intergenerational mobility; the underlying mechanisms; and the macro-level consequences on socioeconomic stratification and mobility.

Finally, beyond rural-urban migration in China, the study echoes the recent literature on the geographic determinants of life chances and inequality. With the changing economic geography and the increasing importance of firms and workplaces, which are spatially embedded, in shaping income distribution in many countries, future research should examine

the role of place and spatial mobility – or the lack thereof – in structuring economic opportunities and inequality (Hanson and Moretti, 2025; J. Song et al., 2019; Tomaskovic-Devey et al., 2020; Card, Rothstein, and M. Yi, 2025). The case of Chinese rural-urban migration, alongside historical research on migration and industrialisation (Long, 2005; MacDonald, 2021; Andersson and Molinder, 2025), contributes to and calls for further historical and comparative evidence to the literature.

Chapter 4

Conclusion

Nowadays, the high-speed train runs between Zhengzhou and Qingdao in less than five hours. The then-young couple, now in their fifties, would travel on these trains from time to time to go back to their home village. They often say that there lie their roots, though the old *yao dong*¹ has been demolished after the flood and landslide in 2021.

Outside the windows on the train, the land remains as flat as Keightley observed in 1981, though nowadays, there are blocks of high-rise residential estates, one much like another, as monuments of the new era of Chinese history. Somewhere else on a different railway, the normal-speed green-skinned train repeats the ten-hour journey from Zhengzhou to Qingdao, with tickets costing one-third as much as the high-speed train, taking many people to the coastal city of better opportunities.

Over forty years after the beginning of the economic reform, China has transformed from a country of widespread starvation and poverty to a global industrial and technological superpower. The rural-urban migrants, from China's immense rural heartland, fuelled this transformation, and many transformed their own lives. Through two studies, this thesis documents how the rural population and migrants have shaped economic inequality and mobility in contemporary China.

¹Sometimes translated as cave dwellings, *yao dong* is a common type of residence in parts of northern China with the loess landscape.

The first study examines the predictors of income variation and the processes shaping income inequality in post-reform China, including the rural agricultural sector that much existing research has excluded. Using nationally representative survey data spanning 1996 to 2023, I show that the rural-urban division and agricultural-industrial divides constitute the most salient dimensions of income variation in contemporary China. Using variance decomposition and counterfactual analysis, I find the change in the between-sector income gap and compositional change in employment sectors – driven by the movement of workers out of agriculture and into urban non-agricultural employment – exerted an equalising effect on income distribution between 1996 and 2010, and more moderately between 2010 and 2015. These effects do not appear in the previous analysis restricted to the urban sample, as rural workers affected by the processes were excluded. The finding highlights the importance of the rural population and structural processes, such as rural-urban migration and industrialisation, in understanding social changes and inequality in contemporary China.

The second study shifts to the individual level, examining the economic consequences of rural-urban migration for migrant workers. Using longitudinal data from the Chinese Family Panel Studies and the fixed effects counterfactual estimators, I estimate a positive income return to rural-urban migration of approximately 40%, persisting for over six years following migration. A further mediation analysis finds suggestive evidence that the transition from agricultural to non-agricultural employment accounts for around one-fifth of the total income effect on average, and a larger share among workers who initially work in agriculture. Together, the results show that, despite structural and institutional disadvantages, rural-urban migration provides an effective channel of upward mobility for rural workers.

These findings invite further research to examine the role of the rural population and related processes in shaping inequality and mobility in contemporary China. For example, future research could examine, from a cohort perspective, how migration as a life course event creates divergent life trajectories, and how changes in migration patterns and return to migration create between-cohort differences in economic trajectories and inequality (Cheng,

2014). It could also examine how these processes narrow or widen disparities between, for example, gender, rural and urban populations and ethnic groups (Lundberg, 2024; Opacic, Wei, and X. Zhou, 2025). Finally, at the micro-level, more research is needed to identify the effect and relative importance of different mechanisms, such as skill accumulation and integration barriers, in shaping migrants' outcomes, using more detailed data and sophisticated causal mediation analysis methods.

Looking forward, as existing research has shown, and as the results of this thesis also suggest, China has entered a new stage characterised by stabilised inequality and accelerated technological changes since the 2010s (Piketty, Yang, and Zucman, 2019; Kanbur, Y. Wang, and X. Zhang, 2021; Hanushek, Y. Wang, and L. Zhang, 2025; Lei, 2023). However, despite these changes, rural-urban migration seems to remain a constant, ongoing process, with the size of the floating population increased by over 88% between 2010 and 2020 (National Bureau of Statistics of China, 2021b). Future research should examine how the new changes and conditions interact with the rural population and migration flows, creating new forms of inequality and mobility. For example, as traditional manufacturing jobs are automated, platform companies become the largest employers of low-skilled migrant workers. This could create new forms of algorithm-mediated labour relationships and labour market stratification (Lei, 2023). As social scientists confront and try to make sense of the new wave of technological and structural changes, the Chinese experience could contribute valuable knowledge, as a unique case where high-modernist technology meets a large, developing rural population and where high-speed rails run parallel with green-skinned trains.

Bibliography

- Aeppli, Clem and Wilmers, Nathan (2022). Rapid Wage Growth at the Bottom Has Offset Rising US Inequality. *Proceedings of the National Academy of Sciences* 119(42), e2204305119.
- Ager, Philipp, Goñi, Marc, and Salvanes, Kjell G. (2026). Gender-Biased Technological Change: Milking Machines and the Exodus of Women from Farming. *American Economic Review* 116(1), pp. 246–286.
- Alisky, Michael, Rozelle, Scott, and Whyte, Martin King (2025). Getting Ahead in Today’s China: From Optimism to Pessimism. *The China Journal* 93, pp. 1–22.
- Allison, Paul D. (1978). Measures of Inequality. *American Sociological Review* 43(6), pp. 865–880.
- Andersson, Jonatan and Molinder, Jakob (2025). Did Cities Increase Skills during Industrialization? Evidence from Rural-Urban Migration. *Journal of Urban Economics* 148, p. 103772.
- Bi, Yufang, Wang, Limin, Xu, Yu, Jiang, Yong, He, Jiang, Zhang, Mei, Zhou, Maigeng, Li, Yichong, Xu, Min, Lu, Jieli, Wang, Tiange, Ding, Lin, Dai, Meng, Zhang, Di, Chen, Chung-Shiuan, Lai, Shenghan, Wang, Weiqing, Wang, Linhong, and Ning, Guang (2016). Diabetes-Related Metabolic Risk Factors in Internal Migrant Workers in China: A National Surveillance Study. *The Lancet Diabetes & Endocrinology* 4(2), pp. 125–135.
- Bian, Yanjie (2002). Chinese Social Stratification and Social Mobility. *Annual Review of Sociology* 28(Volume 28, 2002), pp. 91–116.
- Bian, Yanjie and Logan, John R. (1996). Market Transition and the Persistence of Power: The Changing Stratification System in Urban China. *American Sociological Review* 61(5), pp. 739–758.
- Breen, Richard, Luijkx, Ruud, Müller, Walter, and Pollak, Reinhard (2009). Nonpersistent Inequality in Educational Attainment: Evidence from Eight European Countries. *American Journal of Sociology* 114(5), pp. 1475–1521.
- Cai, Qian (2003). Migrant Remittances and Family Ties: A Case Study in China. *International Journal of Population Geography* 9(6), pp. 471–483.
- Cai, Shu (2020). Migration under Liquidity Constraints: Evidence from Randomized Credit Access in China. *Journal of Development Economics*. Special Issue on Papers from “10th AFD-World Bank Development Conference Held at CERDI, Clermont-Ferrand, on June 30 - July 1, 2017” 142, p. 102247.
- Card, David, Rothstein, Jesse, and Yi, Moises (2025). Location, Location, Location. *American Economic Journal: Applied Economics* 17(1), pp. 297–336.

- Carr, Deborah (2018). “The Linked Lives Principle in Life Course Studies: Classic Approaches and Contemporary Advances”. *Social Networks and the Life Course: Integrating the Development of Human Lives and Social Relational Networks*. Ed. by Duane F. Alwin, Diane H. Felmlee, and Derek A. Kreager. Cham: Springer International Publishing, pp. 41–63.
- Chan, Kam Wing and Buckingham, Will (2008). Is China Abolishing The. *The China Quarterly* 195, pp. 582–606.
- Chan, Kam Wing and Zhang, Li (1999). The Hukou System and Rural-Urban Migration in China: Processes and Changes. *The China Quarterly* 160, pp. 818–855.
- Che, Yi and Zhang, Lei (2018). Human Capital, Technology Adoption and Firm Performance: Impacts of China’s Higher Education Expansion in the Late 1990s. *The Economic Journal* 128(614), pp. 2282–2320.
- Chen, Feinian, Liu, Hui, Vikram, Kriti, and Guo, Yu (2015). For Better or Worse: The Health Implications of Marriage Separation Due to Migration in Rural China. *Demography* 52(4), pp. 1321–1343.
- Chen, Meredith (2026). China Removes Hukou Hurdle for Migrant Workers in Social Insurance Shake-Up. *South China Morning Post*.
- Cheng, Siwei (2014). A Life Course Trajectory Framework for Understanding the Intracohort Pattern of Wage Inequality. *American Journal of Sociology* 120(3), pp. 633–700.
- Choi, Susanne Y. P. and Peng, Yin ni (2016). *Masculine Compromise: Migration, Family, and Gender in China*. Oakland, California: University of California press.
- Chuang, Julia (2016). Factory Girls After the Factory: Female Return Migrations in Rural China. *Gender & Society* 30(3), pp. 467–489.
- de Chaisemartin, Clément and D’Haultfœuille, Xavier (2020). Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects. *American Economic Review* 110(9), pp. 2964–2996.
- DiNardo, John, Fortin, Nicole M., and Lemieux, Thomas (1996). Labor Market Institutions and the Distribution of Wages, 1973-1992: A Semiparametric Approach. *Econometrica* 64(5), pp. 1001–1044.
- Dong, Yige (2023). The Dilemma of Foxconn Moms: Social Reproduction and the Rise of ‘Gig Manufacturing’ in China. *Critical Sociology* 49(7-8), pp. 1231–1249.
- Elder, Glen H., Johnson, Monica Kirkpatrick, and Crosnoe, Robert (2003). “The Emergence and Development of Life Course Theory”. *Handbook of the Life Course*. Ed. by Jeylan T. Mortimer and Michael J. Shanahan. Boston, MA: Springer US, pp. 3–19.
- Elwert, Felix and Winship, Christopher (2014). Endogenous Selection Bias: The Problem of Conditioning on a Collider Variable. *Annual Review of Sociology* 40(Volume 40, 2014), pp. 31–53.
- Erten, Bilge and Leight, Jessica (2021). Exporting Out of Agriculture: The Impact of WTO Accession on Structural Transformation in China. *The Review of Economics and Statistics* 103(2), pp. 364–380.
- Facchini, Giovanni, Liu, Maggie Y., Mayda, Anna Maria, and Zhou, Minghai (2019). China’s “Great Migration”: The Impact of the Reduction in Trade Policy Uncertainty. *Journal of International Economics* 120, pp. 126–144.
- Fan, C. Cindy (2024). *Why Do Rural Chinese Continue to Do Migrant Work: Gender and Social Perspectives*. Social Science Colloquium, The University of Hong Kong.

- Friedman, Eli A. (2022). *The Urbanization of People: The Politics of Development, Labor Markets, and Schooling in the Chinese City*. New York: Columbia University Press.
- Gai, Qingen, Guo, Naijia, Li, Bingjing, Shi, Qinghua, and Zhu, Xiaodong (2025). Rural Pensions, Labor Reallocation, and Aggregate Income: An Empirical and Quantitative Analysis of China. *Econometrica* 93(5), pp. 1663–1696.
- Glaeser, Edward L. and Gottlieb, Joshua D. (2009). The Wealth of Cities: Agglomeration Economies and Spatial Equilibrium in the United States. *Journal of Economic Literature* 47(4), pp. 983–1028.
- Glaeser, Edward L. and Mar, é David C. (2001). Cities and Skills. *Journal of Labor Economics* 19(2), pp. 316–342.
- Gollin, Douglas (2014). The Lewis Model: A 60-Year Retrospective. *Journal of Economic Perspectives* 28(3), pp. 71–88.
- Goodman-Bacon, Andrew (2021). Difference-in-Differences with Variation in Treatment Timing. *Journal of Econometrics*. Themed Issue: Treatment Effect 1 225(2), pp. 254–277.
- Hällsten, Martin (2025). *What We Know and Do Not Know about Social Mobility*.
- Han, Jun, Liu, Runjuan, and Zhang, Junsen (2012). Globalization and Wage Inequality: Evidence from Urban China. *Journal of International Economics* 87(2), pp. 288–297.
- Hannum, Emily, Ishida, Hiroshi, Park, Hyunjoon, and Tam, Tony (2019). Education in East Asian Societies: Postwar Expansion and the Evolution of Inequality. *Annual Review of Sociology* 45(1), pp. 625–647.
- Hannum, Emily, Wang, Meiyan, and Adams, Jennifer (2010). “Rural-Urban Disparities in Access to, Primary and Secondary Education under Market Reforms”. *One Country, Two Societies*. Ed. by Martin King Whyte. 1st ed. Vol. 16. Rural-Urban Inequality in Contemporary China. Harvard University Asia Center, pp. 125–146.
- Hanson, Gordon and Moretti, Enrico (2025). “Where Have All the Good Jobs Gone? Changes in the Geography of Work in the US, 1980-202173”. *Handbook of Regional and Urban Economics*. Ed. by Dave Donaldson and Stephen J. Redding. Vol. 6. Handbook of Regional and Urban Economics. Elsevier, pp. 515–583.
- Hanushek, Eric A., Wang, Yuan, and Zhang, Lei (2025). Understanding Trends in Chinese Skill Premiums, 2007–2018. *Journal of Comparative Economics* 53(2), pp. 584–608.
- Hauser, Seth M. and Xie, Yu (2005). Temporal and Regional Variation in Earnings Inequality: Urban China in Transition between 1988 and 1995. *Social Science Research* 34(1), pp. 44–79.
- He, Congzhi and Ye, Jingzhong (2014). Lonely Sunsets: Impacts of Rural–Urban Migration on the Left-behind Elderly in Rural China. *Population, Space and Place* 20(4), pp. 352–369.
- Hessler, Peter (2007). *Oracle Bones: A Journey through Time in China*. 1st Harper Perennial ed. New York, NY: Harper Collins.
- Hsia, Pei-Hsuan, Tai, An-Shun, Kao, Chu-Lan Michael, Lin, Yu-Hsuan, and Lin, Sheng-Hsuan (2025). Causal Mediation Analysis for Difference-in-Difference Design and Panel Data. *Epidemiologic Methods* 14(1).
- Hsing, You-tien (2010). *The Great Urban Transformation: Politics of Land and Property in China*. 1st ed. Oxford University Press Oxford.
- Hu, Tina and Leahy, Joe (2024). How a Mao-era System Creates Second-Class Citizens in Modern China. *Financial Times*.

- Hu, Xiaojiang, Cook, Sarah, and Salazar, Miguel A. (2008). Internal Migration and Health in China. *The Lancet* 372(9651), pp. 1717–1719.
- Huang, Philip C. (1985). *The Peasant Economy and Social Change in North China*. Stanford, California: Stanford University Press.
- Imai, Kosuke, Keele, Luke, and Tingley, Dustin (2010). A General Approach to Causal Mediation Analysis. *Psychological Methods* 15(4), pp. 309–334.
- Imai, Kosuke, Keele, Luke, and Yamamoto, Teppei (2010). Identification, Inference and Sensitivity Analysis for Causal Mediation Effects. *Statistical Science* 25(1).
- Imbert, Clement, Seror, Marlon, Zhang, Yifan, and Zylberberg, Yanos (2022). Migrants and Firms: Evidence from China. *American Economic Review* 112(6), pp. 1885–1914.
- Kanbur, Ravi, Wang, Yue, and Zhang, Xiaobo (2021). The Great Chinese Inequality Turnaround. *Journal of Comparative Economics* 49(2), pp. 467–482.
- Kanbur, Ravi and Zhang, Xiaobo (2005). Fifty Years of Regional Inequality in China: A Journey Through Central Planning, Reform, and Openness. *Review of Development Economics* 9(1), pp. 87–106.
- Kartsonaki, Christiana, Jackson, Michelle, and Cox, David R. (2013). “Primary and Secondary Effects: Some Methodological Issues”. *Determined to Succeed?* Ed. by Michelle Jackson. Stanford University Press, pp. 34–55.
- Kaufman, Harrold F., Dudley, Duncan Otis, Gross, Neal, and Sewell, William H. (1953). Problems of Theory and Method in the Study of Social Stratification in Rural Society. *Rural Sociology* 18(1), pp. 12–24.
- Langley, William (2026). Chinese Migrant Workers Return Home as Urban Jobs Grow Scarcer. *Financial Times*.
- Lee, Ching Kwan (1995). Engendering the Worlds of Labor: Women Workers, Labor Markets, and Production Politics in the South China Economic Miracle. *American Sociological Review* 60(3), pp. 378–397.
- Lei, Ya-Wen (2023). *The Gilded Cage: Technology, Development, and State Capitalism in China*. Princeton ; Oxford: Princeton University Press.
- Lemieux, Thomas (2006). Increasing Residual Wage Inequality: Composition Effects, Noisy Data, or Rising Demand for Skill? *American Economic Review* 96(3), pp. 461–498.
- Lewis, W. Arthur (1954). Economic Development with Unlimited Supplies of Labor. *The Manchester School* 22(2).
- Li, Shi and Luo, Chuliang (2010). “Reestimating the Income Gap between Urban and Rural Households in China”. *One Country, Two Societies*. Ed. by Martin King Whyte. 1st ed. Vol. 16. Rural-Urban Inequality in Contemporary China. Harvard University Asia Center, pp. 105–122.
- Liang, Ying, Yi, Yingying, and Sun, Qiufen (2014). The Impact of Migration on Fertility under China’s Underlying Restrictions: A Comparative Study Between Permanent and Temporary Migrants. *Social Indicators Research* 116(1), pp. 307–326.
- Liang, Zai (2016). China’s Great Migration and the Prospects of a More Integrated Society. *Annual Review of Sociology* 42(Volume 42, 2016), pp. 451–471.
- Liang, Zai, Li, Zhen, and Ma, Zhongdong (2014). Changing Patterns of the Floating Population in China, 2000–2010. *Population and Development Review* 40(4), pp. 695–716.

- Liu, Licheng, Wang, Ye, and Xu, Yiqing (2024). A Practical Guide to Counterfactual Estimators for Causal Inference with Time-Series Cross-Sectional Data. *American Journal of Political Science* 68(1), pp. 160–176.
- Long, Jason (2005). Rural-Urban Migration and Socioeconomic Mobility in Victorian Britain. *The Journal of Economic History* 65(1), pp. 1–35.
- Lu, Hanchao (2010). “Small-Town China: A Historical Perspective on Rural-Urban Relations”. *One Country, Two Societies*. Ed. by Martin King Whyte. 1st ed. Vol. 16. Rural-Urban Inequality in Contemporary China. Harvard University Asia Center, pp. 29–54.
- Lu, Jingliang (2017). Capital Accumulation and Skill Premium: Evidence from China (zi ben ji lei yu jin neng gong zi cha ju: lai zi zhong guo de zheng ju). *China Economic Quarterly (Jing Ji Xue: Ji Kan)* 16(2), pp. 577–598.
- (2018). Evolution of Urban Wage Inequality: 1995–2013 (cheng zhen lao dong zhe gong zi bu ping deng de yan hua: 1995–2013). *China Economic Quarterly (Jing Ji Xue: Ji Kan)* 17(4), pp. 1305–1328.
- Lu, Jingliang and Feng, Shuaizhang (2025). Evolution of China’s Urban Labor Market: 1984–2024 (zhong guo cheng zhen lao dong li shi chang de yan hua: 1984–2024). *China Economic Quarterly (Jing Ji Xue: Ji Kan)* 25(3), pp. 547–564.
- Lu, Yao and Qin, Lijian (2014). Healthy Migrant and Salmon Bias Hypotheses: A Study of Health and Internal Migration in China. *Social Science & Medicine* 102, pp. 41–48.
- Lundberg, Ian (2024). The Gap-Closing Estimand: A Causal Approach to Study Interventions That Close Disparities Across Social Categories. *Sociological Methods & Research* 53(2), pp. 507–570.
- Luo, Wenfang, Jiang, Yanfeng, Qian, Junxi, and Feng, Dedong (2026). The Impact of Social Integration on the Fertility Intentions of Migrants: Evidence From China. *Population, Space and Place* 32(3), e70247.
- Maas, Ineke and van, Leeuwen Marco H. D. (2016). Toward Open Societies? Trends in Male Intergenerational Class Mobility in European Countries during Industrialization1. *American Journal of Sociology* 122(3), pp. 838–885.
- MacDonald, Daniel (2021). Internal Migration and Sectoral Shift in the Nineteenth-Century United States. *Social Science History* 45(4), pp. 843–862.
- Moretti, Enrico (2013). Real Wage Inequality. *American Economic Journal: Applied Economics* 5(1), pp. 65–103.
- Murphy, Rachel (2020). *The Children of China’s Great Migration*. Cambridge: Cambridge University Press.
- National Bureau of Statistics of China (2015). *China Statistical Yearbook, 2015*.
- (2021a). *China Population Census Yearbook, 2020*.
- (2021b). *Rural Urban Populations and the Floating Population*. The Seventh Population Census Report No. 7. National Bureau of Statistics of China.
- Nee, Victor (1989). A Theory of Market Transition: From Redistribution to Markets in State Socialism. *American Sociological Review* 54(5), pp. 663–681.
- (1991). Social Inequalities in Reforming State Socialism: Between Redistribution and Markets in China. *American Sociological Review* 56(3), pp. 267–282.
- (1996). The Emergence of a Market Society: Changing Mechanisms of Stratification in China. *American Journal of Sociology* 101(4), pp. 908–949.

- Oaxaca, Ronald L. and Ransom, Michael R. (1999). Identification in Detailed Wage Decompositions. *The Review of Economics and Statistics* 81(1), pp. 154–157.
- Obermaier, Amber and Hastings, Orestes P. (2025). Growing up Different(Ly than Last Time We Asked): Social Status and Changing Reports of Childhood Income Rank. *Social Indicators Research* 180(2), pp. 923–942.
- Opacic, Aleksei, Wei, Lai, and Zhou, Xiang (2025). Disparity Analysis: A Tale of Two Approaches. *Journal of the Royal Statistical Society Series A: Statistics in Society*, qnaf008.
- Pan, Zehan and Dong, Weizhen (2020). Can Money Substitute Adult Children’s Absence? Measuring Remittances’ Compensation Effect on the Health of Rural Migrants’ Left-behind Elderly Parents. *Journal of Rural Studies* 79, pp. 216–225.
- Piketty, Thomas, Yang, Li, and Zucman, Gabriel (2019). Capital Accumulation, Private Property, and Rising Inequality in China, 1978–2015. *American Economic Review* 109(7), pp. 2469–2496.
- Roca, Jorge De La and Puga, Diego (2017). Learning by Working in Big Cities. *The Review of Economic Studies* 84(1), pp. 106–142.
- Rozelle, Scott and Hell, Natalie (2022). *Invisible China: How the Urban-Rural Divide Threatens China’s Rise*. Chicago, IL: University of Chicago Press.
- Shi, Xinjie (2022). Moving out but Not for the Better: Health Consequences of Interprovincial Rural-Urban Migration in China. *Health Economics* 31(4), pp. 555–573.
- Sicular, Terry, Yue, XIMING, Gustafsson, Björn A., and SHI, LI (2010). “How Large Is China’s Rural-Urban Income Gap?” *One Country, Two Societies*. Ed. by Martin King Whyte. 1st ed. Vol. 16. Rural-Urban Inequality in Contemporary China. Harvard University Asia Center, pp. 85–104.
- Song, Jae, Price, David J, Guvenen, Fatih, Bloom, Nicholas, and von Wachter, Till (2019). Firming Up Inequality. *The Quarterly Journal of Economics* 134(1), pp. 1–50.
- Song, Qian (2017). Aging, and Separation from Children: The Health Implications of Adult Migration for Elderly Parents in Rural China. *Demographic Research* 37, pp. 1761–1792.
- Song, Xi, Massey, Catherine G., Rolf, Karen A., Ferrie, Joseph P., Rothbaum, Jonathan L., and Xie, Yu (2020). Long-Term Decline in Intergenerational Mobility in the United States since the 1850s. *Proceedings of the National Academy of Sciences* 117(1), pp. 251–258.
- Sun, Jingfang (2017). Changes to Hukou Discrimination in China’s Labor Market: Employment and Wages of Rural Migrant Workers. *Economic Research Journal (Jing Ji Yan Jiu)* (8), pp. 171–186.
- Tomaskovic-Devey, Donald, Rainey, Anthony, Avent-Holt, Dustin, Bandelj, Nina, Boza, István, Cort, David, Godechot, Olivier, Hajdu, Gergely, Hällsten, Martin, Henriksen, Lasse Folke, Hermansen, Are Skeie, Hou, Feng, Jung, Jiwook, Kanjuo-Mrčela, Aleksandra, King, Joe, Kodama, Naomi, Kristal, Tali, Křížková, Alena, Lippényi, Zoltán, Melzer, Silvia Maja, Mun, Eunmi, Penner, Andrew, Petersen, Trond, Poje, Andreja, Safi, Mirna, Thaning, Max, and Tufail, Zaibu (2020). Rising Between-Workplace Inequalities in High-Income Countries. *Proceedings of the National Academy of Sciences* 117(17), pp. 9277–9283.
- Tong, Yuying, Chen, Feinian, and Shu, Binbin (2019). Spousal Migration and Married Adults’ Psychological Distress in Rural China: The Roles of Intimacy, Autonomy and Responsibility. *Social Science Research* 83, p. 102312.

- Tong, Yuying and Piotrowski, Martin (2012). Migration and Health Selectivity in the Context of Internal Migration in China, 1997–2009. *Population Research and Policy Review* 31(4), pp. 497–543.
- Waight, Hannah and Goldstein, Adam (2025). The Polarization of Inequality Perceptions in the New Gilded Age¹. *American Journal of Sociology* 131(3), pp. 586–629.
- Walder, Andrew G. (1986). *Communist Neo-Traditionalism: Work and Authority in Chinese Industry*. University of California Press.
- (2023). China’s Extreme Inequality: The Structural Legacies of State Socialism. *The China Journal* 90, pp. 1–26.
- Wang, Fei-Ling (2004). Reformed Migration Control and New Targeted People: China’s Hukou System in the 2000s. *The China Quarterly* 177, pp. 115–132.
- Wang, Lamei and Mesman, Judi (2015). Child Development in the Face of Rural-to-Urban Migration in China. *Perspectives on Psychological Science*.
- Wang, Senhu and Hu, Yang (2019). Migration and Health in China: Linking Sending and Host Societies. *Population, Space and Place* 25(6), e2231.
- Wang, Xiaxin and Shen, Yan (2014). The Effect of China’s Agricultural Tax Abolition on Rural Families’ Incomes and Production. *China Economic Review* 29, pp. 185–199.
- Wang, Zhenwei (2026). Remittances as Moral Battlegrounds: Navigating the Qualitative Meanings of Money, Power, and Intimacy in Translocal Chinese Families. *Journal of Ethnic and Migration Studies* 52(5), pp. 1380–1397.
- Western, Bruce and Bloome, Deirdre (2009). Variance Function Regressions for Studying Inequality. *Sociological Methodology* 39(1), pp. 293–326.
- Whyte, Martin King (2010a). *Myth of the Social Volcano: Perceptions of Inequality and Distributive Injustice in Contemporary China*. Stanford, Calif: Stanford University Press.
- ed. (2010b). *One Country, Two Societies: Rural-Urban Inequality in Contemporary China*. 1st ed. Harvard University Asia Center.
- (2024). “Xi Jinping Confronts Inequality: Bold Leadership or Modest Steps?” *The Xi Jinping Effect*. Ed. by Ashley Esarey and Rongbin Han. University of Washington Press, pp. 93–128.
- Williams, Martin J. (2025). Causal Inference, Agency, and the Problem of Inherent Endogeneity. *Annual Review of Political Science*.
- World Bank (2025). *World Urbanization Prospects 2025*. UN DESA/POP/2025/TR No. 12. World Bank.
- Wu, Guogang (2022). *China’s Common Prosperity Program: Causes, Challenges, and Implications*. Policy Report. Asia Society Policy Institute.
- Wu, Jieh-Min (2010). “Rural Migrant Workers and China’s Differential Citizenship: A Comparative Institutional Analysis”. *One Country, Two Societies*. Ed. by Martin King Whyte. 1st ed. Vol. 16. Rural-Urban Inequality in Contemporary China. Harvard University Asia Center, pp. 55–82.
- Wu, Xiaogang (2019). Inequality and Social Stratification in Postsocialist China. *Annual Review of Sociology* 45(Volume 45, 2019), pp. 363–382.
- Wu, Xiaogang and Treiman, Donald (2007). Inequality and Equality under Chinese Socialism: The Hukou System and Intergenerational Occupational Mobility¹. *American Journal of Sociology*.

- Wu, Xiaogang and Xie, Yu (2003). Does the Market Pay off? Earnings Returns to Education in Urban China. *American Sociological Review* 68(3), p. 425.
- Wu, Xiaogang and Zheng, Bingdao (2018). Household Registration, Urban Status Attainment, and Social Stratification in China. *Research in Social Stratification and Mobility* 53, pp. 40–49.
- Xie, Yu, Dong, Hao, Zhou, Xiang, and Song, Xi (2022). Trends in Social Mobility in Postrevolution China. *Proceedings of the National Academy of Sciences* 119(7), e2117471119.
- Xie, Yu and Hannum, Emily (1996). Regional Variation in Earnings Inequality in Reform-Era Urban China. *American Journal of Sociology* 101(4), pp. 950–992.
- Xie, Yu and Wu, Xiaogang (2008). Danwei Profitability and Earnings Inequality in Urban China. *The China Quarterly* 195, pp. 558–581.
- Xie, Yu and Zhou, Xiang (2014). Income Inequality in Today’s China. *Proceedings of the National Academy of Sciences* 111(19), pp. 6928–6933.
- Xiong, Wanru (2023). Love Is Elsewhere: Internal Migration and Marriage Prospects in China. *European Journal of Population* 39(1), p. 6.
- Yi, Yang and Shangguan, Yiwen (2024). The Impact of the New Trend of Urban-to-Urban Migration on Fertility Behaviors. *Population, Space and Place* 30(6), e2772.
- Yip, Winnie (2010). “Disparities in Health Care and Health Status:” *One Country, Two Societies*. Ed. by Martin King Whyte. 1st ed. Vol. 16. Rural-Urban Inequality in Contemporary China. Harvard University Asia Center, pp. 147–165.
- You, Xijia (2026). Fertility Norms as Unifying Framework: Coordinating Migration and Fertility Decisions in China’s Internal Migration. *Population, Space and Place* 32(2), e70230.
- Yuan, Li (2023). They Propelled China’s Rise. Now They Have Nothing to Fall Back On. *The New York Times*.
- Yue, Ai, Bai, Yu, Shi, Yaojiang, Luo, Renfu, Rozelle, Scott, Medina, Alexis, and Sylvia, Sean (2020). Parental Migration and Early Childhood Development in Rural China. *Demography* 57(2), pp. 403–422.
- Yun, Myeong-Su (2005). A Simple Solution to the Identification Problem in Detailed Wage Decompositions. *Economic Inquiry* 43(4), pp. 766–772.
- Zhou, Dongxu (2025). Commentary: Why China Is Worried About Migrants Returning Home - Caixin Global. *Caixin*.
- Zhou, Min and Guo, Wei (2021). Comparison of Second-Child Fertility Intentions between Local and Migrant Women in Urban China: A Blinder–Oaxaca Decomposition. *Journal of Ethnic and Migration Studies* 47(11), pp. 2417–2438.
- Zhou, Xiang (2014). Increasing Returns to Education, Changing Labor Force Structure, and the Rise of Earnings Inequality in Urban China, 1996–2010. *Social Forces* 93(2), pp. 429–455.
- Zhou, Xiang and Wodtke, Geoffrey T. (2019). A Regression-with-Residuals Method for Estimating Controlled Direct Effects. *Political Analysis* 27(3), pp. 360–369.
- Zhou, Xiang and Xie, Yu (2019). Market Transition, Industrialization, and Social Mobility Trends in Postrevolution China. *American Journal of Sociology* 124(6), pp. 1810–1847.
- Zhou, Xue-Guang (2000). Economic Transformation and Income Inequality in Urban China: Evidence from Panel Data. *American Journal of Sociology* 105(4), pp. 1135–1174.

Zhuo, Yue and Liang, Zai (2015). “Migration and Wellbeing of the Elderly in Rural China”. *Handbook of Chinese Migration*. Edward Elgar Publishing. Chap. Handbook of Chinese Migration, pp. 126–147.

Appendix A

Supplementary Materials for Chapter 2

A.1 LHSC and CGSS Data Processing

In Chapter 2, to ensure comparability with X. Zhou (2014)'s results based on the urban sample, I construct the LHSC and CGSS samples following the original procedure in the replication materials generously provided by Xiang Zhou. Besides including the rural, agricultural observations, I make two modifications to X. Zhou (2014)'s original procedure.

First, in X. Zhou (2014)'s original procedure, he uses only labour income in the 2010 sample, while the 1996 measure includes labour income and average household profits. For consistency, I use the total income measure, which includes income from labour and other sources.

Second, X. Zhou (2014) excludes observations with annual income less than 100 yuan. I changed the threshold to 50 yuan.

A.2 Post-weighting Balance Measures

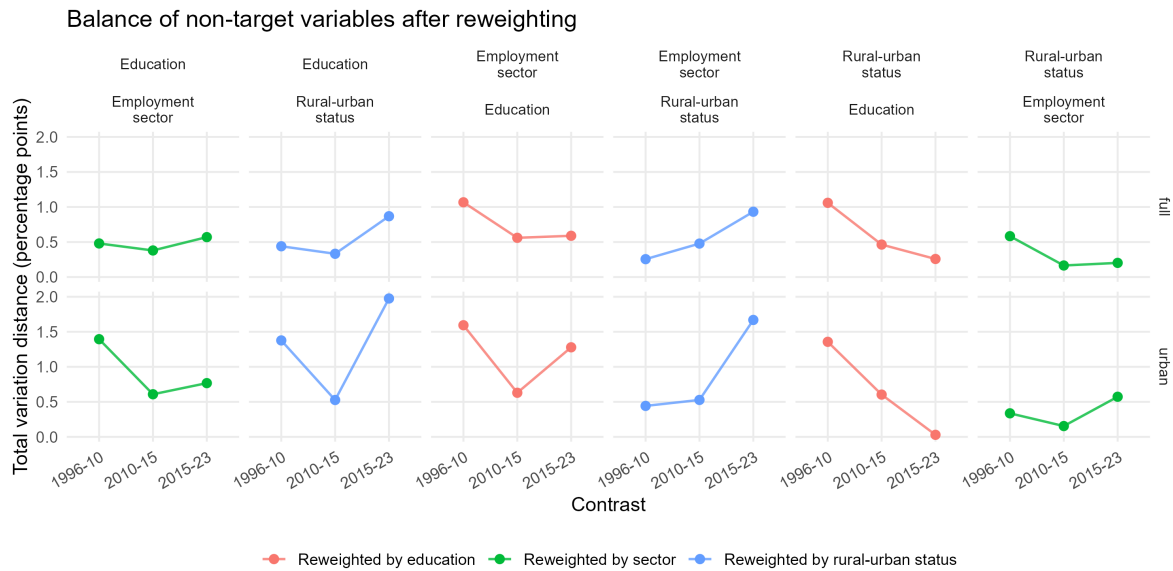


Figure A.1: Total Variance Distance of the Post-weighting Distributions of Non-target Variables

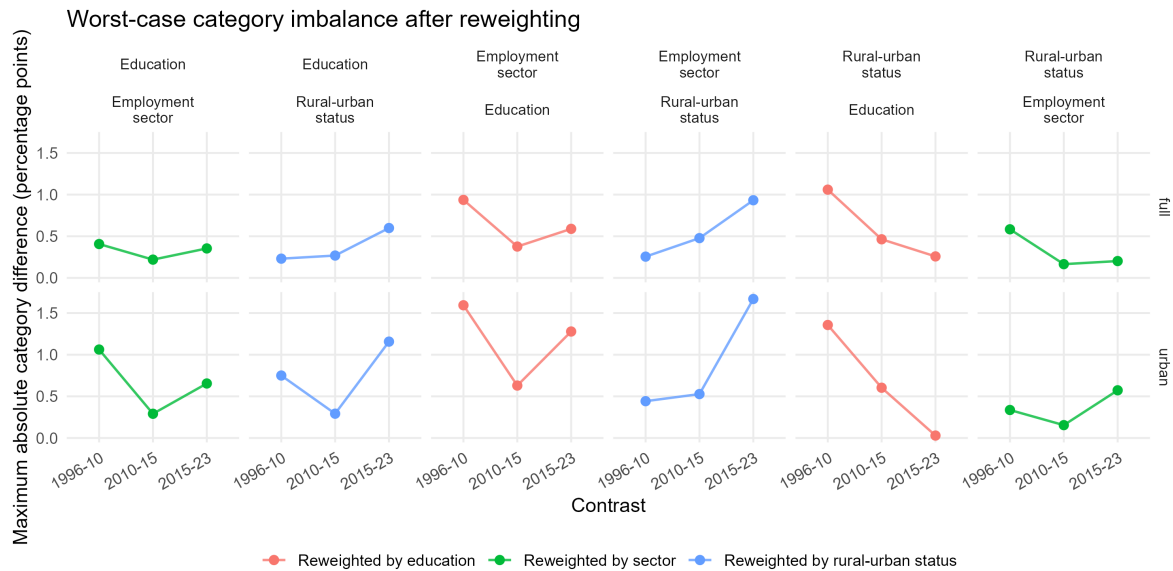


Figure A.2: Worst Proportional Difference of the Post-weighting Distributions of Non-target Variables

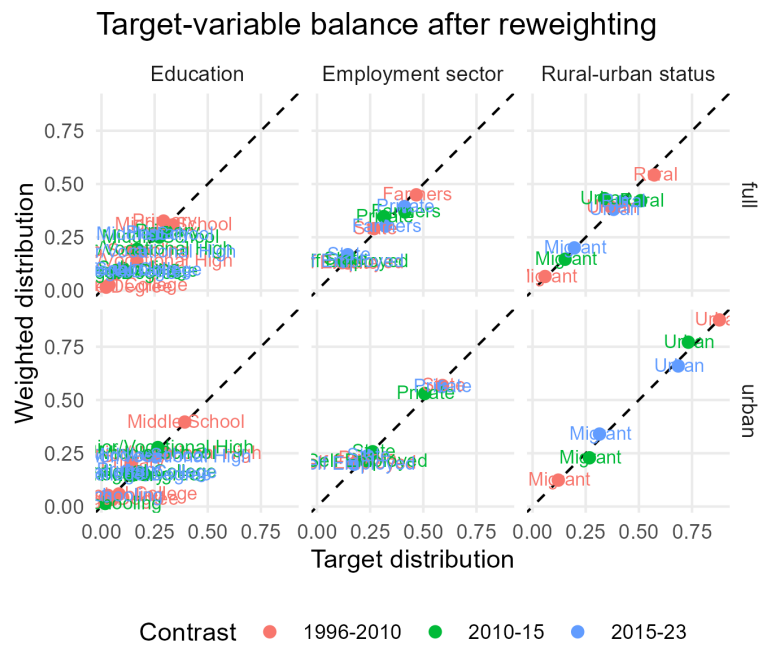


Figure A.3: True and Post-weighted Proportions of the Target Variable

A.3 Additional Regression Results

	1996	2010	2015	2023
Female	−0.146 (0.017)	−0.369 (0.019)	−0.359 (0.021)	−0.369 (0.032)
Han	0.026 (0.035)	0.017 (0.044)	0.064 (0.049)	−0.042 (0.067)
Age	0.043 (0.006)	0.048 (0.007)	0.057 (0.007)	0.072 (0.013)
Age squared	0.000 (0.000)	−0.001 (0.000)	−0.001 (0.000)	−0.001 (0.000)
CCP member	0.077 (0.021)	0.103 (0.028)	0.094 (0.028)	0.011 (0.045)
Rural	−0.087 (0.030)	−0.051 (0.030)	−0.150 (0.034)	−0.216 (0.049)
Migrant	0.071 (0.043)	−0.036 (0.032)	0.012 (0.030)	−0.048 (0.041)
No schooling	−0.303 (0.042)	−0.390 (0.045)	−0.589 (0.067)	−0.292 (0.094)
Primary school	−0.050 (0.029)	−0.200 (0.027)	−0.173 (0.035)	−0.222 (0.057)
Senior/Vocational	0.065 (0.022)	0.208 (0.029)	0.171 (0.030)	0.228 (0.047)
Vocational college	0.162 (0.028)	0.587 (0.036)	0.468 (0.037)	0.367 (0.052)
College degree	0.309 (0.034)	0.878 (0.040)	0.693 (0.038)	0.598 (0.064)
Private sector	−0.069 (0.031)	−0.126 (0.025)	−0.006 (0.025)	−0.013 (0.043)
Self-employed	−0.348 (0.075)	0.257 (0.047)	0.318 (0.046)	0.317 (0.057)
Farmers	−1.119 (0.037)	−0.684 (0.036)	−0.516 (0.044)	−0.644 (0.066)
Never married	0.022 (0.030)	−0.106 (0.044)	−0.119 (0.035)	−0.064 (0.055)
Other	−0.078 (0.042)	0.003 (0.054)	−0.019 (0.037)	−0.018 (0.049)
Num.Obs.	4777	6198	5334	2378
R2	0.504	0.557	0.476	0.422
Province FE	Yes	Yes	Yes	Yes

Table A.1: Regression Results for Mean Logged Income by Year, Full Sample

	1996	2010	2015	2023
Female	−0.141 (0.020)	−0.299 (0.023)	−0.293 (0.026)	−0.312 (0.037)
Han	0.011 (0.069)	0.066 (0.062)	0.024 (0.070)	−0.057 (0.140)
Age	0.023 (0.009)	0.048 (0.010)	0.052 (0.010)	0.029 (0.020)
Age squared	0.000 (0.000)	−0.001 (0.000)	−0.001 (0.000)	0.000 (0.000)
CCP member	0.080 (0.023)	0.103 (0.030)	0.104 (0.032)	−0.044 (0.055)
Migrant	−0.021 (0.045)	−0.004 (0.033)	0.030 (0.033)	−0.006 (0.044)
No schooling	−0.320 (0.073)	−0.518 (0.108)	−0.798 (0.188)	0.027 (0.108)
Primary school	−0.048 (0.050)	−0.281 (0.044)	−0.208 (0.057)	−0.283 (0.095)
Senior/Vocational	0.060 (0.026)	0.249 (0.033)	0.170 (0.036)	0.277 (0.062)
Vocational college	0.178 (0.031)	0.630 (0.037)	0.470 (0.042)	0.407 (0.064)
College degree	0.301 (0.038)	0.938 (0.042)	0.711 (0.043)	0.722 (0.080)
Private sector	−0.064 (0.039)	−0.136 (0.026)	0.002 (0.028)	−0.013 (0.047)
Self-employed	−0.293 (0.120)	0.263 (0.057)	0.314 (0.056)	0.373 (0.068)
Never married	−0.003 (0.041)	−0.034 (0.055)	−0.072 (0.039)	−0.099 (0.074)
Other	−0.087 (0.056)	0.148 (0.067)	−0.064 (0.056)	−0.076 (0.062)
Num.Obs.	2021	3106	2801	1292
R2	0.262	0.471	0.358	0.304
Province FE	Yes	Yes	Yes	Yes

Table A.2: Regression Results for Mean Logged Income by Year, Urban Sample

	1996	2010	2015	2023
Female	0.006 (0.058)	−0.080 (0.049)	−0.021 (0.065)	0.005 (0.078)
Han	−0.031 (0.108)	−0.210 (0.073)	−0.075 (0.112)	0.058 (0.131)
Age	−0.039 (0.018)	−0.028 (0.015)	−0.075 (0.019)	−0.046 (0.027)
Age squared	0.001 (0.000)	0.000 (0.000)	0.001 (0.000)	0.001 (0.000)
CCP member	−0.158 (0.096)	−0.035 (0.079)	−0.199 (0.107)	0.048 (0.122)
Rural	−0.129 (0.098)	−0.032 (0.077)	0.076 (0.099)	0.199 (0.117)
Migrant	−0.068 (0.130)	0.007 (0.080)	−0.034 (0.095)	0.044 (0.106)
No schooling	−0.197 (0.101)	0.171 (0.093)	0.365 (0.131)	−0.014 (0.185)
Primary school	−0.057 (0.075)	0.094 (0.063)	0.210 (0.089)	0.111 (0.112)
Senior/Vocational	−0.199 (0.085)	0.174 (0.075)	−0.012 (0.098)	−0.208 (0.120)
Vocational college	−0.540 (0.158)	0.083 (0.105)	0.008 (0.127)	−0.309 (0.149)
College degree	−0.545 (0.195)	0.150 (0.115)	0.040 (0.131)	0.044 (0.157)
Private sector	0.914 (0.098)	0.283 (0.081)	0.216 (0.098)	0.120 (0.135)
Self-employed	1.788 (0.108)	1.031 (0.096)	0.696 (0.126)	0.500 (0.153)
Farmers	1.242 (0.116)	0.733 (0.100)	0.656 (0.128)	0.542 (0.160)
Never married	−0.183 (0.107)	0.064 (0.110)	−0.033 (0.107)	0.270 (0.125)
Other	−0.228 (0.141)	0.102 (0.118)	0.063 (0.103)	−0.078 (0.128)
Num.Obs.	4777	6198	5334	2378
Log.Lik.	−250.903	−1427.014	−1418.069	−782.974

Table A.3: Regression Results for Logged Income Variance by Year, Full Sample

	1996	2010	2015	2023
Female	−0.094 (0.097)	−0.243 (0.068)	−0.109 (0.095)	0.077 (0.105)
Han	−0.478 (0.243)	−0.225 (0.123)	−0.218 (0.200)	−0.242 (0.269)
Age	−0.113 (0.034)	−0.071 (0.025)	−0.120 (0.031)	−0.091 (0.040)
Age squared	0.002 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
CCP member	−0.303 (0.134)	−0.053 (0.093)	−0.341 (0.142)	0.148 (0.162)
Migrant	−0.240 (0.157)	0.089 (0.086)	−0.063 (0.114)	0.106 (0.115)
No schooling	−0.404 (0.293)	0.273 (0.257)	0.812 (0.334)	−0.516 (0.358)
Primary school	0.046 (0.153)	0.254 (0.110)	0.418 (0.165)	0.074 (0.185)
Senior/Vocational	−0.380 (0.115)	0.270 (0.094)	0.040 (0.138)	−0.305 (0.156)
Vocational college	−0.651 (0.183)	0.213 (0.114)	0.217 (0.160)	−0.413 (0.183)
College degree	−0.655 (0.222)	0.337 (0.123)	0.235 (0.164)	0.041 (0.189)
Private sector	0.796 (0.122)	0.332 (0.083)	0.234 (0.117)	0.085 (0.152)
Self-employed	1.857 (0.140)	1.213 (0.102)	0.787 (0.155)	0.536 (0.175)
Never married	−0.092 (0.186)	0.184 (0.137)	−0.143 (0.143)	0.387 (0.159)
Other	−0.119 (0.258)	0.153 (0.178)	0.344 (0.160)	0.040 (0.174)
Num.Obs.	2021	3106	2801	1292
Log.Lik.	1018.904	−143.516	104.330	−147.147

Table A.4: Regression Results for Logged Income Variance by Year, Urban Sample

A.4 Total Decomposition Result

Table A.5: Total Decomposition Results

Component	All: Est (SE)	All: % (SE)	Urban: Est (SE)	Urban: % (SE)
1996-2010, Ref.=2010				
$\hat{\Delta}$	0.317 (0.014)		0.368 (0.017)	
$\delta^{\hat{B}}$	0.071 (0.009)	22.5 (2.9)	0.131 (0.007)	35.6 (1.9)
$\delta^{\hat{W}}$	0.018 (0.010)	5.8 (3.2)	0.024 (0.015)	6.6 (4.2)
$\delta^{\hat{D}}$	0.201 (0.007)	63.4 (2.3)	0.121 (0.006)	33.0 (1.6)
$\delta^{\hat{A}}$	0.026 (0.005)	8.3 (1.6)	0.091 (0.008)	24.8 (2.2)
$\delta^{\hat{B}} + \delta^{\hat{W}}$	0.090 (0.014)	28.4 (4.3)	0.156 (0.017)	42.3 (4.7)
$\delta^{\hat{D}} + \delta^{\hat{A}}$	0.227 (0.009)	71.6 (2.7)	0.213 (0.009)	57.7 (2.6)
2010-15, Ref.=2015				
$\hat{\Delta}$	0.090 (0.023)		-0.084 (0.026)	
$\delta^{\hat{B}}$	-0.050 (0.016)	-56.1 (17.8)	-0.108 (0.011)	127.8 (12.7)
$\delta^{\hat{W}}$	0.104 (0.015)	116.2 (17.2)	-0.033 (0.019)	39.3 (22.6)
$\delta^{\hat{D}}$	0.050 (0.008)	55.7 (8.9)	0.038 (0.006)	-44.8 (6.9)
$\delta^{\hat{A}}$	-0.014 (0.005)	-15.8 (6.0)	0.019 (0.006)	-22.3 (7.2)
$\delta^{\hat{B}} + \delta^{\hat{W}}$	0.054 (0.022)	60.1 (24.9)	-0.141 (0.025)	167.2 (29.4)
$\delta^{\hat{D}} + \delta^{\hat{A}}$	0.036 (0.011)	39.9 (12.0)	0.057 (0.009)	-67.2 (10.2)
2015-23, Ref.=2023				
$\hat{\Delta}$	-0.083 (0.038)		-0.062 (0.033)	
$\delta^{\hat{B}}$	0.036 (0.028)	-43.6 (33.8)	-0.042 (0.014)	68.2 (22.7)
$\delta^{\hat{W}}$	-0.012 (0.026)	14.2 (30.8)	-0.045 (0.024)	73.7 (39.5)
$\delta^{\hat{D}}$	-0.112 (0.017)	135.2 (20.7)	-0.024 (0.008)	38.6 (13.5)
$\delta^{\hat{A}}$	0.005 (0.011)	-5.7 (13.1)	0.050 (0.011)	-80.5 (18.4)
$\delta^{\hat{B}} + \delta^{\hat{W}}$	0.024 (0.038)	-29.5 (46.2)	-0.088 (0.031)	141.9 (50.3)
$\delta^{\hat{D}} + \delta^{\hat{A}}$	-0.108 (0.021)	129.5 (25.4)	0.026 (0.015)	-41.9 (23.9)

Notes: Percent contributions are computed as (component/total change) $\times$ 100. Standard errors (SE) in parentheses are calculated based on 250 bootstrap simulation.

$\hat{\Delta}$: Total change in logged income variance; $\delta^{\hat{B}}$: Change due to changes in between-group income variance; $\delta^{\hat{W}}$: Change due to changes in within-group income variance; $\delta^{\hat{D}}$: changes in between-group variance due compositional to changes; $\delta^{\hat{A}}$: changes in within-group variance due compositional to changes.

A.5 Results without Including Employment Sectors

A.5.1 Regression Results

	1996	2010	2015	2023
Female	−0.177 (0.022)	−0.417 (0.021)	−0.392 (0.023)	−0.355 (0.036)
Han	0.014 (0.044)	0.038 (0.052)	0.104 (0.055)	−0.049 (0.074)
Age	0.035 (0.008)	0.056 (0.007)	0.070 (0.008)	0.102 (0.013)
Age ²	0.000 (0.000)	−0.001 (0.000)	−0.001 (0.000)	−0.001 (0.000)
CCP member	0.159 (0.027)	0.132 (0.029)	0.050 (0.031)	0.049 (0.053)
Rural	−0.810 (0.030)	−0.416 (0.031)	−0.412 (0.034)	−0.465 (0.053)
Migrant	−0.097 (0.059)	−0.079 (0.037)	−0.017 (0.032)	−0.098 (0.048)
No schooling	−0.573 (0.046)	−0.528 (0.051)	−0.698 (0.068)	−0.470 (0.095)
Primary school	−0.208 (0.035)	−0.272 (0.031)	−0.266 (0.037)	−0.334 (0.056)
Senior/Vocational	0.154 (0.030)	0.239 (0.034)	0.167 (0.033)	0.249 (0.054)
Vocational college	0.250 (0.040)	0.578 (0.037)	0.448 (0.040)	0.304 (0.061)
College degree	0.403 (0.041)	0.863 (0.041)	0.670 (0.041)	0.510 (0.068)
Never married	0.006 (0.041)	−0.147 (0.044)	−0.150 (0.036)	−0.049 (0.059)
Other	0.018 (0.052)	0.067 (0.066)	−0.071 (0.038)	0.029 (0.056)
Num.Obs.	4777	6198	5334	2378
R^2	0.444	0.545	0.463	0.414
Province FE	Yes	Yes	Yes	Yes

Table A.6: Regression Results for Mean Logged Income by Year, Without Sector, Full Sample

	1996	2010	2015	2023
Female	−0.117 (0.024)	−0.306 (0.027)	−0.308 (0.027)	−0.327 (0.040)
Han	0.005 (0.085)	0.036 (0.078)	−0.012 (0.079)	−0.031 (0.146)
Age	0.008 (0.011)	0.048 (0.012)	0.057 (0.010)	0.055 (0.020)
Age ²	0.000 (0.000)	−0.001 (0.000)	−0.001 (0.000)	−0.001 (0.000)
CCP member	0.142 (0.027)	0.107 (0.032)	0.031 (0.034)	−0.093 (0.056)
Migrant	−0.093 (0.056)	−0.011 (0.040)	0.004 (0.034)	−0.020 (0.047)
No schooling	−0.579 (0.097)	−0.576 (0.122)	−0.748 (0.196)	−0.003 (0.126)
Primary school	−0.149 (0.065)	−0.288 (0.054)	−0.198 (0.060)	−0.355 (0.101)
Senior/Vocational	0.112 (0.034)	0.220 (0.039)	0.127 (0.038)	0.296 (0.070)
Vocational college	0.225 (0.041)	0.598 (0.041)	0.436 (0.046)	0.311 (0.069)
College degree	0.343 (0.043)	0.944 (0.045)	0.686 (0.047)	0.574 (0.081)
Never married	−0.047 (0.052)	−0.077 (0.057)	−0.091 (0.041)	−0.085 (0.076)
Other	−0.013 (0.063)	0.139 (0.089)	−0.165 (0.057)	−0.007 (0.068)
Num.Obs.	2021	3106	2801	1292
R^2	0.254	0.442	0.328	0.277
Province FE	Yes	Yes	Yes	Yes

Table A.7: Regression Results for Mean Logged Income by Year, Without Sector, Urban Sample

	1996	2010	2015	2023
Female	−0.091 (0.066)	−0.087 (0.052)	0.013 (0.063)	0.012 (0.073)
Han	0.122 (0.124)	−0.224 (0.078)	−0.164 (0.109)	0.074 (0.123)
Age	−0.046 (0.020)	−0.017 (0.017)	−0.066 (0.019)	−0.044 (0.025)
Age ²	0.001 (0.000)	0.000 (0.000)	0.001 (0.000)	0.001 (0.000)
CCP member	−0.498 (0.108)	−0.070 (0.085)	−0.246 (0.104)	−0.002 (0.114)
Rural	0.359 (0.080)	0.160 (0.071)	0.316 (0.085)	0.318 (0.103)
Migrant	0.323 (0.144)	0.203 (0.085)	0.064 (0.091)	0.148 (0.099)
No schooling	−0.150 (0.115)	0.140 (0.099)	0.313 (0.126)	−0.005 (0.173)
Primary school	−0.068 (0.086)	0.061 (0.067)	0.198 (0.086)	0.090 (0.104)
Senior/Vocational	−0.370 (0.097)	0.037 (0.080)	−0.080 (0.095)	−0.204 (0.112)
Vocational college	−0.569 (0.181)	−0.173 (0.110)	−0.111 (0.122)	−0.476 (0.136)
College degree	−0.975 (0.222)	−0.192 (0.121)	−0.094 (0.125)	−0.132 (0.140)
Never married	−0.124 (0.123)	−0.021 (0.118)	−0.066 (0.104)	0.185 (0.116)
Other	−0.154 (0.163)	0.137 (0.127)	−0.039 (0.101)	−0.168 (0.120)
Num.Obs.	4777	6198	5334	2378
Log.Lik.	−1107.688	−2073.712	−1840.971	−997.100

Table A.8: Regression Results for Logged Income Variance by Year, Without Sector, Full Sample

	1996	2010	2015	2023
Female	−0.250 (0.121)	−0.361 (0.080)	−0.143 (0.095)	0.032 (0.103)
Han	−0.440 (0.303)	−0.309 (0.144)	−0.280 (0.199)	−0.204 (0.263)
Age	−0.113 (0.043)	−0.071 (0.030)	−0.093 (0.031)	−0.047 (0.039)
Age ²	0.001 (0.001)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
CCP member	−0.913 (0.163)	−0.125 (0.108)	−0.468 (0.141)	0.094 (0.159)
Migrant	0.030 (0.189)	0.178 (0.100)	−0.007 (0.113)	0.107 (0.113)
No schooling	−0.216 (0.364)	0.058 (0.303)	0.756 (0.333)	−0.453 (0.350)
Primary school	0.139 (0.190)	0.149 (0.129)	0.362 (0.165)	−0.003 (0.180)
Senior/Vocational	−0.604 (0.141)	0.049 (0.110)	−0.064 (0.137)	−0.280 (0.153)
Vocational college	−0.687 (0.226)	−0.137 (0.131)	0.080 (0.158)	−0.610 (0.175)
College degree	−1.102 (0.273)	−0.142 (0.141)	0.106 (0.161)	−0.230 (0.175)
Never married	−0.125 (0.230)	0.024 (0.161)	−0.176 (0.143)	0.398 (0.155)
Other	−0.078 (0.321)	0.237 (0.209)	0.243 (0.160)	0.066 (0.170)
Num.Obs.	2021	3106	2801	1292
Log.Lik.	677.512	−302.331	21.639	−145.773

Table A.9: Regression Results for Logged Income Variance by Year, Without Sector, Urban Sample

A.5.2 Counterfactual Results

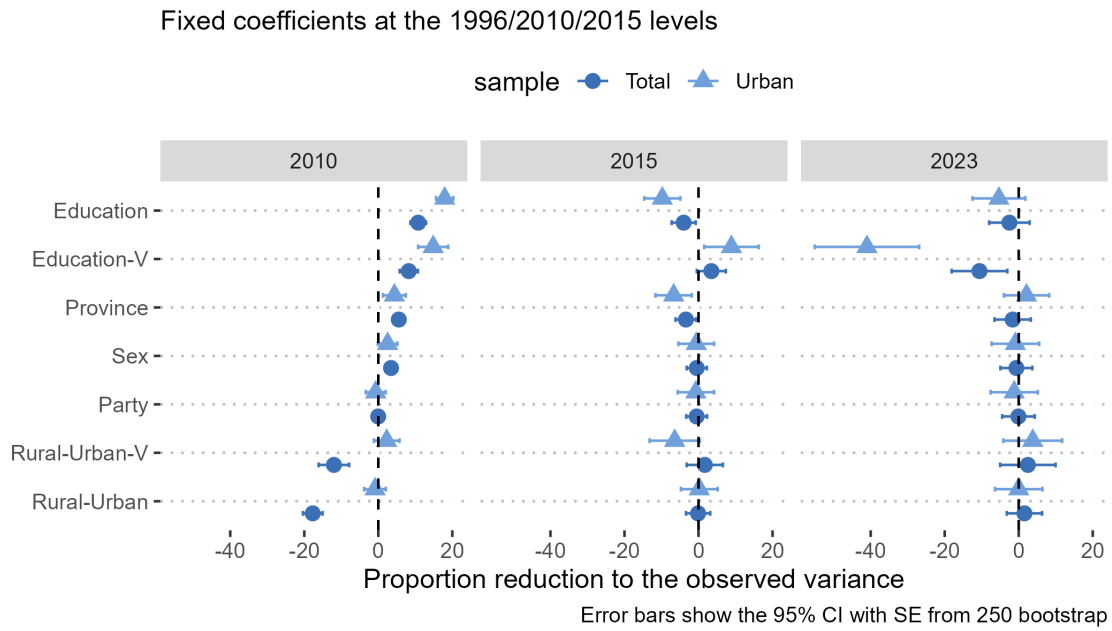


Figure A.4: Variance under Counterfactual Conditional Mean and Variance, without Employment Sector

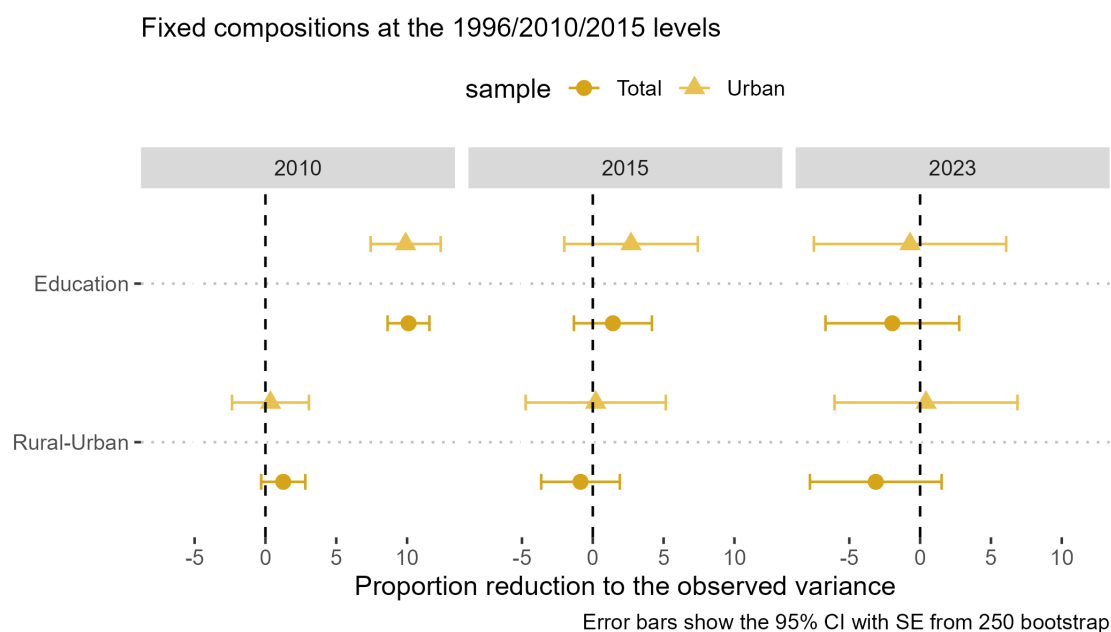


Figure A.5: Variance under Counterfactual Compositions - Between, without Employment Sector

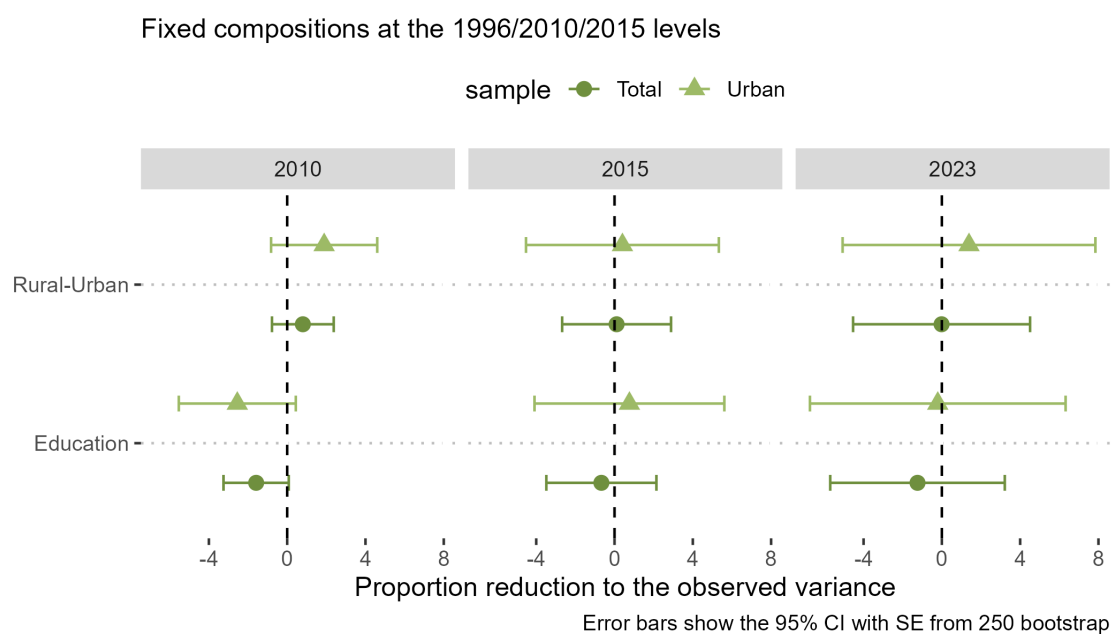


Figure A.6: Variance under Counterfactual Compositions - Within, without Employment Sector

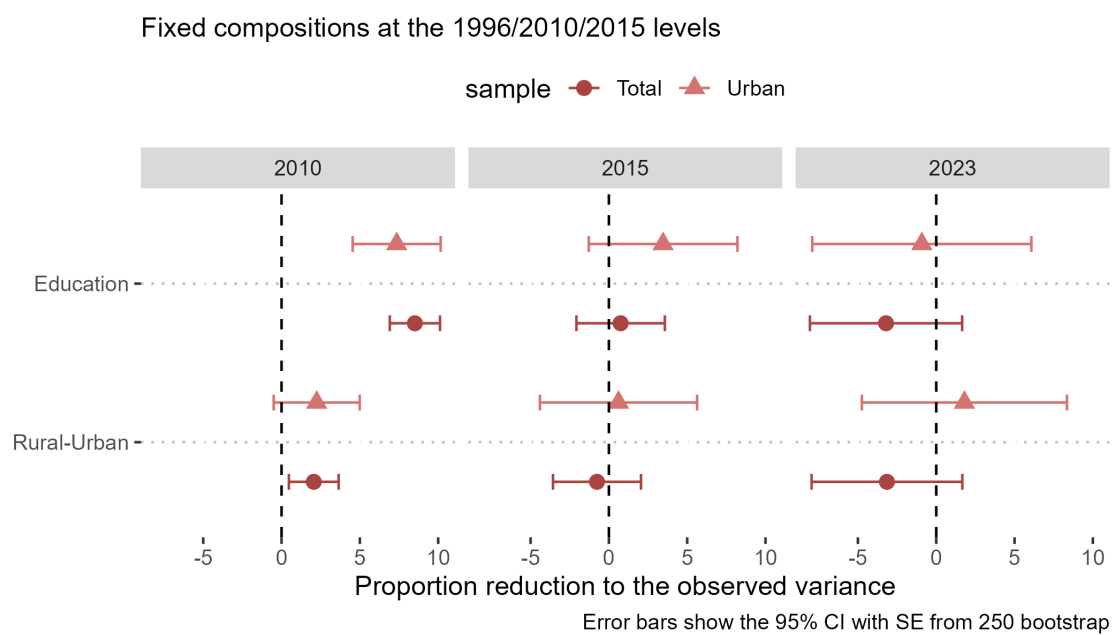


Figure A.7: Variance under Counterfactual Compositions - Total, without Employment Sector

Appendix B

Supplementary Materials for Chapter 3

B.1 Agricultural Employment Trajectory

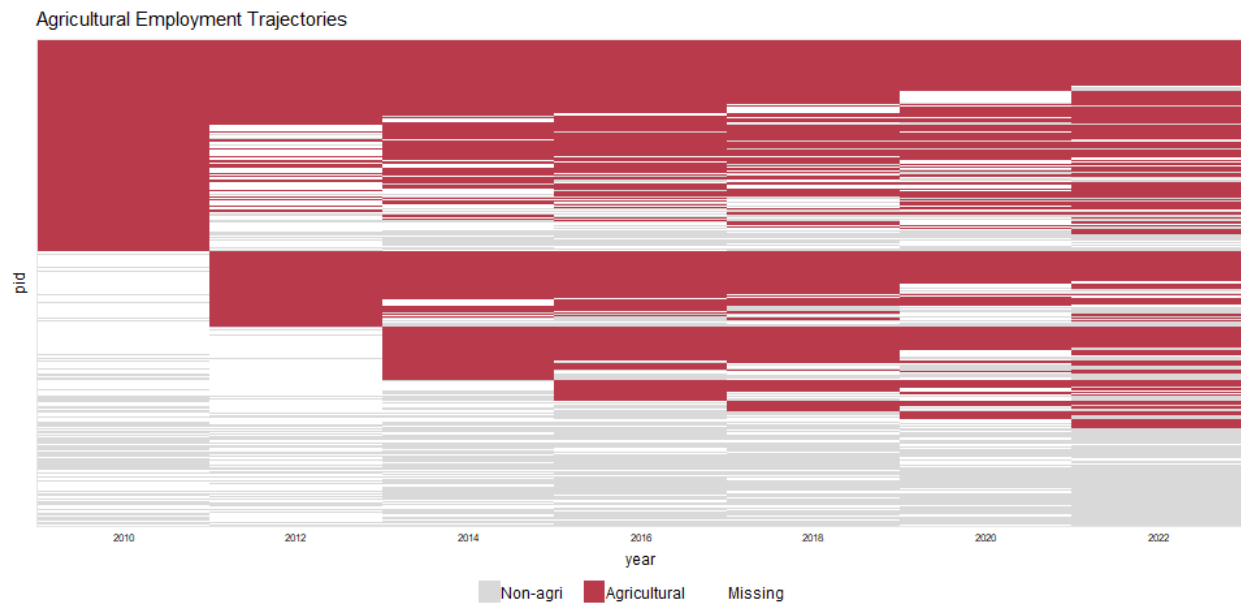
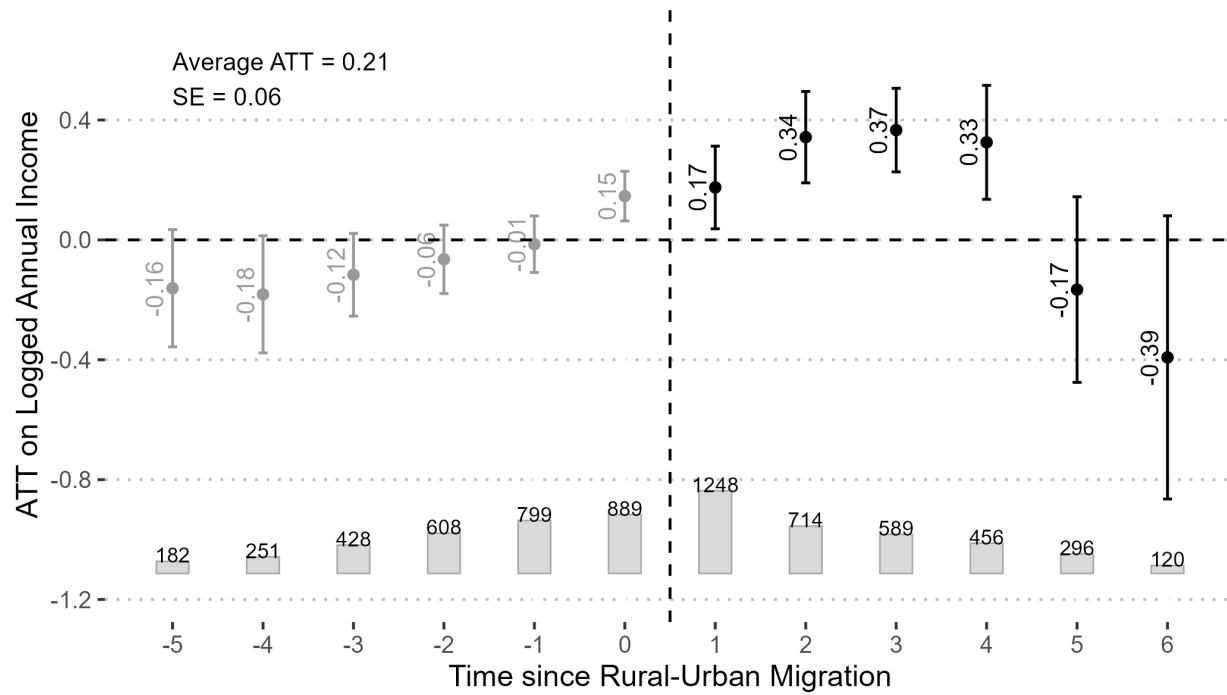


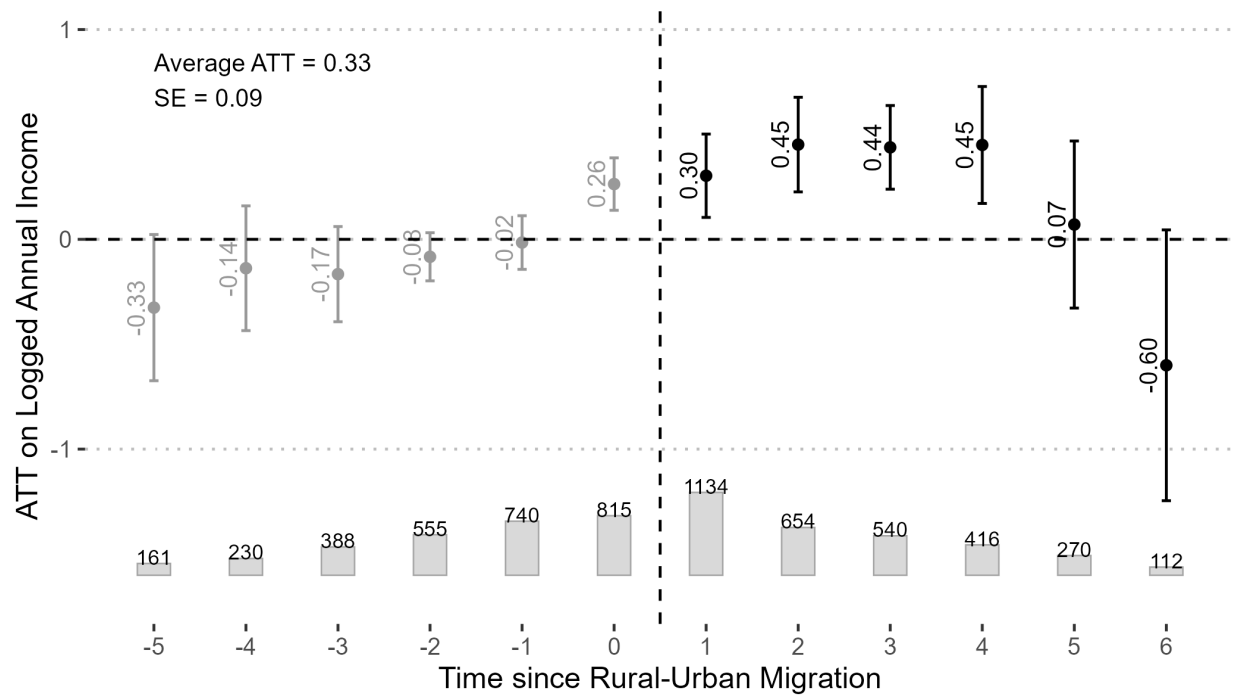
Figure B.1: Agricultural Employment Trajectory

B.2 Additional Results



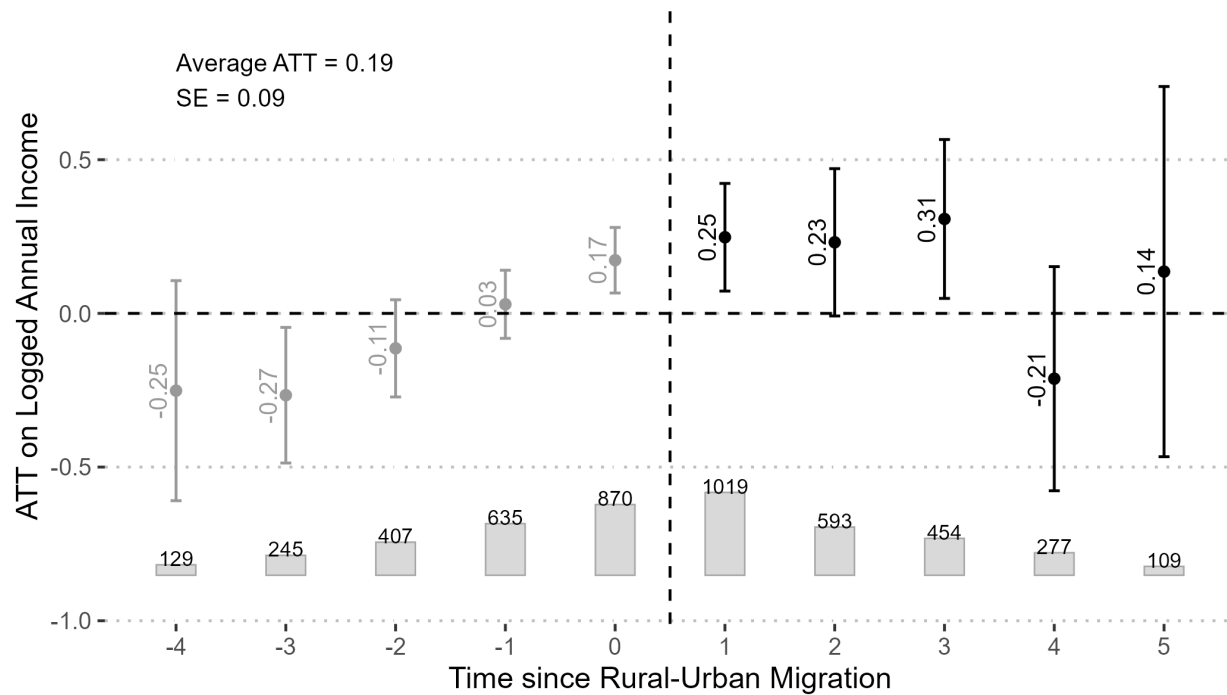
Notes: $t = 1$ indicates the first treated period. Error bars show the 95% confidence interval with S.E. simulated from bootstraps with 500 iterations. The bars and numbers at the bottom show the number of units used to estimate the period-specific effect. Panel weights are used in model estimation and result aggregation.

Figure B.2: Dynamic ATT of Rural-Urban Migration on Logged Annual Income, Unweighted



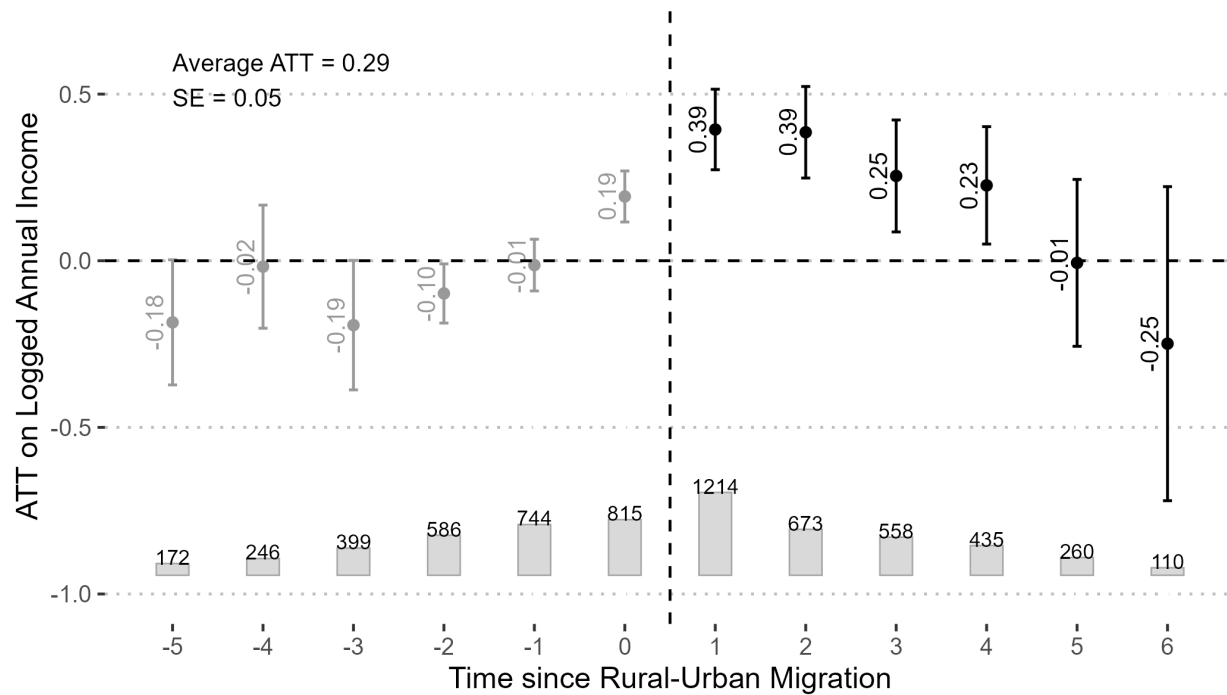
Notes: $t = 1$ indicates the first treated period. Error bars show the 95% confidence interval with S.E. simulated from bootstraps with 500 iterations. The bars and numbers at the bottom show the number of units used to estimate the period-specific effect. Panel weights are used in model estimation and result aggregation.

Figure B.3: Dynamic ATT of Rural-Urban Migration on Logged Annual Income, with Family Controls



Notes: $t = 1$ indicates the first treated period. Error bars show the 95% confidence interval with S.E. simulated from bootstraps with 500 iterations. The bars and numbers at the bottom show the number of units used to estimate the period-specific effect. Panel weights are used in model estimation and result aggregation.

Figure B.4: Dynamic ATT of Rural-Urban Migration on Logged Annual Income, Lagged Treatment



Notes: $t = 1$ indicates the first treated period. Error bars show the 95% confidence interval with S.E. simulated from bootstraps with 500 iterations. The bars and numbers at the bottom show the number of units used to estimate the period-specific effect. Panel weights are used in model estimation and result aggregation.

Figure B.5: Dynamic ATT of Rural-Urban Migration on Logged Annual Income, with Trimmed Income